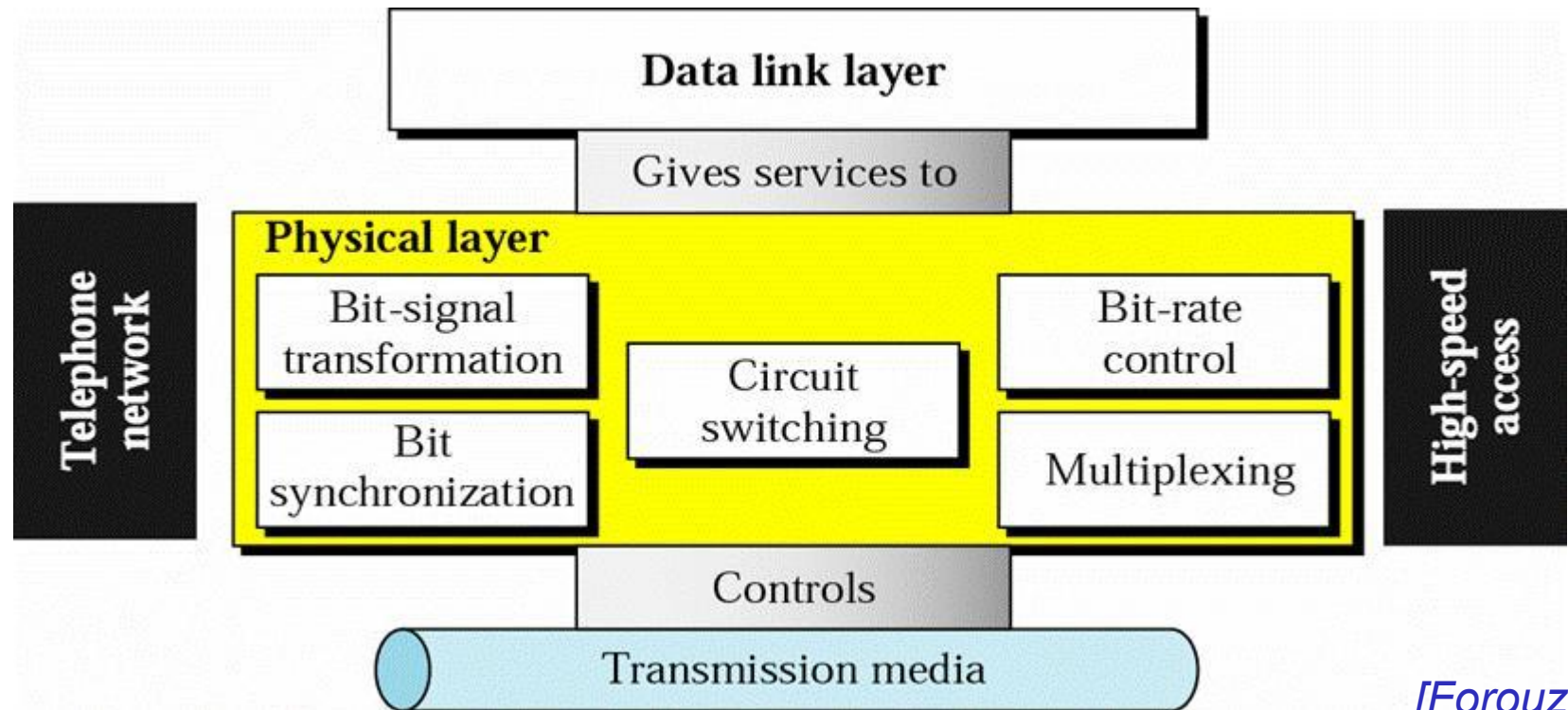


Chapter 2

The Physical Layer

Position and Function of Physical Layer

- The lowest layer in our protocol model.
- It defines the electrical, timing and other interfaces by which bits are sent as signals over channels.



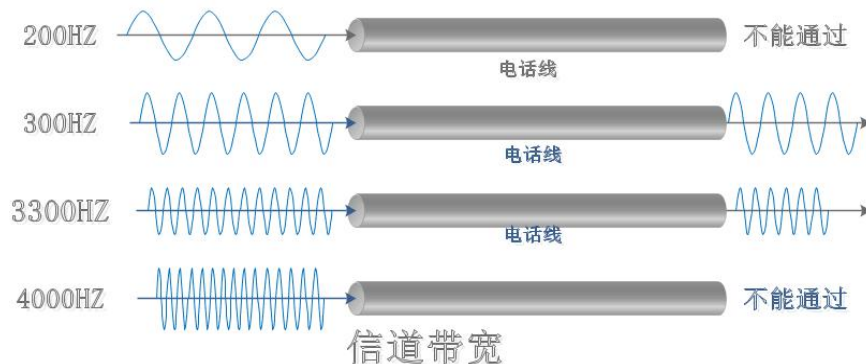
[Forouzan]

Outline

- 2.0 Basic Concepts on Data Communications
- 2.1 Theoretical Basis for Data Communication
- 2.2 Guided Transmission Media(Self-study)
- 2.3 Wireless Transmission(Self-study)
- 2.4 Communication Satellites(Self-study)
- 2.5 Digital Modulation and Multiplexing
- 2.6 The Public Switched Telephone Network
- (补充) Protocol Example: RS-232

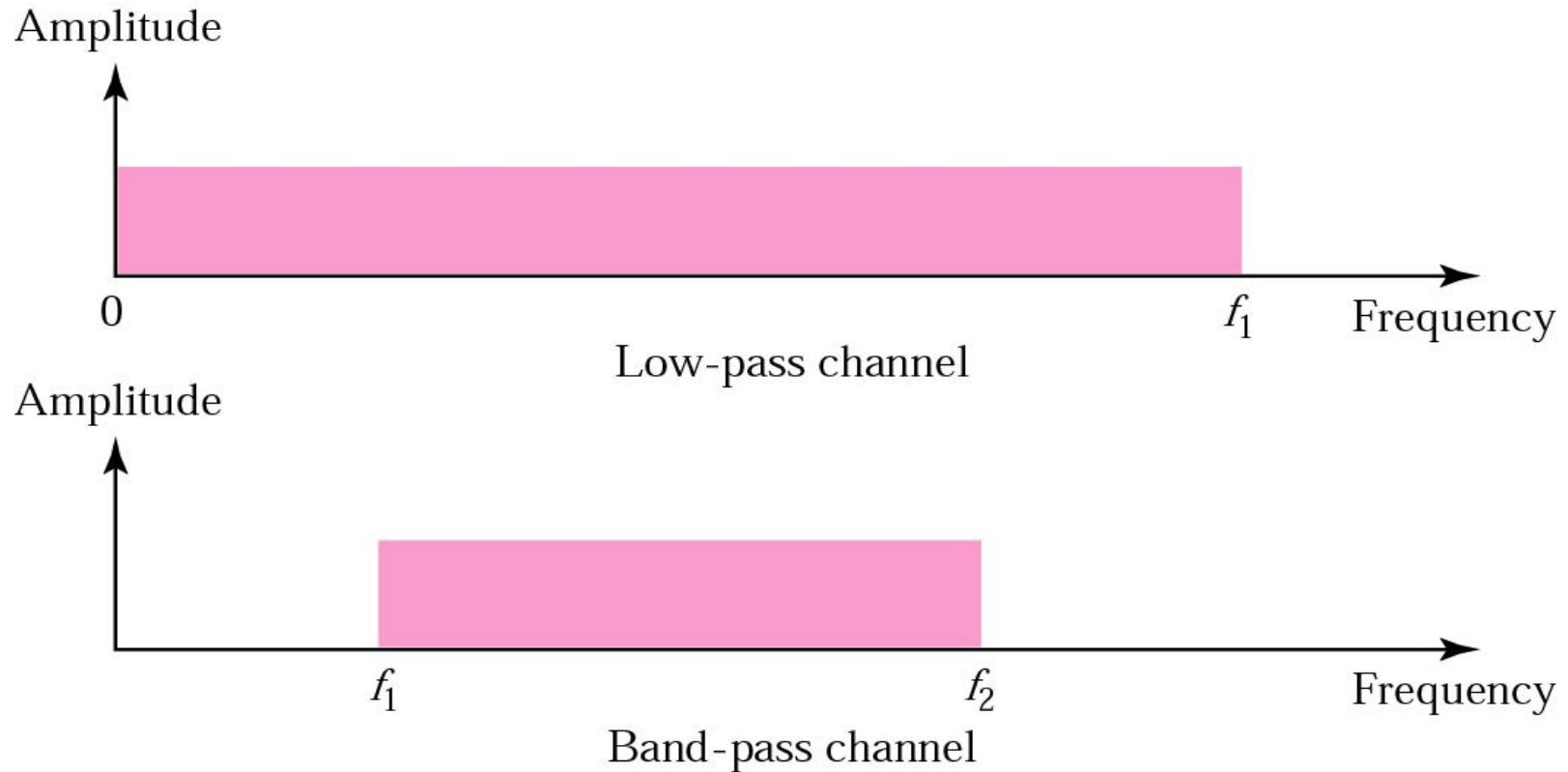
2.0 Basic Concepts on Data Communications (1)

- **Communication**: The transmission of **information** between two distant points
- **Message**: The information(data) to be communicated
- **Signal**: Involving a single independent variable (time)
➤ a function $s(t)$ that varies with time (t stands for time)
- **Channel**: 传送信息的介质（媒体）
- **Bandwidth(Hz)**: The Width of the frequency range transmitted without being strongly attenuated

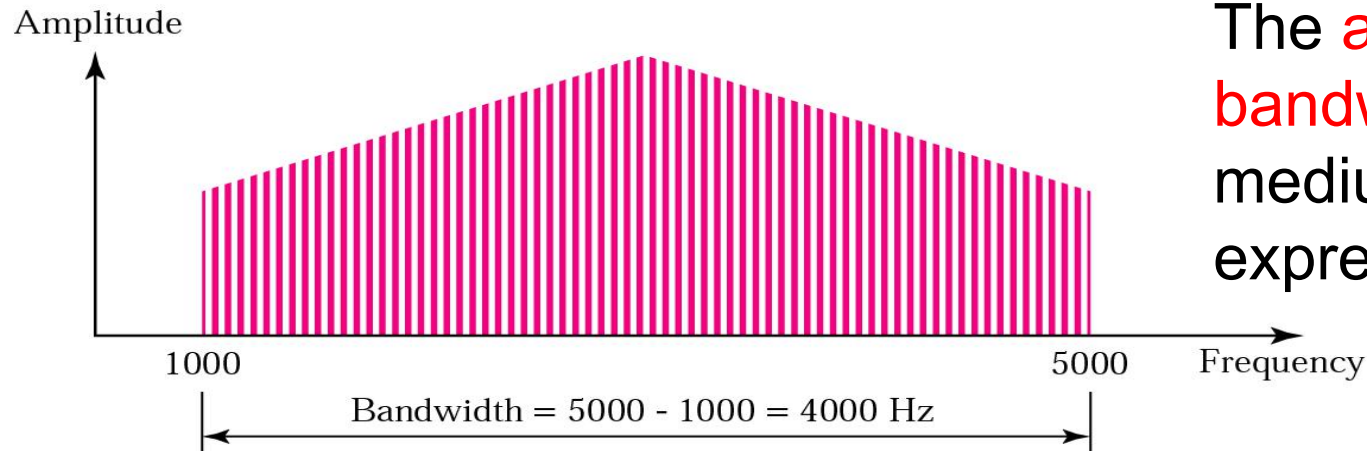


信道能够有效传输
信号的频率范围

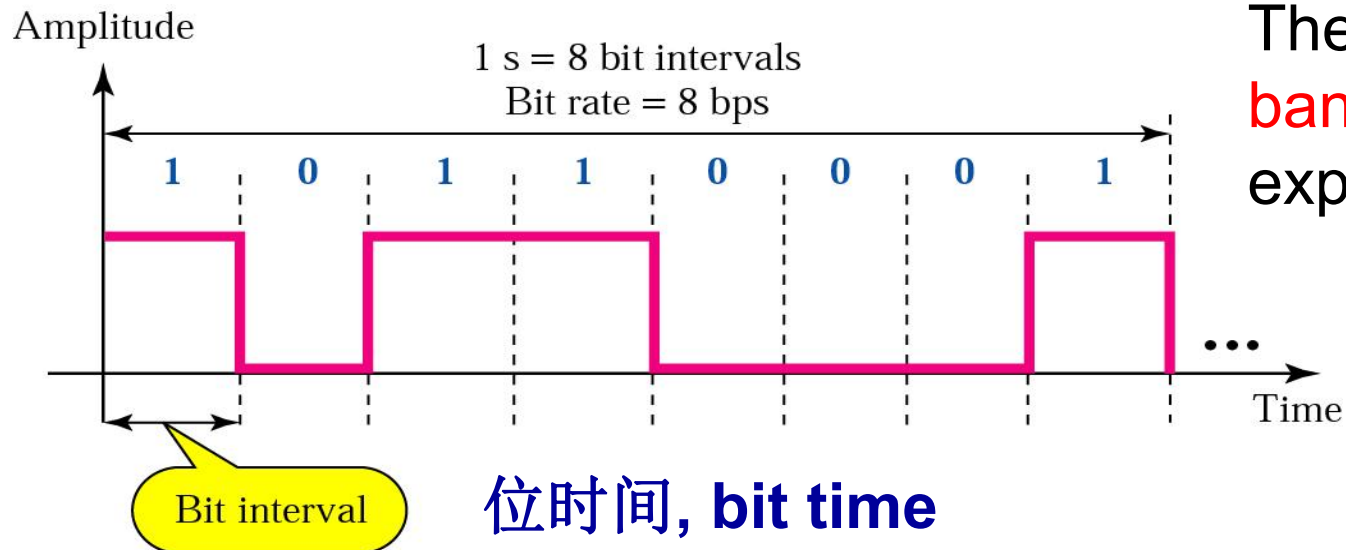
Bandwidth: Low-pass and Band-pass



Bandwidth: Hz vs bps



The **analog bandwidth** of a medium is expressed in **Hz**

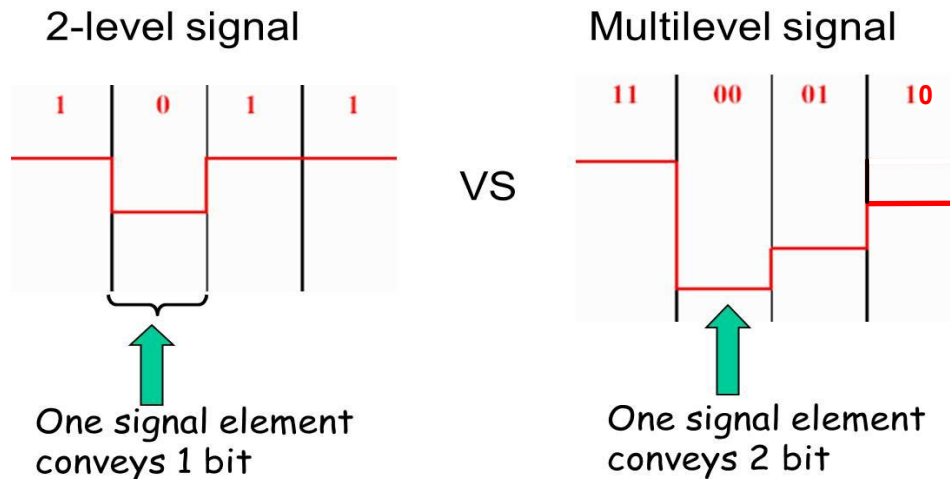


The **digital bandwidth** is expressed in **bps**

Basic Concepts on Data Communications (2)

- **Bit Rate**(bps): The number of **bits** transmitted per second
- **Baud Rate**(Baud): The number of **signal units** per second required to represent bits

Baud Rate= $1/T$ (T is the period of a signal unit)



one signal unit $\rightarrow \log_2 V$ bit
(V是信号的有效状态数/电平级数)

■ **$Bit Rate = Baud Rate * (\log_2 V)$** **公式1**

Q: tow signal unit $\rightarrow$ one bit ?

A: Bit Rate = Baud Rate/2

➡ Manchester

Basic Concepts on Data Communications (3)

- **信道容量(Channel Capacity)**: 信道的最大数据率
- **吞吐量(Throughput)**: 网络容量的度量, 表示单位时间内网络可以传送的数据位数 (bps)
- **负载(Load)**: 表示单位时间内注入(进入)网络的数据位数 (bps)
- **误码率BER(Bit Error Rate)**: 信道传输可靠性指标

$$P = \text{传送错的位数} / \text{总的传送位数}$$

Basic Concepts on Data Communications (4)

- **时延 (Delay)**：从向网络中发送数据块的第一位开始，到最后一位数据被接收所经历的时间
- **时延的组成**：发送时延、传播时延、结点处理时延、排队时延
 - ◆ **发送时延 (Transmission Delay)**：设备发送一个数据块所需要的时间
 - ◆ **传播时延 (Propagation Delay)**：信号通过传输介质的时间。
 - ◆ **节点处理时延 (Nodal Processing Delay)**：交换机/路由器检查数据、选路的时间
 - ◆ **排队时延 (Queuing Delay)**：在交换机/路由器中排队等待的时间

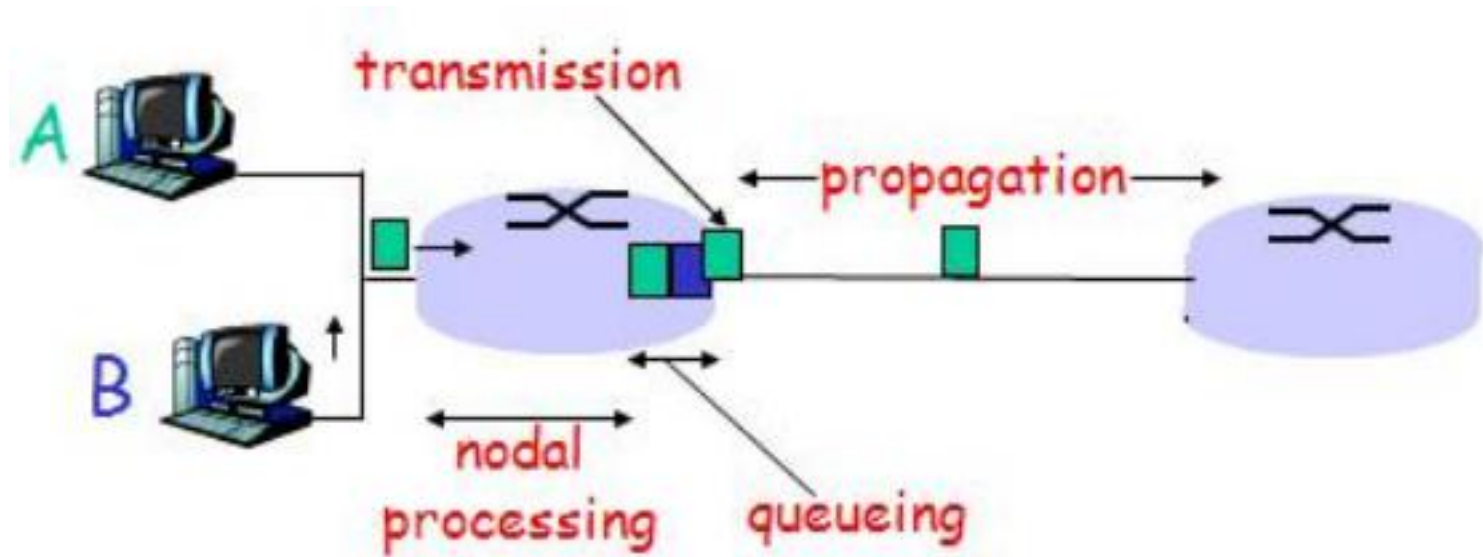
Sources of Delay

1. Transmission delay:

- ◆ B = link bandwidth (bps)
- ◆ M = message length (bits)
- ◆ time to send bits into link = M/B

2. Propagation delay:

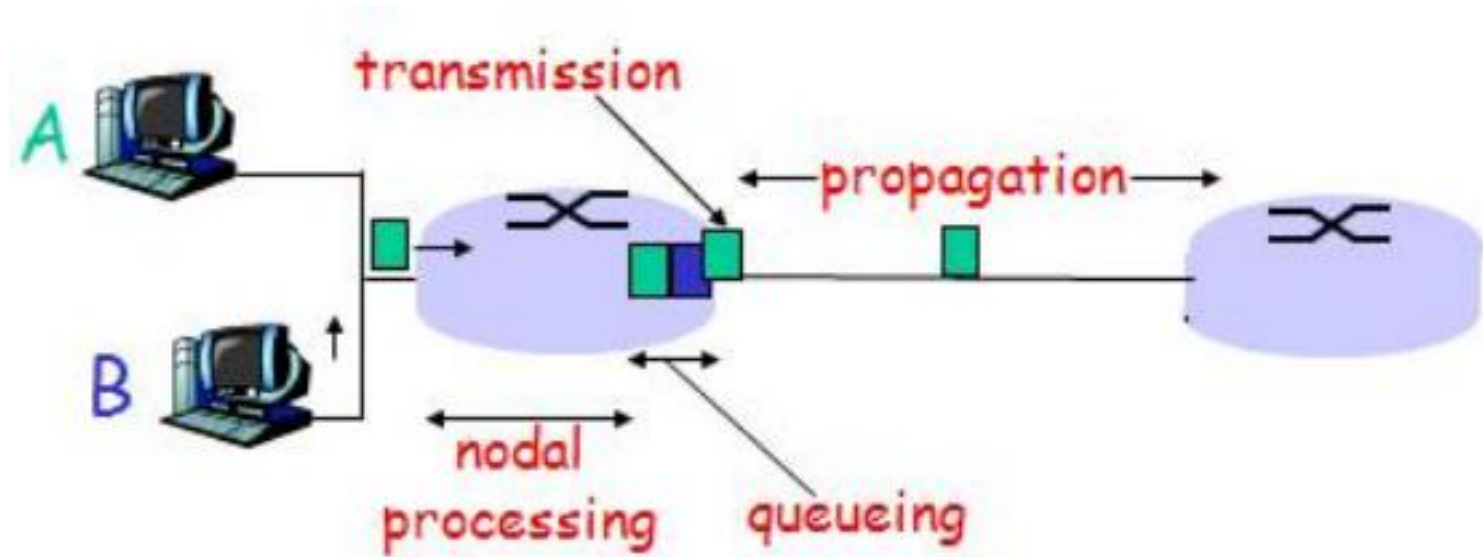
- ◆ d = length of physical link
- ◆ s = propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- ◆ propagation delay = d/s



[Kurose]

Sources of Delay

- 3. Nodal processing delay:
 - ◆ check bit errors
 - ◆ determine output link
- 4. Queuing delay:
 - ◆ time waiting at output link for transmission
 - ◆ depends on congestion level of router



[Kurose]

Basic Concepts on Data Communications (5)

- **带宽时延积 (bandwidth-delay product)**：表示充满整个链路的比特数，是一种网络性能指标。

$$\text{Bandwidth-Delay Product} = \text{propagation delay} \times \text{bandwidth}$$



[谢]

Simplex, Half-duplex, Full-duplex

■ Simplex（单工）

- ◆ Transmission occurs in only one direction
- ◆ eg. FM Radio broadcasting

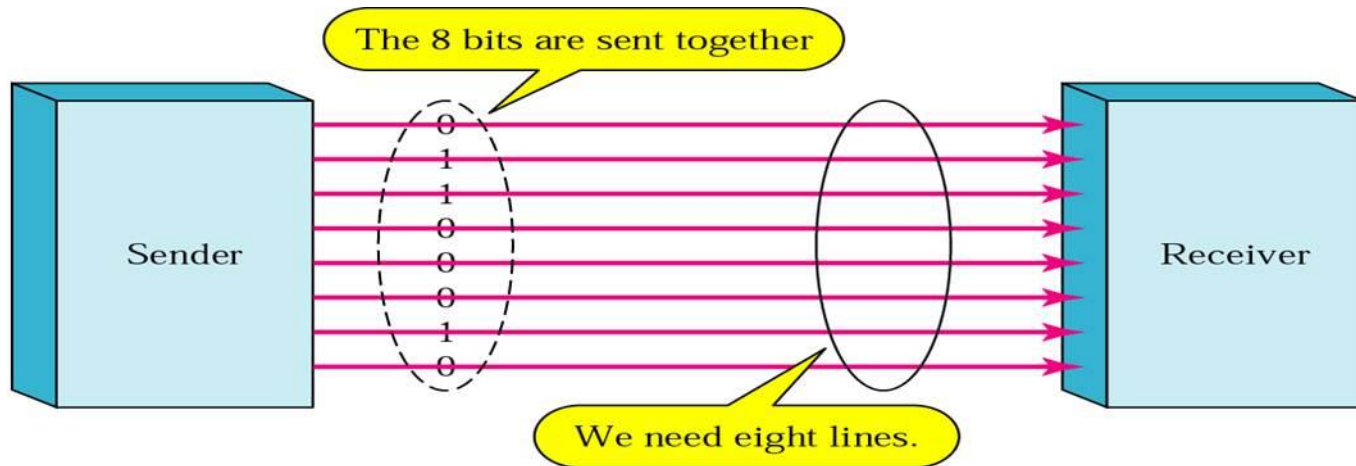
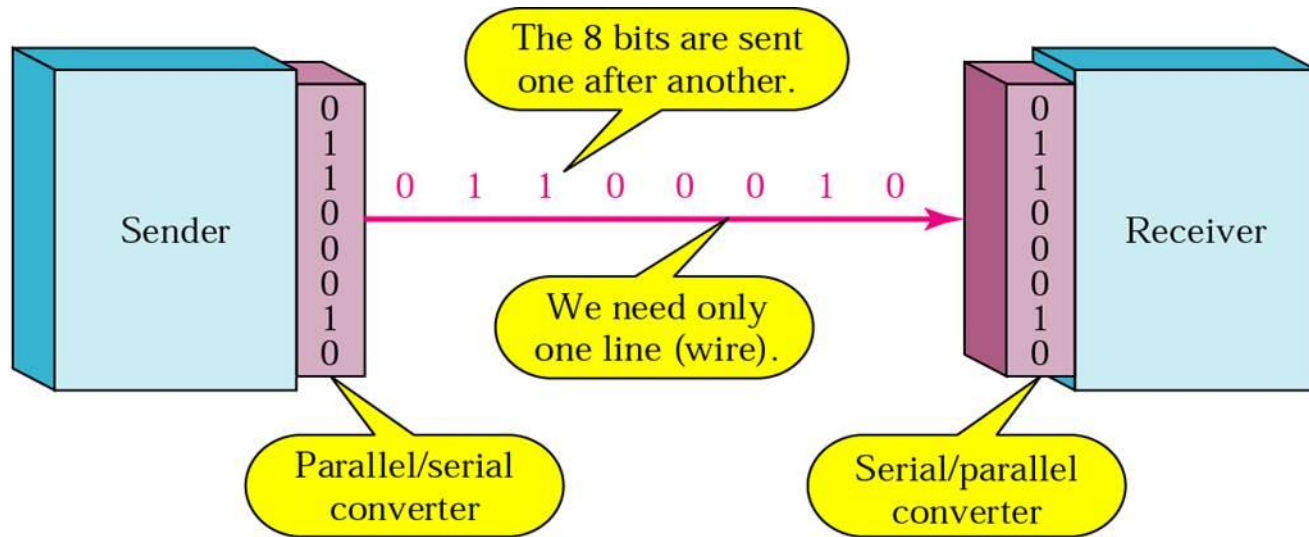
■ Half-duplex（半双工）

- ◆ Transmission can be made in both direction, but at one time only one direction is available
- ◆ eg. talkie and walkie（对讲机）

■ Full-duplex（全双工）

- ◆ Transmission can be made in both directions simultaneously
- ◆ eg. Phone conversation

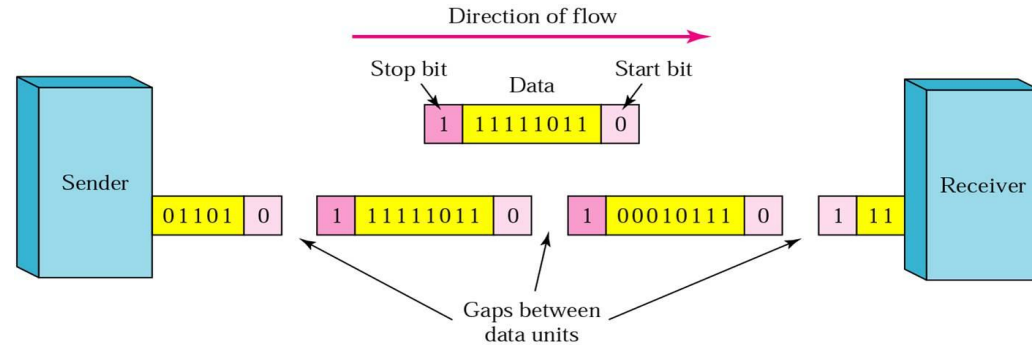
Serial transmission(串行传输) vs. Parallel transmission (并行传输)



Asynchronous Transmission(异步传输) vs. Synchronous Transmission (同步传输)

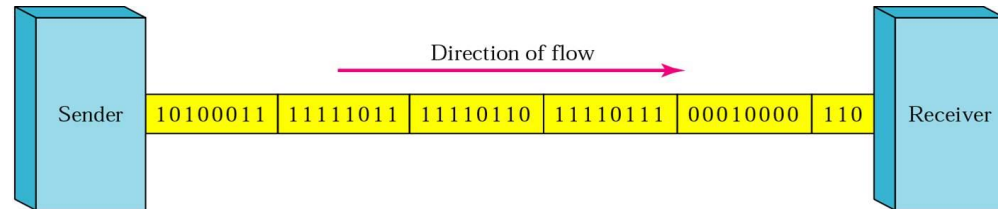
异步串行传输

- ◆ 独立时钟，无须同步
- ◆ 以**字符为单位**进行传输
- ◆ 字符之间的**间隔是任意的**
- ◆ 接收方依靠字符中的起始位和停止位来同步



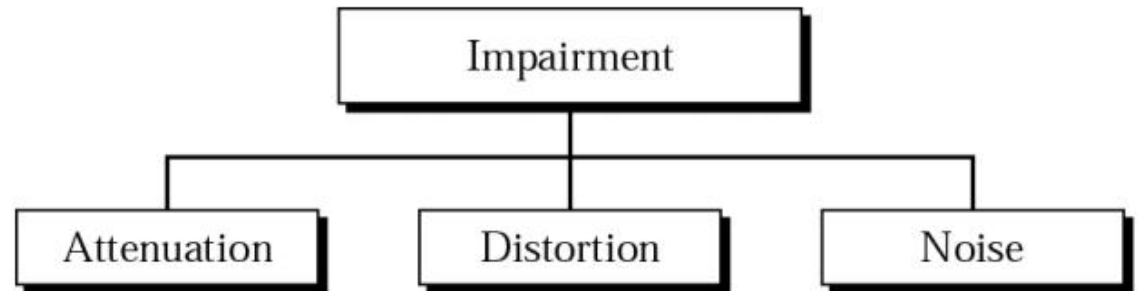
同步串行传输

- ◆ 收发双方时钟同步
- ◆ 以**数据块（帧或分组）为单位**传输
- ◆ 字符连续传输
- ◆ 接收方根据时钟对传输的数据线上的信号进行比特同步



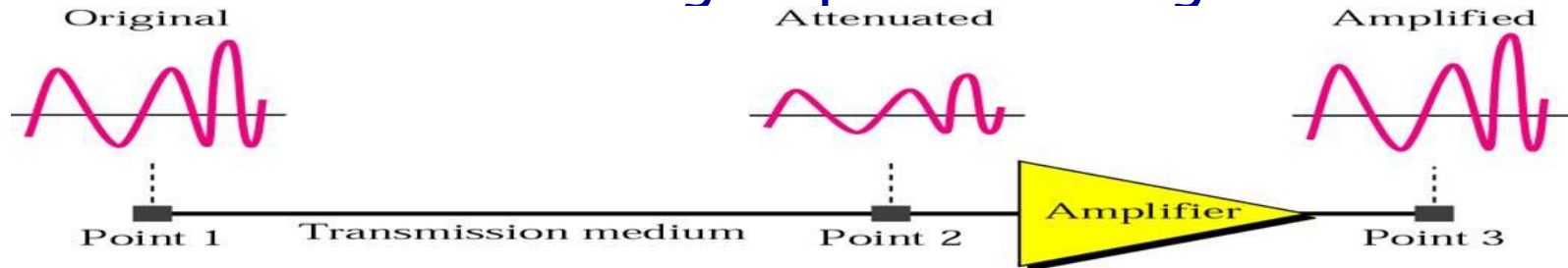
Transmission Impairment(1)

- Signal received may differ from signal transmitted
 - ◆ Analog – degradation(劣化) of signal quality
 - ◆ Digital - bit errors(误码)
- Caused by
 - ◆ Attenuation(衰减) and attenuation distortion(失真)
 - ◆ Delay distortion(时延失真)
 - ◆ Noise(噪声)

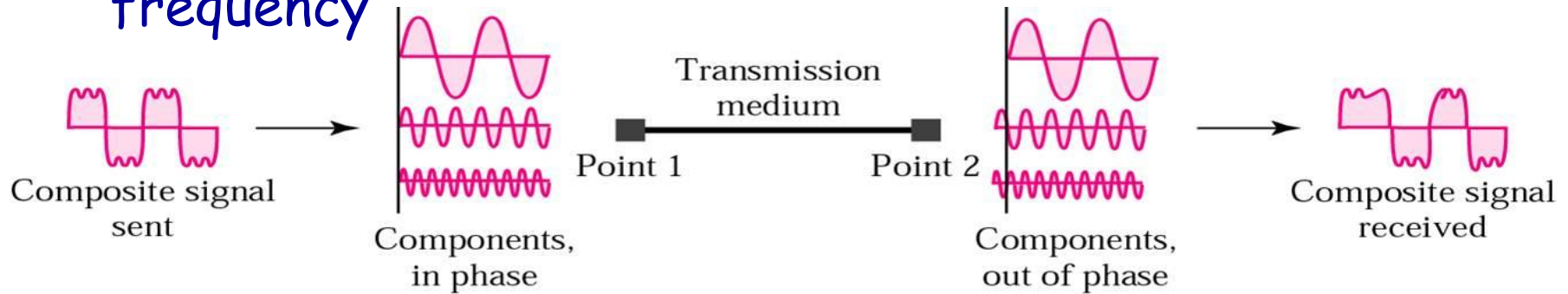


Transmission Impairment(2)

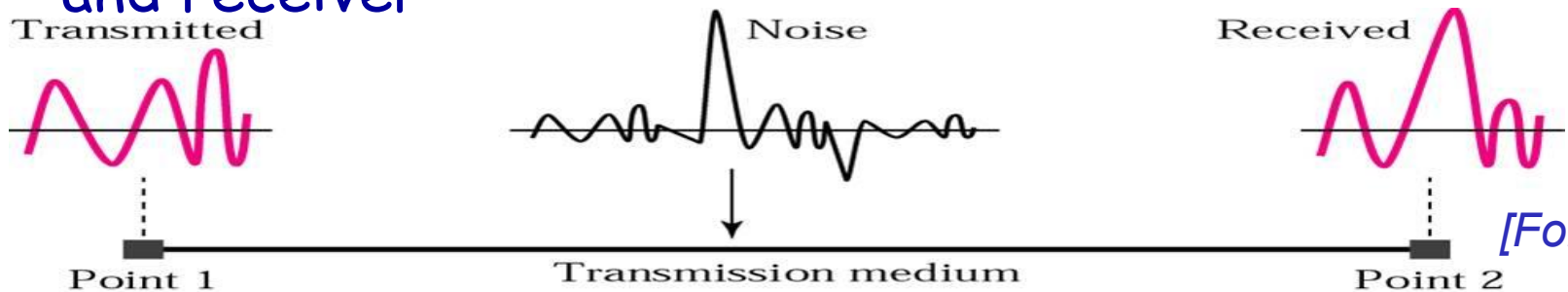
- **Attenuation:** loss of signal power through distance



- **Delay Distortion:** Propagation velocity varies with frequency



- **Noise:** Additional signals inserted between transmitter and receiver



[Forouzan]

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Fourier Transform (傅里叶变换)

- Any **periodic signal** $g(t)$ with **period** T ($=1/f$) can be constructed by summing a (possibly infinite) number of sines and cosines

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- To construct signal $g(t)$ we need to compute the values 振幅 $a_0, a_1, \dots, b_0, b_1, \dots$, and 常数 c
- Energy**: proportional to root-mean-square amplitude $\sqrt{a_n^2 + b_n^2}$
- Often the magnitude of the a_n 's and b_n 's get smaller as the frequency ($2\pi n f$) gets higher (频率越高, 能量越小)

Bandwidth

- No **transmission facility** can transmit signal without **losing some power**
- All transmission facilities diminish **different Fourier components** by **different amounts**
- Range of frequencies transmitted without being strongly attenuated(0.5) is called **bandwidth**
 - ◆ Telephone wire's bandwidth: 1 MHz
 - ◆ UTP 5: 100MHz
- bandwidth is a **physical property** of the transmission medium, depends on
 - ◆ construction, thickness, length
- bandwidth: **Hz, bps**
- **Conclusion:** Higher data rates require more bandwidth.
Limiting bandwidth limits data rate.

Summary

- 任何实际信道的带宽都是有限的
 - ◆ 传输介质的物理特性
 - ◆ 发送器对带宽的故意限制
- 目标：尽可能有效利用给定的带宽
 - ◆ 信号质量控制在可接受的范围内
 - ◆ 传输速率尽量高
- 需要研究：传输速率 c 、信道带宽 B 之间的关系
- 带宽受限信道的最大传输速率
 - ◆ 理想信道
 - ◆ 有噪声信道--随机噪声

Capacity for noiseless channel(无噪声信道)

- Nyquist proved that: if an arbitrary signal has been run through a low-pass filter of bandwidth B , the filtered signal can be completely reconstructed by making only $2B$ samples per second
- Given binary signal, data rate supported by B Hz is $2B$ bps
- Nyquist Bit Rate(奈奎斯特公式, 1924)

$$bit\ rate = 2 \times bandwidth \times \log_2 V$$

V is the number of signal levels used to represent data.

- 例1: 无噪信道带宽3kHz, 传二进制信号(即信号级数为2), 信道最大数据率是多少?

$$C = 2B \log_2 V = 2 * 3000 * \log_2 2 = 6000 \text{ bps}$$

- 例2: 无噪信道带宽3kHz, 信号每秒采样2000次, 二进制信号, 该信号的传输率是多少?

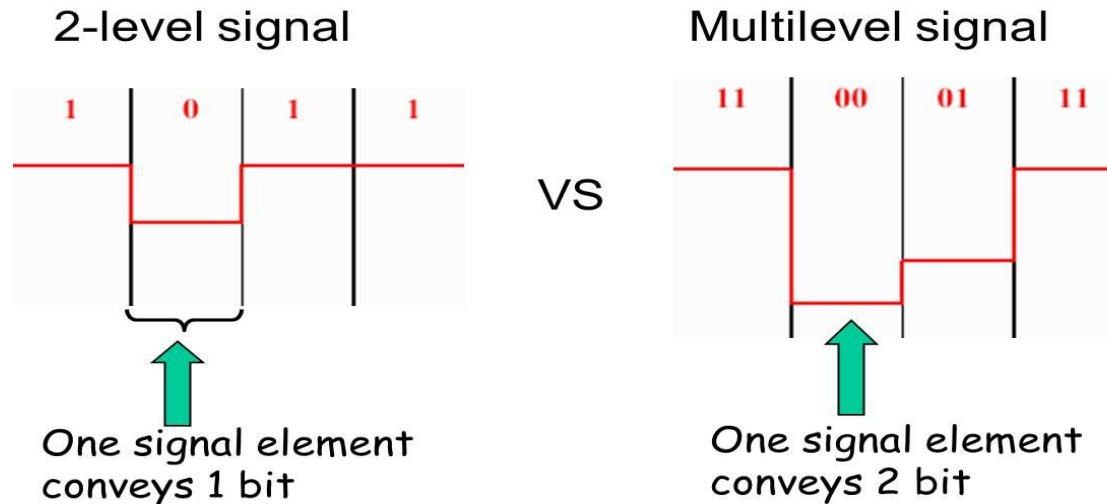
$$C = \text{波特率} * \log_2 V = 2000 * \log_2 2 = 2000 \text{ bps}$$

- 例3: 无噪信道带宽3kHz, 信道最大传输速率是多少?

$$C = 2B \log_2 V = 2 * 3000 * \log_2 V \quad (V \text{ 足够大, } C \text{ 无穷大})$$

Capacity for noiseless channel(无噪声信道)

- Increasing the levels of a signal may increase the bit rate of the channel.



- 信号振幅范围确定的条件下，级数越多，码元(信号单元)的电压差越小
 - Noisy Channel: Increasing the levels of a signal may reduce the reliability of the channel.
 - 有噪信道的最大传输速率与信道带宽和信道质量有关
- ◆ SNR (Signal-to-Noise Ratio)

Noisy Channel: Shannon Capacity

(香农公式, 1948)

- The theoretical highest data rate for a noisy channel
Shannon's theorem

$$Capacity = bandwidth \times \log_2 \left(1 + \frac{signal_power}{noise_power} \right)$$

- where capacity is in bits/second, bandwidth is in hertz, and signal and noise powers are measured in the same physical units, such as watts
- Usually, the **SNR (Signal-to-Noise Ratio)** is expressed on a log scale as:

$$S/N_{db} = 10 \log_{10} \underline{S/N}$$

- ◆ The units of this log scale are called **decibels(dB)**

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Example of Computing Channel Capacity

(1) A channel has $B = 1\text{MHz}$ and $\text{SNR}_{\text{dB}} = 24\text{dB}$,

1) what is the channel capacity limit?

$$\diamond \text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \quad \text{SNR} = 251$$

$$\diamond C = B \log_2(1 + \text{SNR}) \approx 8 \text{ Mbps}$$

2) Assume we can achieve the theoretical C ,
how many signal levels are required?

$$\diamond C = 2 B \log_2 V$$

$$\diamond V = 16 \text{ levels}$$

(2) Voice phone line **300Hz-3300Hz**, $S/N = 30\text{dB}$

$$\diamond C = B \log_2(1 + \text{SNR}) = 3000 \log_2(1 + 1000) = 30\text{kbps}$$

Review

◆ The Physical Layer

- ◆ the lowest layer
- ◆ defines the interfaces (bits are sent as signals over channels)

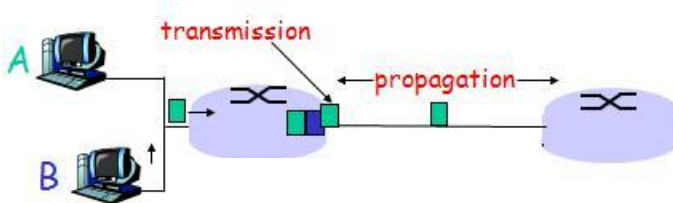
◆ Basic Concepts on Data Communications

- ◆ Channel, bandwidth(Hz or bps)
- ◆ Bit rate(bps) vs. Baud rate(Baud)
- ◆ Delay

$$\text{Bit Rate} = \text{Baud Rate} * (\log_2 V)$$

$$\text{Transmission delay} = M/B$$

$$\text{Propagation delay} = d/s$$



◆ Bandwidth

- ◆ Signal bandwidth: Fourier Transform 傅里叶变换
- ◆ Channel bandwidth: Physical property

◆ 带宽受限信道的最大传输速率

- ◆ Noiseless channel: $C_{\text{Nyquist}} = 2B \times \log_2 V$

- ◆ Noisy channel: $C_{\text{Shannon}} = B \times \log_2 (1 + S/N)$

$$S/N_{\text{dB}} = 10 \log_{10} S/N$$

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Transmission Media(翻转课堂)

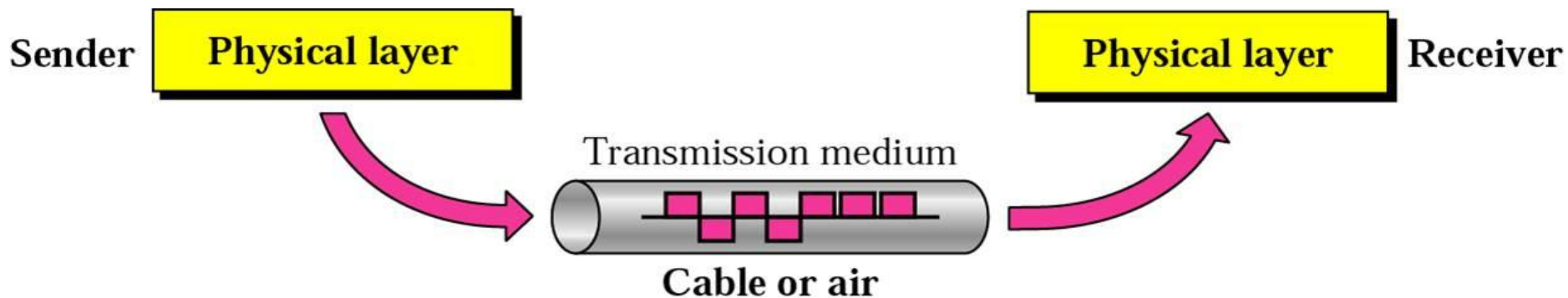
- Magnetic Media

- Guided

 - ◆ Twisted Pair, Coaxial Cable, Fiber Optics, Power Lines

- Unguided (wireless transmission)

 - ◆ Radio, Microwave and Satellite



- 相关资料

 - ◆ 传输介质教案、讲义PPT

 - ◆ 视频

 - ◆ 补充材料：光纤之父-高锟

2025-4-29

2.2.1 Magnetic Media

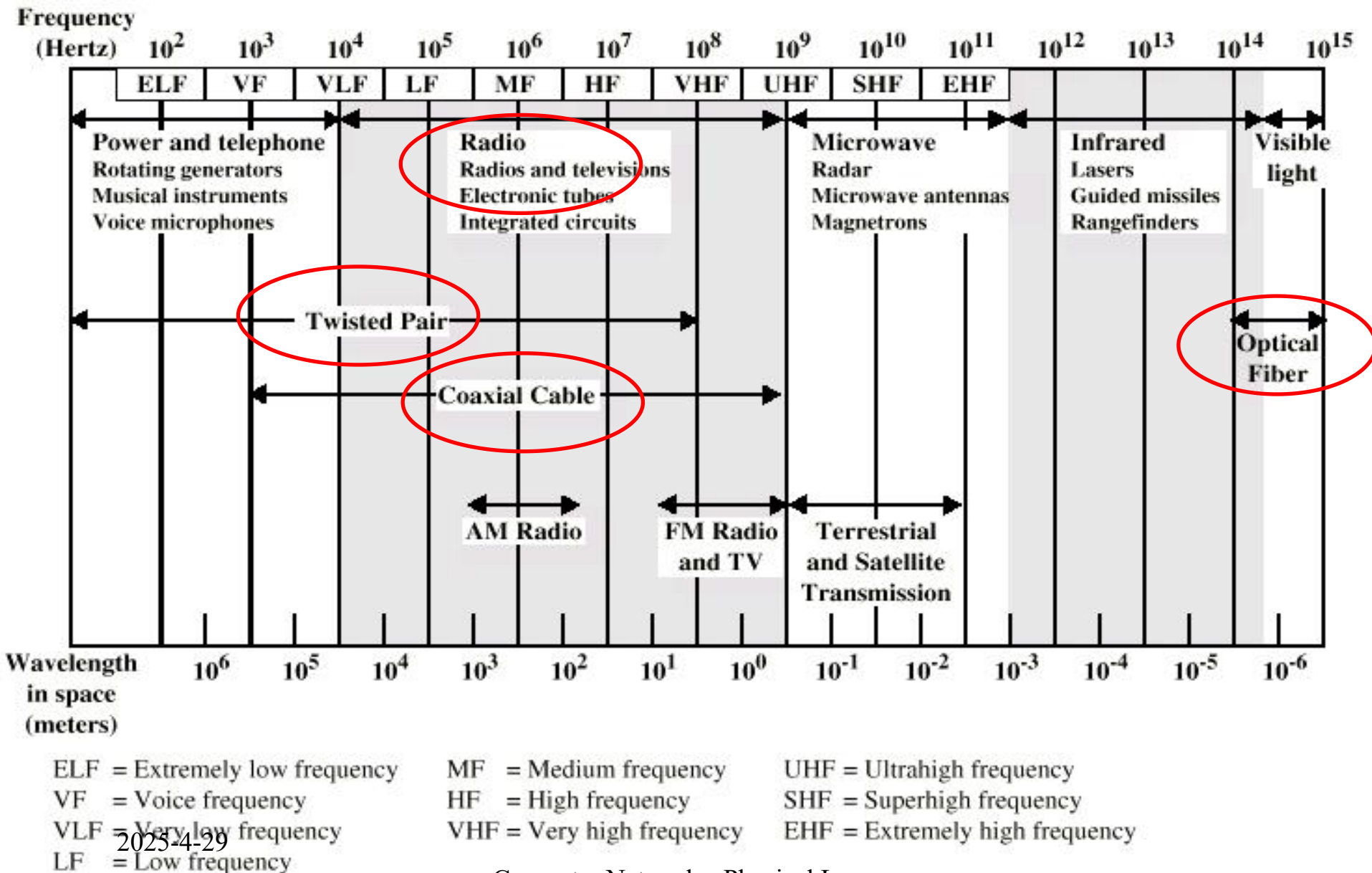
■ Bandwidth (bps)

- ◆ An industry standard Ultrium tape can hold 800 GB
- ◆ A box can hold 1000 tapes: total capacity of 800 TB
- ◆ Assumed that it takes 1 day to send the box by train from Beijing to Shenzhen
- ◆ The **effective bandwidth** of this transmission is:
$$800\text{T} \times 8\text{bits} / 86,400\text{ sec} = 70\text{Gbps}.$$

■ Cost

- ◆ tape cost \$4000 per box per usage
- ◆ \$1000 for shipping
- ◆ \$5000 for 800TB, so 1 GB for under 0.5 cents.

2.2.2 The Electromagnetic Spectrum



传输介质的特性(补)

■ 不同传输介质的特性不同

◆ 带宽

◆ 传播时延

◆ 最大传输距离

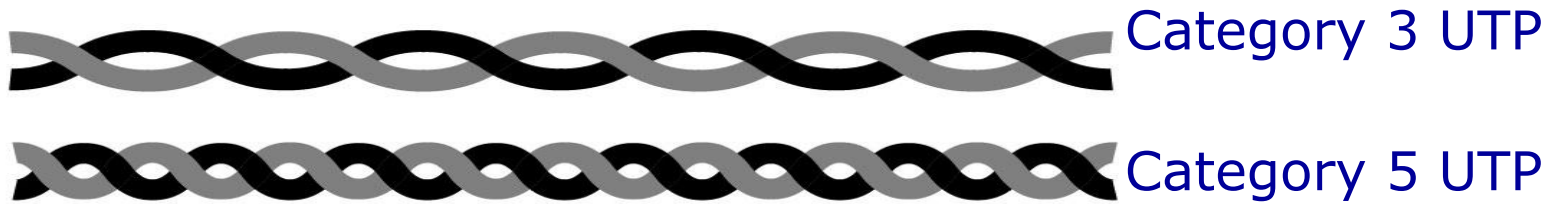
◆ 抗干扰能力

◆ 安全性

◆ 安装与维护的难易程度

◆ 成本

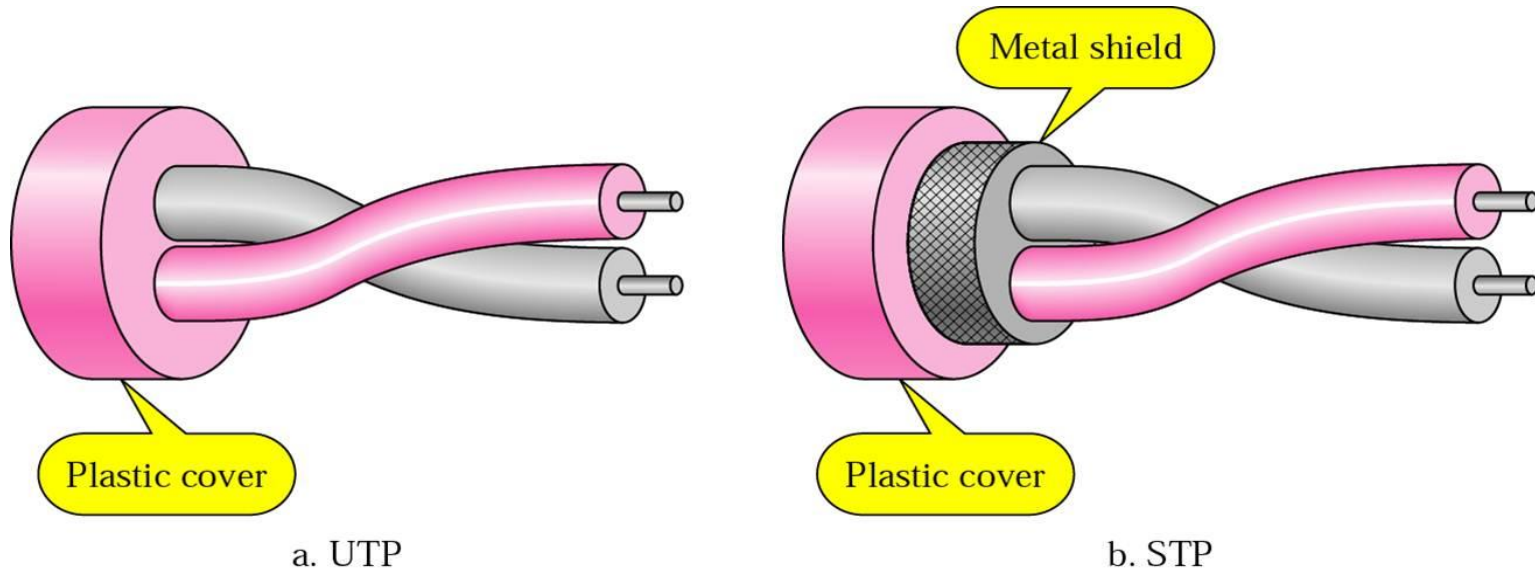
Twisted Pair(双绞线)



(b)

- Most common transmission media, Can be used for transmitting either **analog** or **digital** signals
- consists of **two insulated copper wires**, about 1 mm thick, the wires are **twisted together** in a helical form (螺旋状)
- When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively
- The bandwidth depends on
 - the thickness of the wire, the purity of the copper core
 - the distance traveled
 - the twists per meter
- several **megabits/sec** can be achieved for a few kilometers

UTP and STP

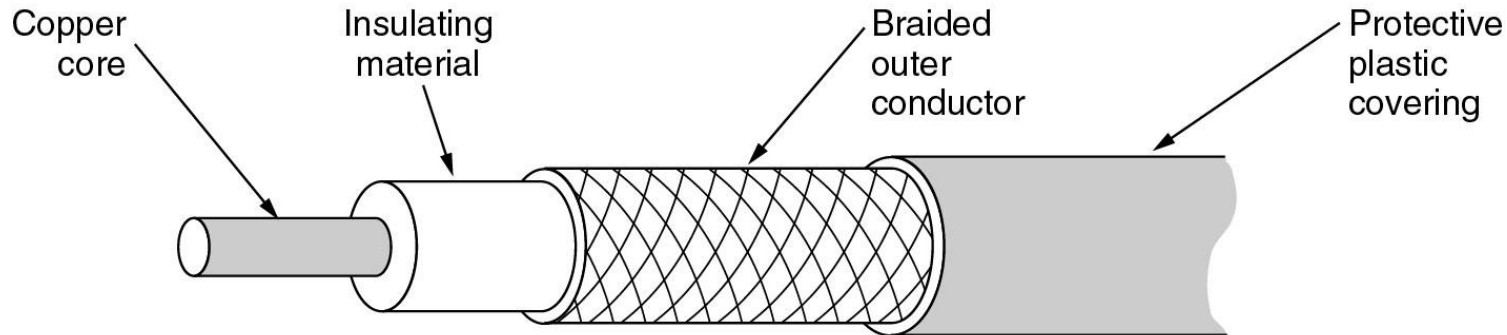


Unshielded TP

Shielded TP

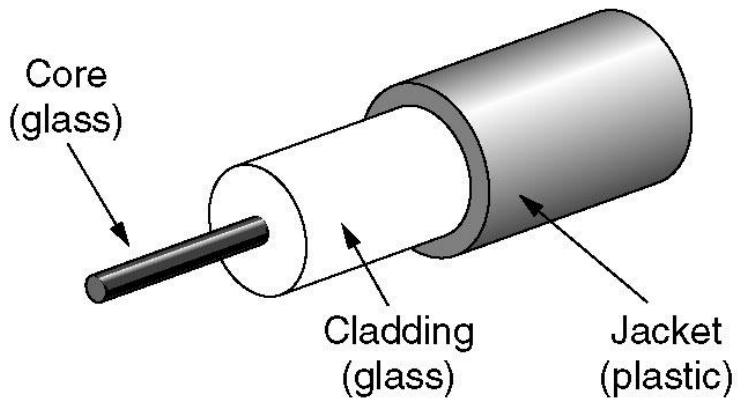
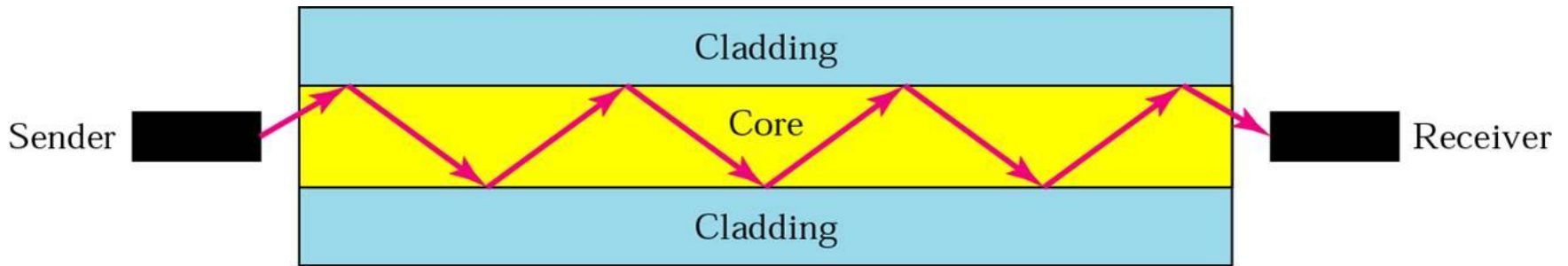
- ◆ Category 3 UTP: 16MHz
- ◆ Category 5 UTP: 100MHz
- ◆ Category 6 UTP: 250MHz
- ◆ Category 7 UTP: 600MHz

2.2.3 Coaxial Cable (同轴电缆)

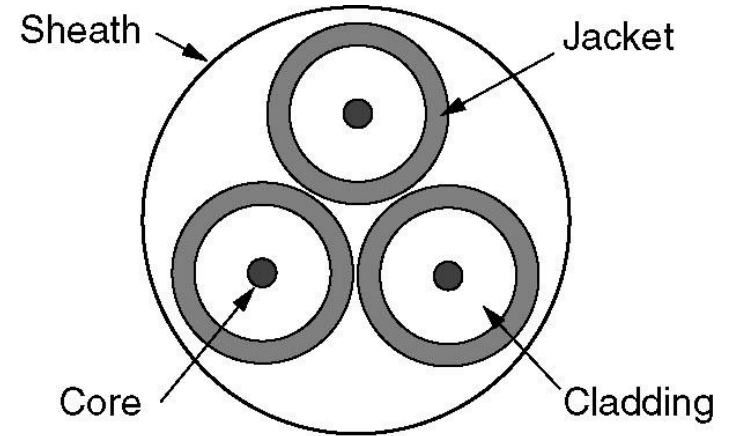


- It has **better shielding** and **greater bandwidth** than UTP(span longer distances at higher speed)sulv
- Two kinds of coaxial cable are widely used
 - ◆ **50-ohm cable**, is commonly used when it is intended for **digital** transmission
 - ◆ **75-ohm cable**, is commonly used for **analog** transmission and cable television
- high bandwidth(1GHz)
- excellent noise immunity

2.2.5 Fiber Cables



(a)

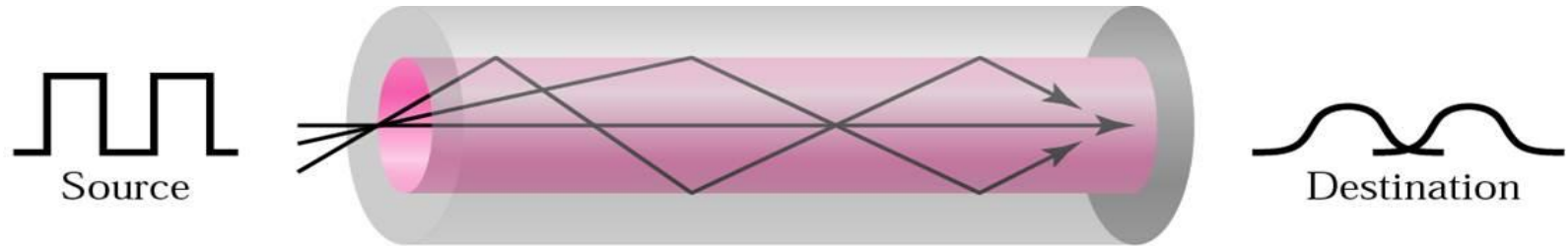


(b)

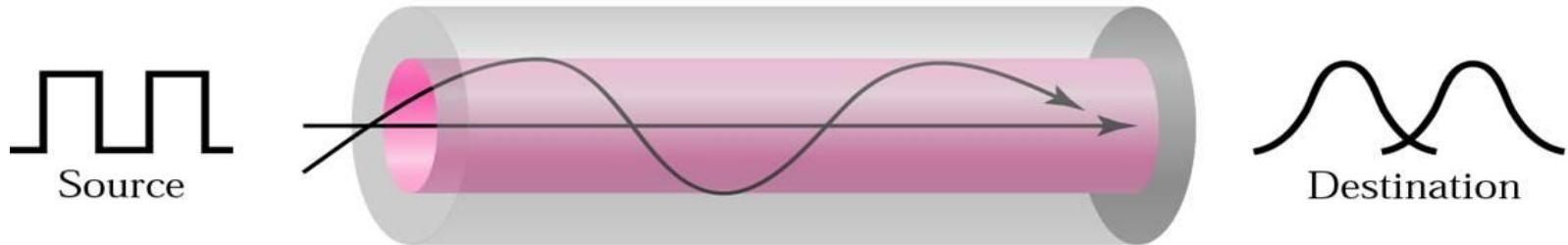
(a) Side view of a single fiber.

(b) End view of a sheath with three fibers.

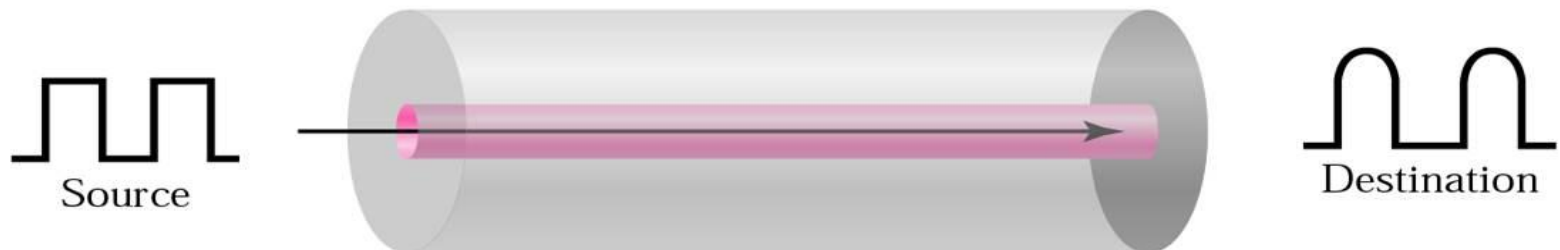
Modes of Fiber



a. Multimode, step-index

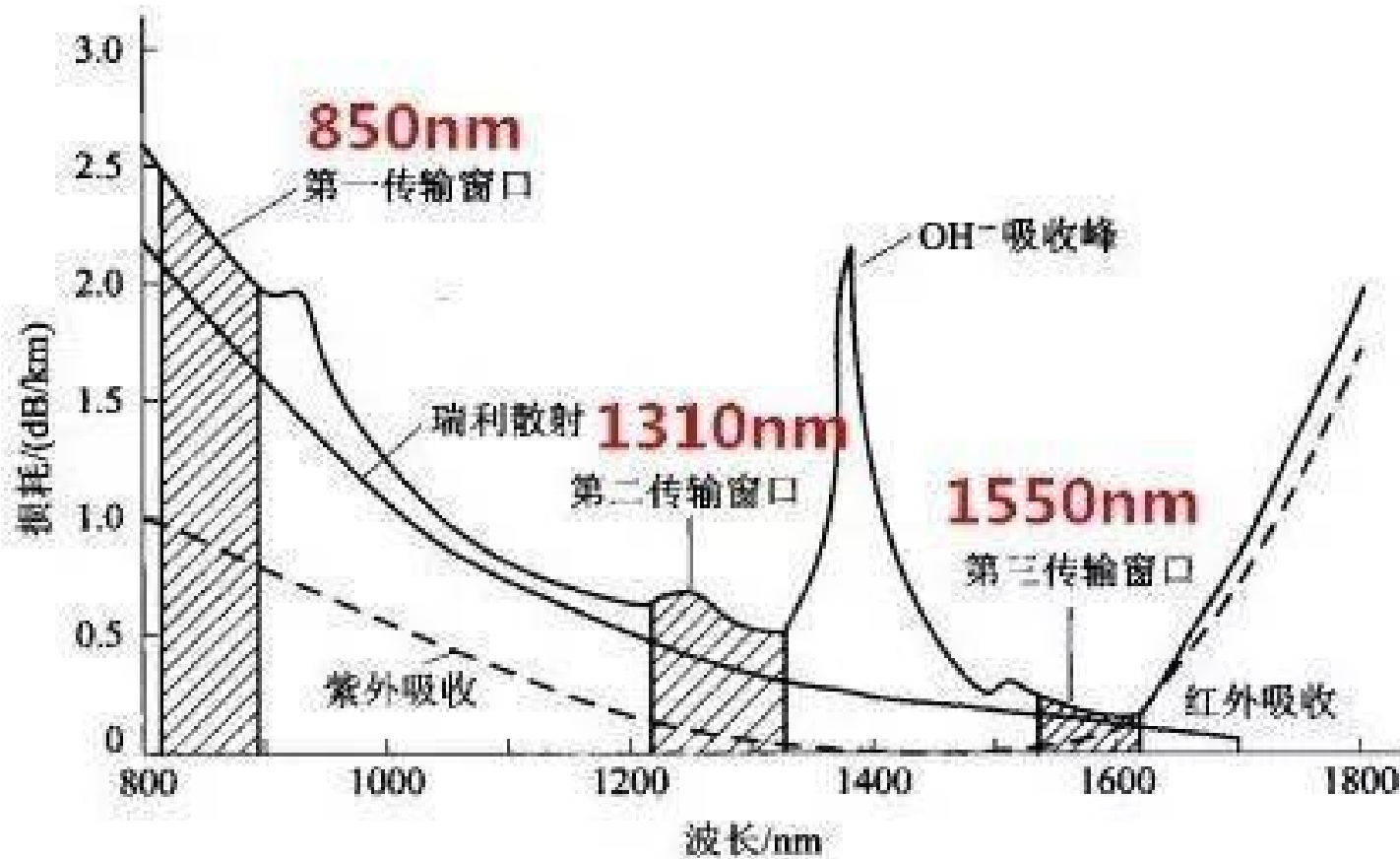


b. Multimode, graded-index



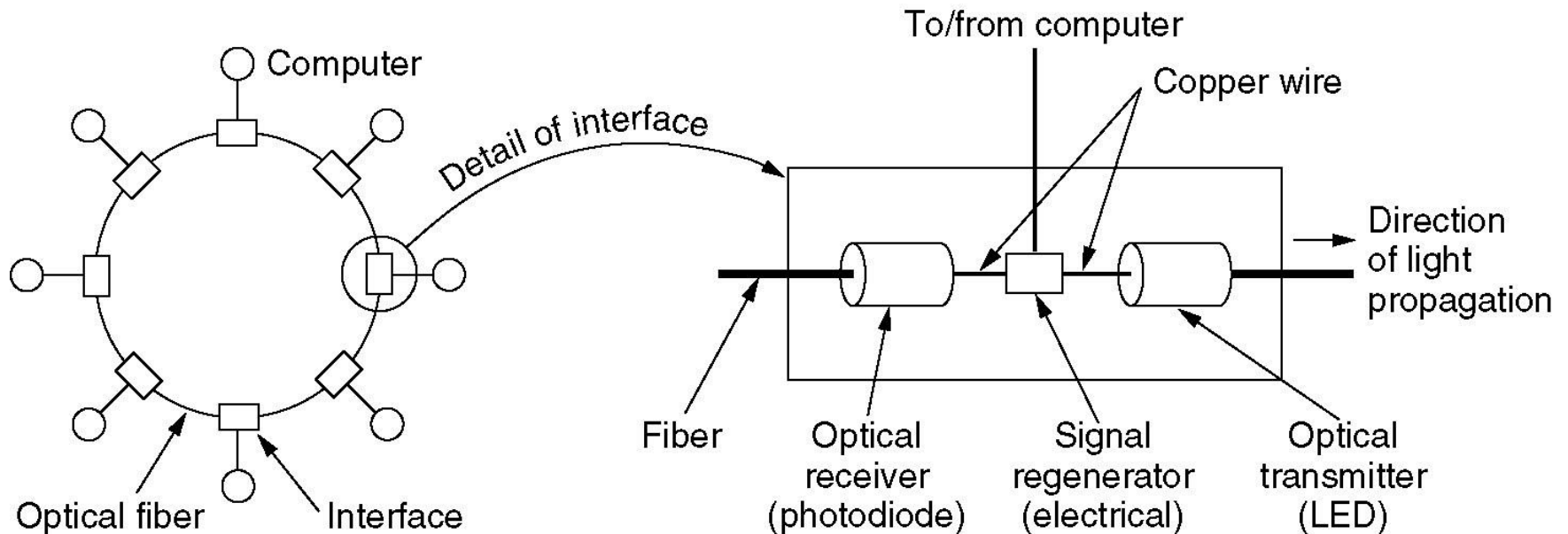
c. Single-mode

光纤的工作窗口



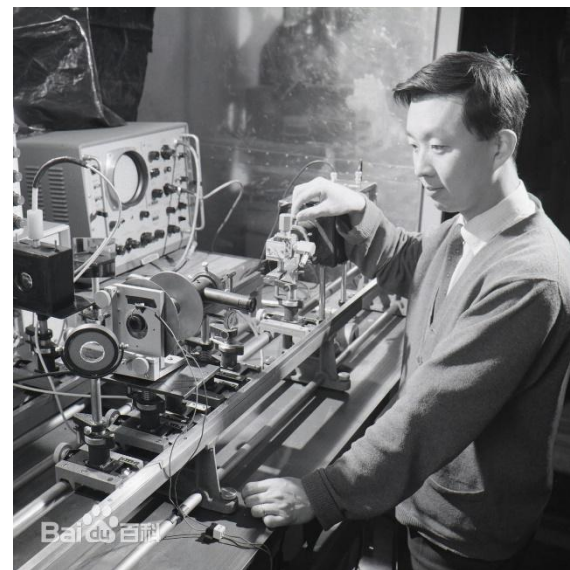
Fiber Optic Networks

A fiber optic ring with active repeaters.



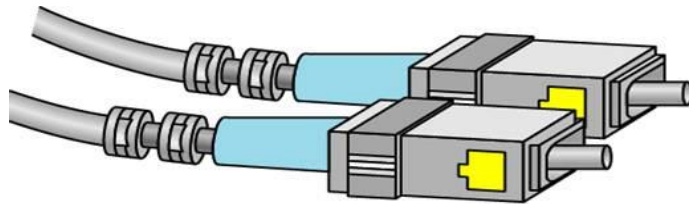
Father of Fiber Optic Communications---高锟

- 1964年提出在电话网络中以光代替电流，以玻璃代替导线
- 1965年，得到结论：玻璃光衰减基本限制在 20dB/km 以下，这是光通信的关键阈值，开启了寻找低损耗材料和合适纤维以达到这一标准的过程
- 1966年，发表《Dielectric-fibre surface waveguides for optical frequencies》，开创性地描述：长距离和高信息光通信所需要的绝缘性纤维的结构和材料特性。
- 2009年获得诺贝尔物理学奖

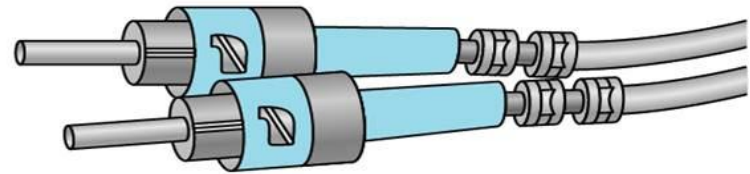


Advantages of Fiber

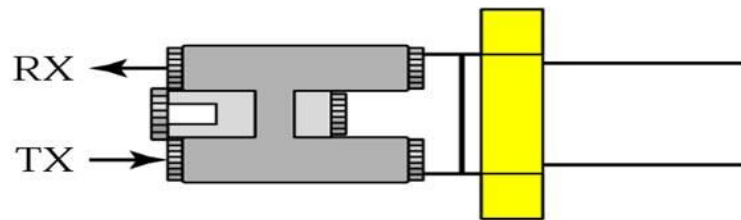
- High bandwidth
- Lightweight
- Security



SC connector



ST connector



MT-RJ connector

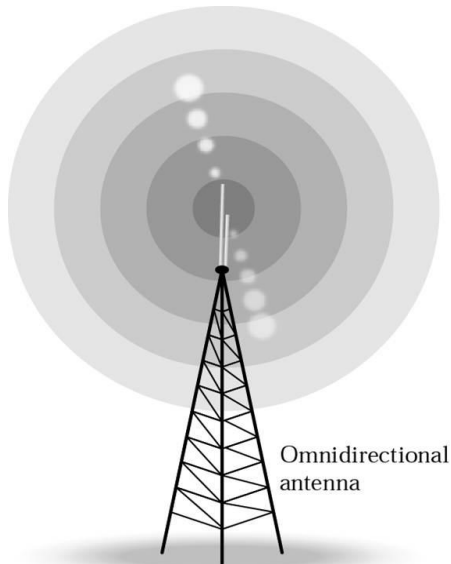
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- 2.4 Communication Satellites(Self-study)
- 2.5 Digital Modulation and Multiplexing
- 2.6 The Public Switched Telephone Network
- (补充) Protocol Example: RS-232

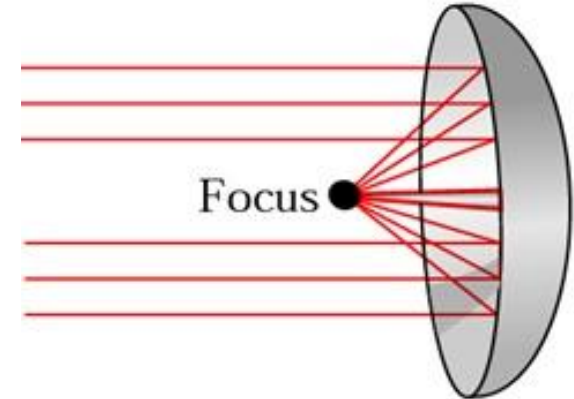
Wireless Transmission

■ Features

- ◆ Comparatively higher bit error rate
- ◆ Longer propagation delay
- ◆ Omni-direction vs. Uni-direction



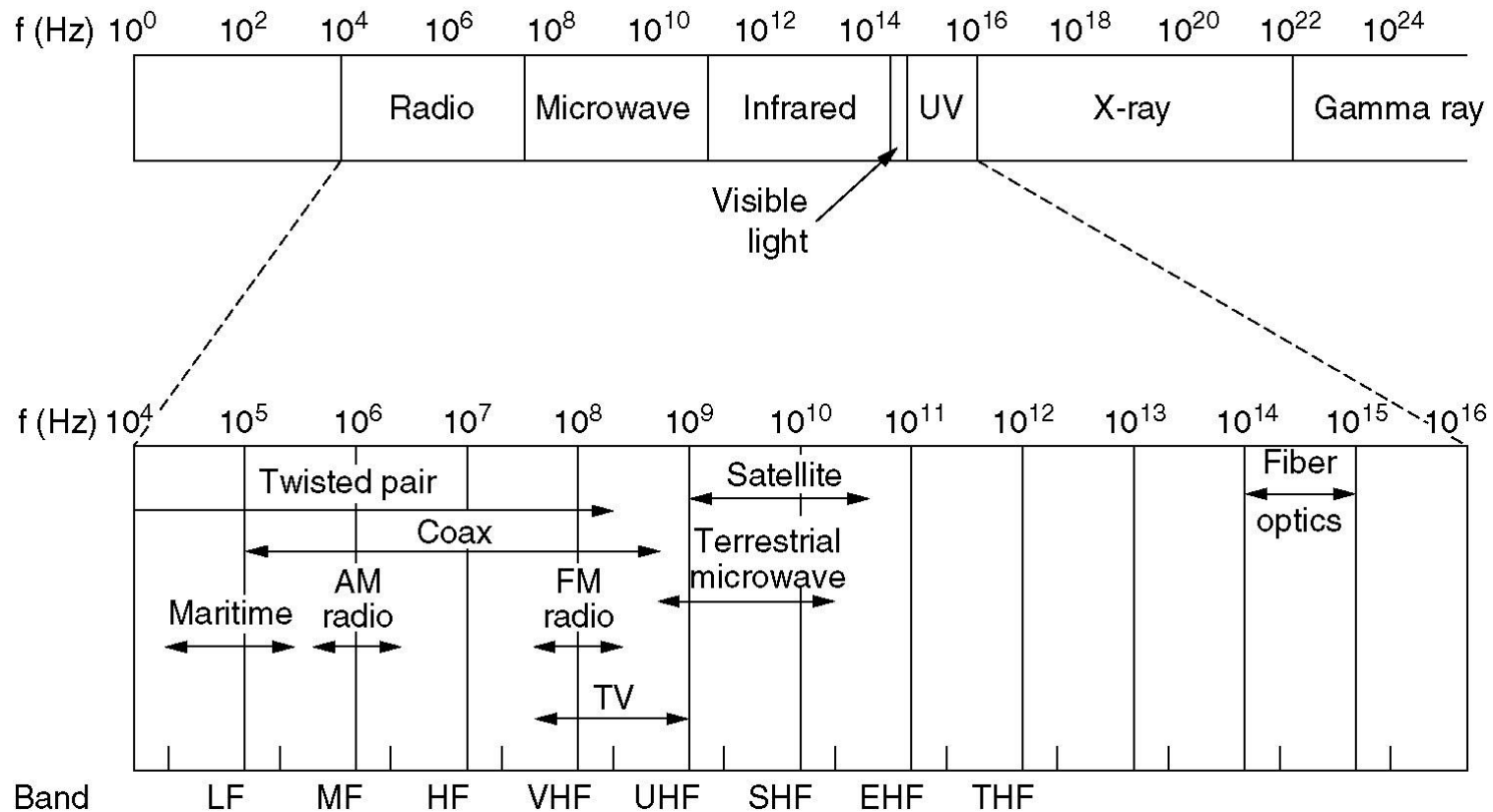
Omnidirectional antennas



Unidirectional antennas

2.3.1 The Electromagnetic Spectrum

Electromagnetic spectrum and its uses for communication



MF = Medium frequency

HF = High frequency

VHF = Very high frequency

UHF = Ultrahigh frequency

SHF = Superhigh frequency

EHF = Extremely high frequency

Transmission Frequency Band

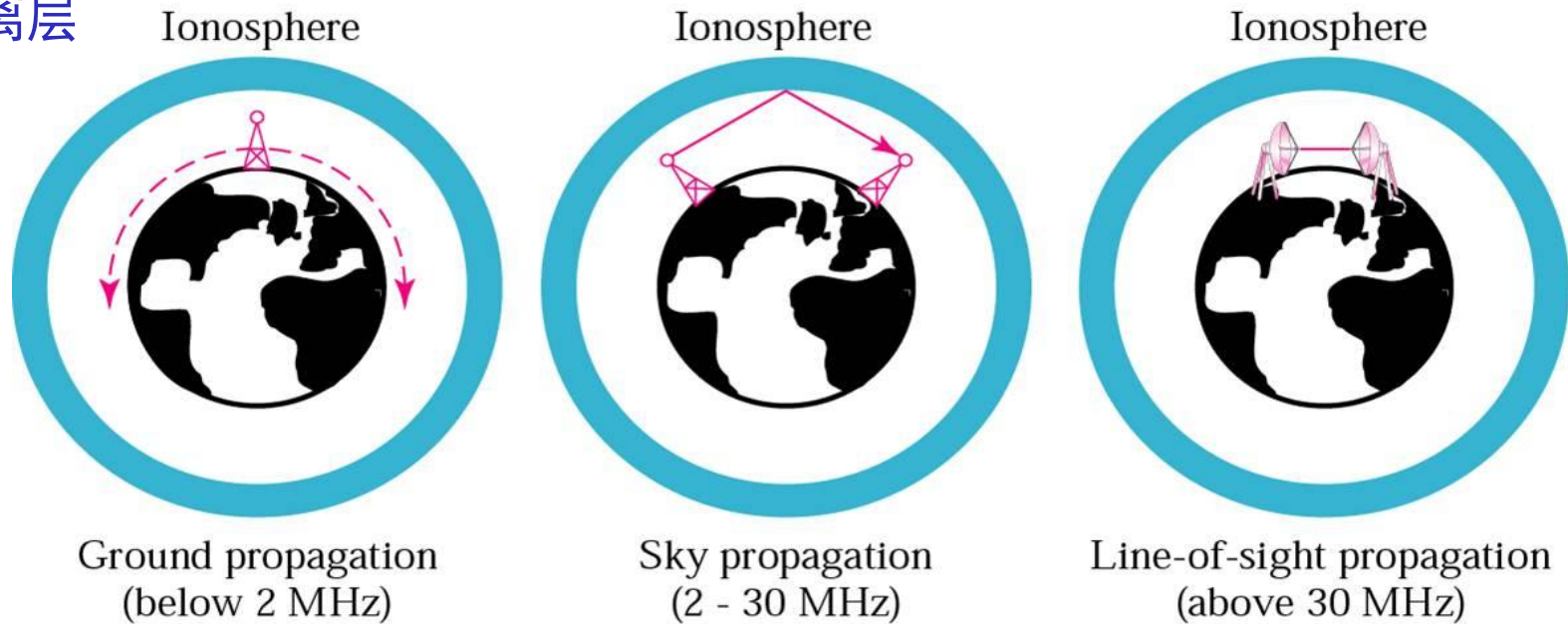
- Using narrow frequency band
 - ◆ Most transmission use narrow frequency band to get the best reception
- Spread Spectrum (bluetooth, wlan, cdma, ...)
 - ◆ FHSS: Frequency hopping spread spectrum
 - ◆ DSSS: Direct sequence spread spectrum
 - ◆ UWB: Ultra-WideBand

Category of Wireless Transmission

- Radio Transmission
- Microwave Transmission
- Communication **Satellites**
- Infrared and Millimeter Waves
- Light wave Transmission

2.3.2 Radio Propagation

电离层



- (a) In the VLF, LF, and MF bands, radio waves follow the **curvature of the earth**.
- (b) In the HF band, they **bounce off the ionosphere**.
- (c) In the VHF band and above, they use **line-of-sight** (视距) transmission, not affected by ionosphere.

Bands and Applications

Band	Range	Propagation	Application
VLF	3–30 KHz	Ground	Long-range radio navigation
LF	30–300 KHz	Ground	Radio beacons and navigational locators
MF	300 KHz–3 MHz	Sky	AM radio
HF	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF	3–30 GHz	Line-of-sight	Satellite communication
EHF	30–300 GHz	Line-of-sight	Long-range radio navigation

2.3.3 Microwave Transmission

(100M~10GHz)

- ◆ Before fiber optic, microwaves formed the heart of the **long-distance** telephone transmission system
- ◆ waves travel in nearly **straight lines**, the transmitting & receiving **antennas must be accurately aligned**
- ◆ Distance between repeaters: **80km**
- ◆ **Multipath fading** is often a serious problem (weather and frequency dependent)
(At about 4GHz, absorbed by rain)

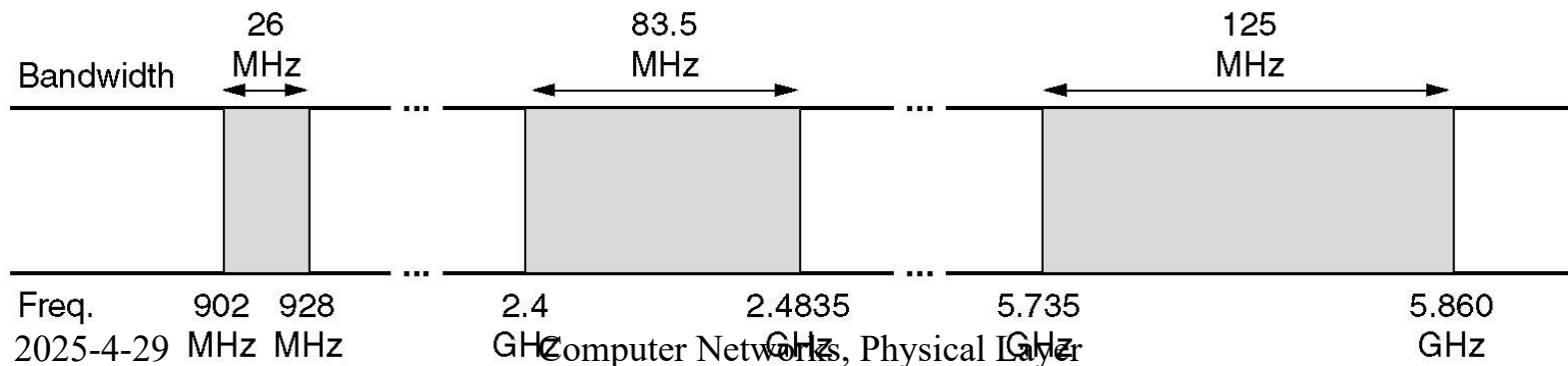
Politics of Electromagnetic Spectrum

■ National governments allocate spectrum

- ◆ Worldwide: ITU-R; USA: FCC; **China**: Radio administration of the Ministry of industry and information technology (National Radio Office)

■ ISM (Industrial, Scientific, Medical) bands

- ◆ some frequency bands for **unlicensed usage**
- ◆ **功率和带外辐射满足一定约束(power is under 1 watt)** (限制频率范围, 调制技术, 信号强度)
- ◆ FCC mandates that all devices in ISM bands use **spread spectrum techniques** to minimize interference between uncoordinated devices



ISM bands in USA

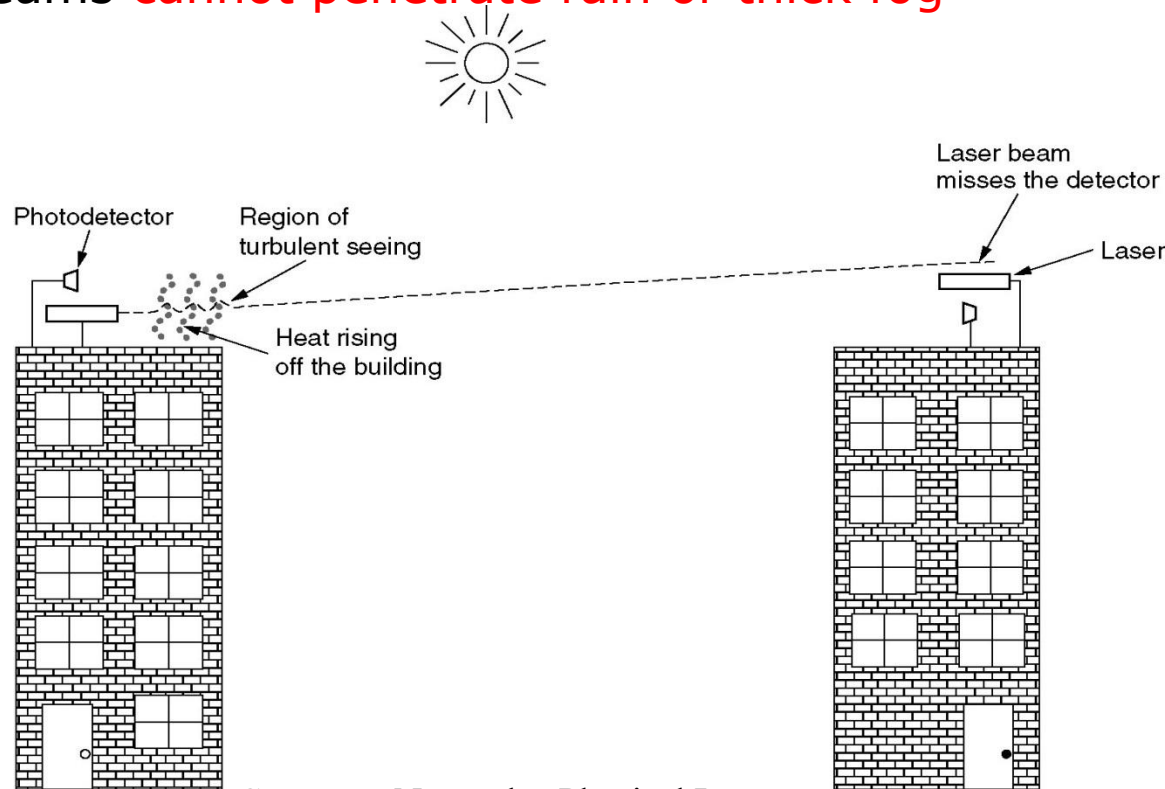
2.3.4 Infrared Transmission

- widely used for short-range communication
- relatively directional, cheap
- do not pass through solid objects
- no government license is needed to operate an infrared system

2.3.5 Lightwave Transmission

(Unguided optical signaling)

- ◆ **Very narrow beam:** Aiming a laser beam 1mm wide at a target the size of a pin head 500m away
- ◆ Wind and temperature changes can **distort the beam**
- ◆ Laser beams **cannot penetrate rain or thick fog**



2025-4-29

Computer Networks, Physical Layer

Convection currents interfere with laser communication systems

Outline

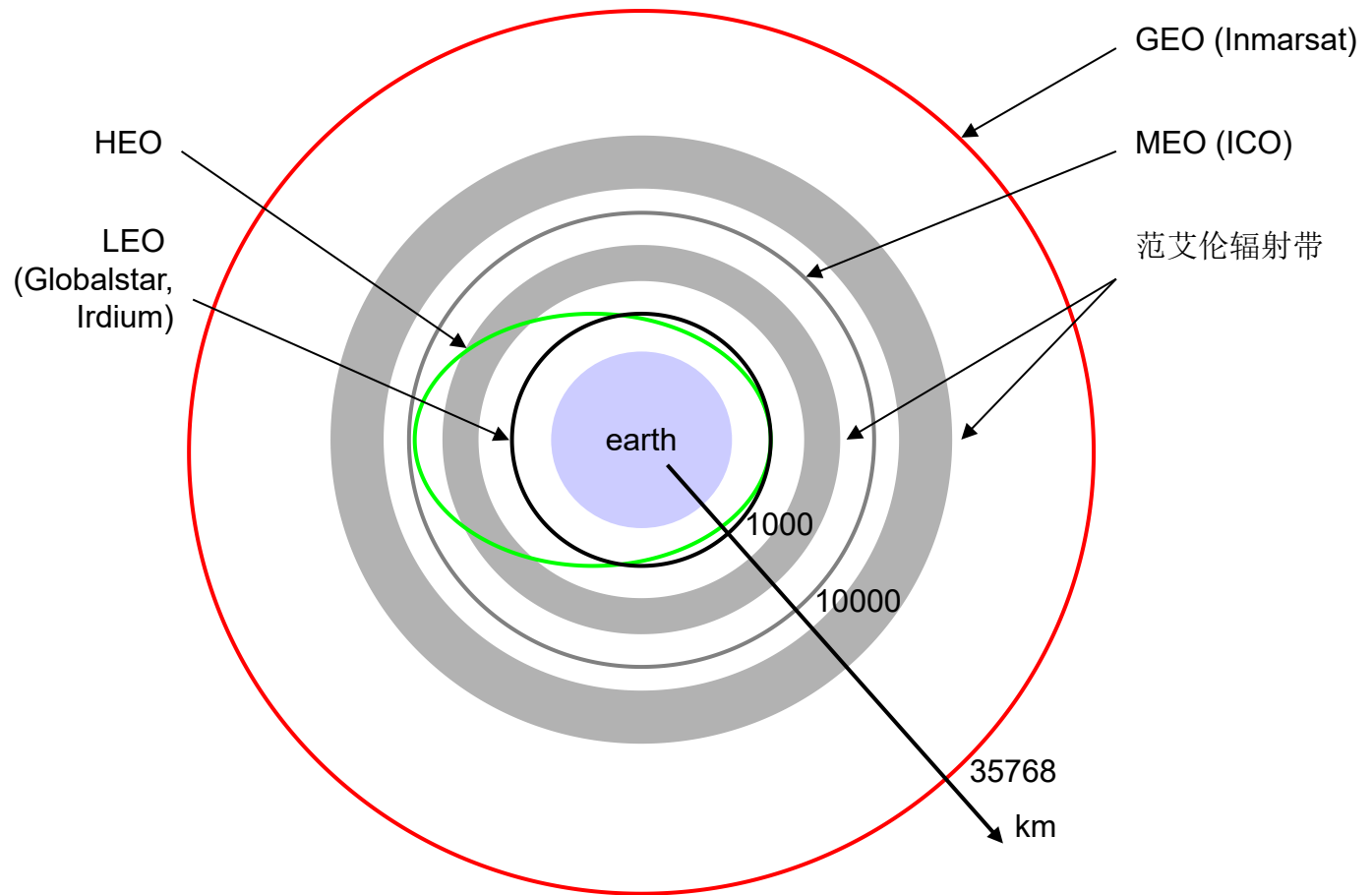
- 2.0 Basic Concepts on Data Communications
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Communication Satellites

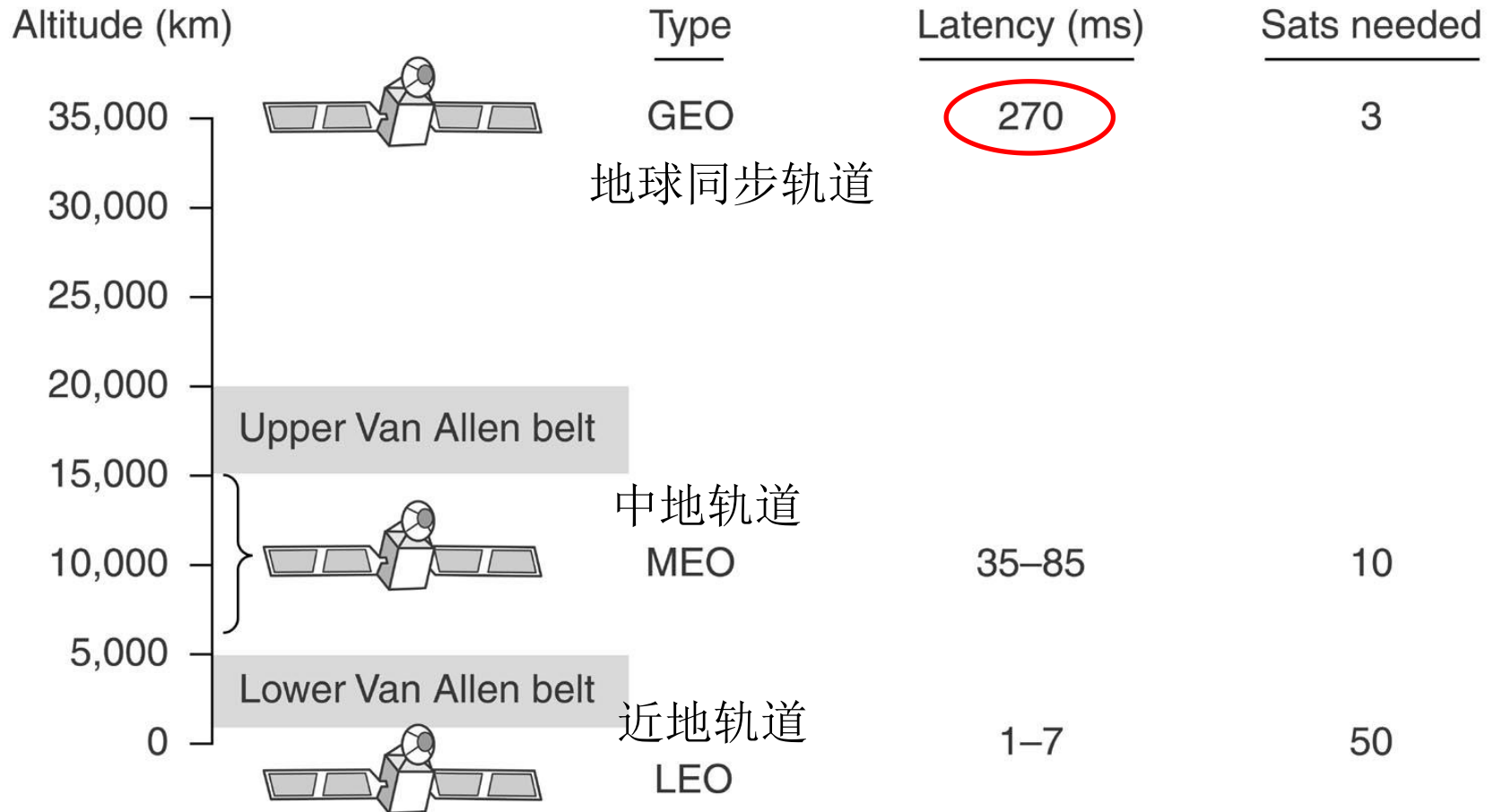
- Act as a big microwave repeater in the sky
 - ◆ Contains multiple antennas and multiple transponders
- Transponder
 - ◆ listen to some portion of the spectrum
 - ◆ amplifies the incoming signals
 - ◆ rebroadcasts it at another frequency (downward beams)—Bent pipe
- Downward beams
 - ◆ Broad: a substantial fraction of the earth's surface
 - ◆ Narrow: spot beams cover an area hundreds of kms in diameter (bent pipe)

Communication satellites and some of their properties

- GEO
- MEO
- LEO



Communication satellites and some of their properties



altitude above the earth, round-trip delay time and number of satellites needed for global coverage

Features of Satellite Communication

- The long round-trip distance introduces a substantial delay for GEO satellites.
- ◆ Long propagation delay
- They are inherently broadcast media.
- The cost of transmitting a message is independent of the distance traversed.

The Principal Satellite Bands

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

A modern satellite has around 40 transponders, most often with a 36-MHz bandwidth

小结

■ 传输介质

◆ 传输介质特性：带宽、时延、最大传输距离、抗干扰性

...

◆ 有线传输介质：原理、结构、分类、特性、应用

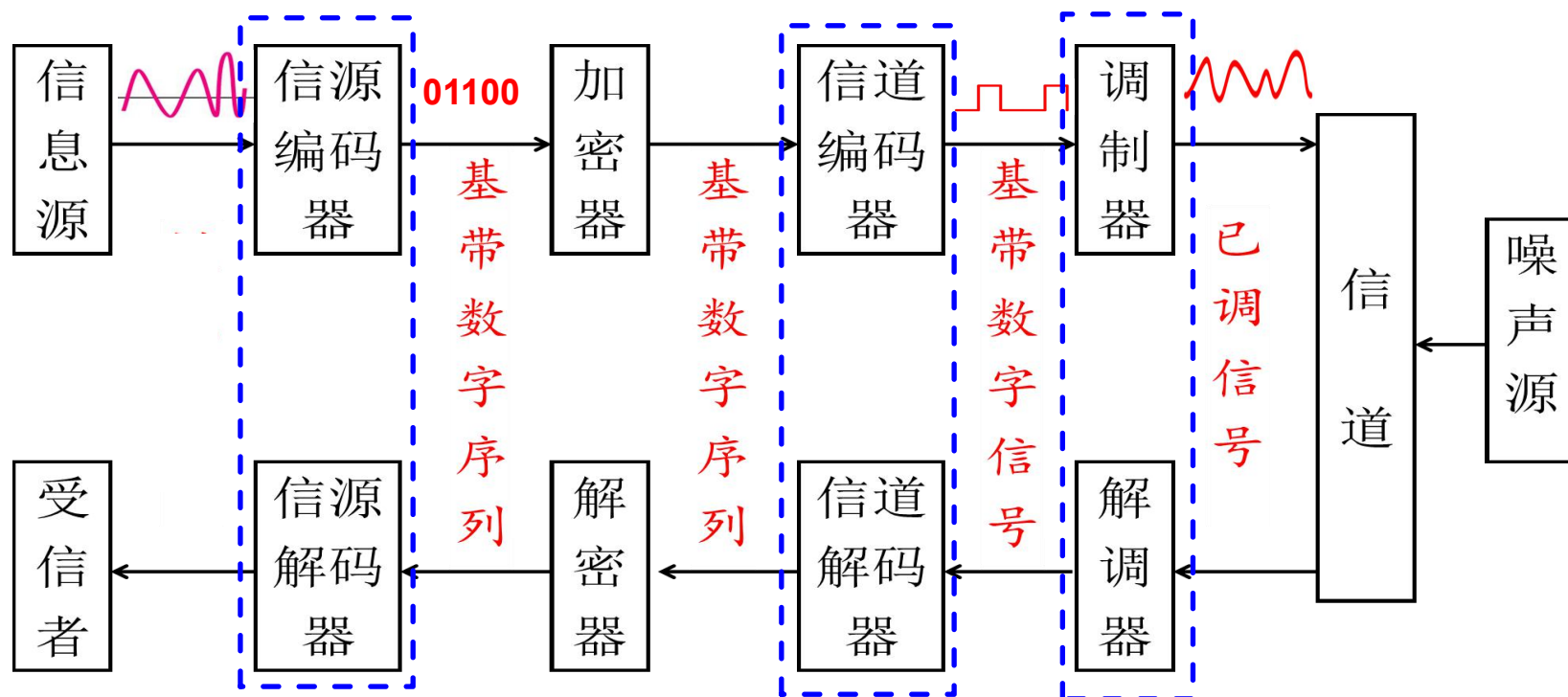
◆ 无线传输介质：特性、分类、应用

◆ 通信卫星

Outline

- 2.0 Basic Concepts on Data Communications
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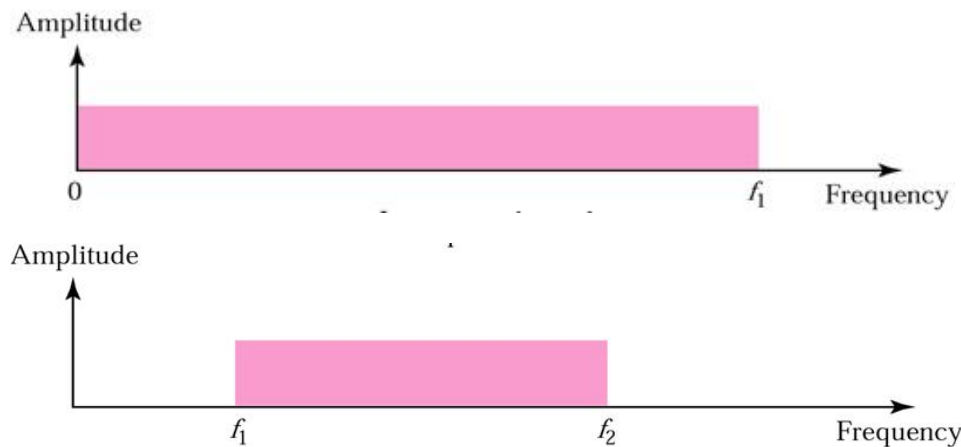
(补充)数字通信系统基本模型



- **信源编/译码器**——模拟信号数字化、压缩编码（减少冗余，提高编码效率）
- **加/解密器**——保密传输
- **信道编/译码器**——以信息在信道上的正确传输为目标，通过差错控制，提高信号的抗干扰能力
- **调制/解调器**——使信源发出的信号适合传输信道传输

Baseband transmission vs Passband transmission

- **Baseband signal:** Signals that run from 0 up to a maximum frequency
- **Passband signal:** Signals that are shifted to occupy a higher range of frequencies. eg. wireless transmissions

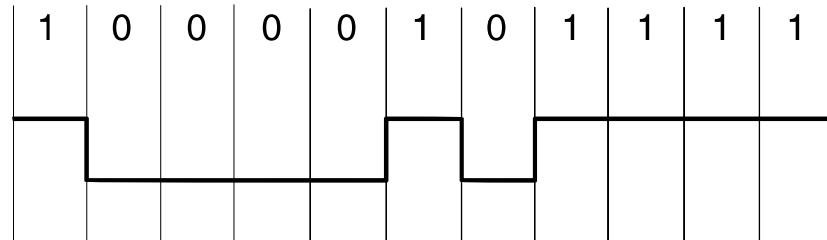


- **Baseband transmission:** In which the signals occupies frequencies from **0 to a maximum frequency**.
- **Passband transmission :** In which the signals occupies a band of frequencies around the frequency of the **carrier signal**.

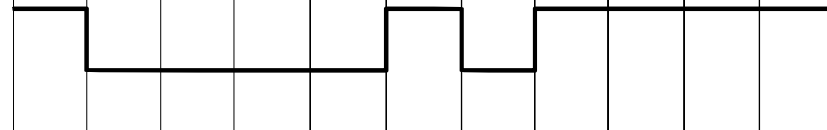
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2.5.1 Baseband transmission

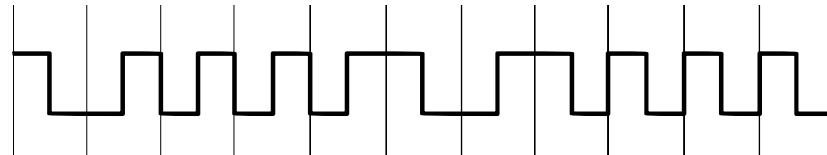
(a) Bit stream



(b) Non-Return to Zero (NRZ)



(d) Manchester



(e) Bipolar encoding
(also Alternate Mark
Inversion, AMI)

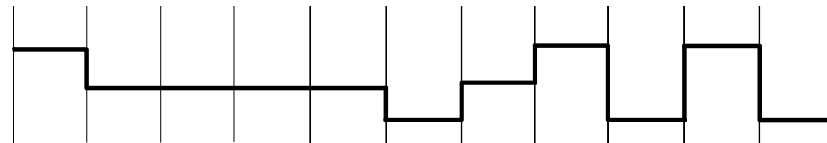
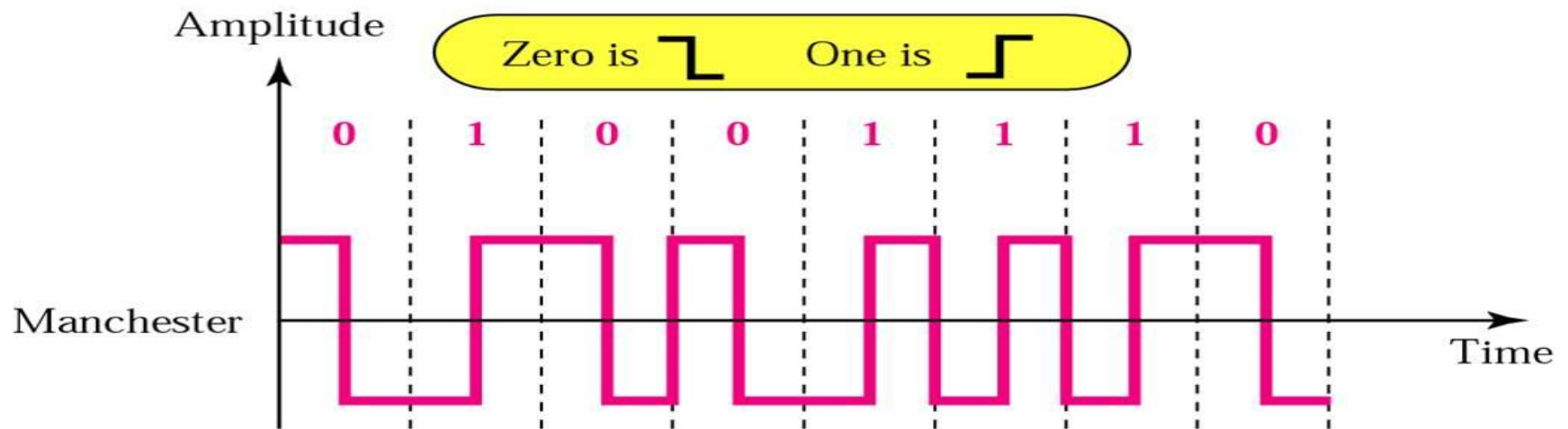


Figure 2-20. Line codes: (a) Bits, (b) NRZ, (c) NRZI, (d) Manchester, (e) Bipolar or AMI.

Manchester

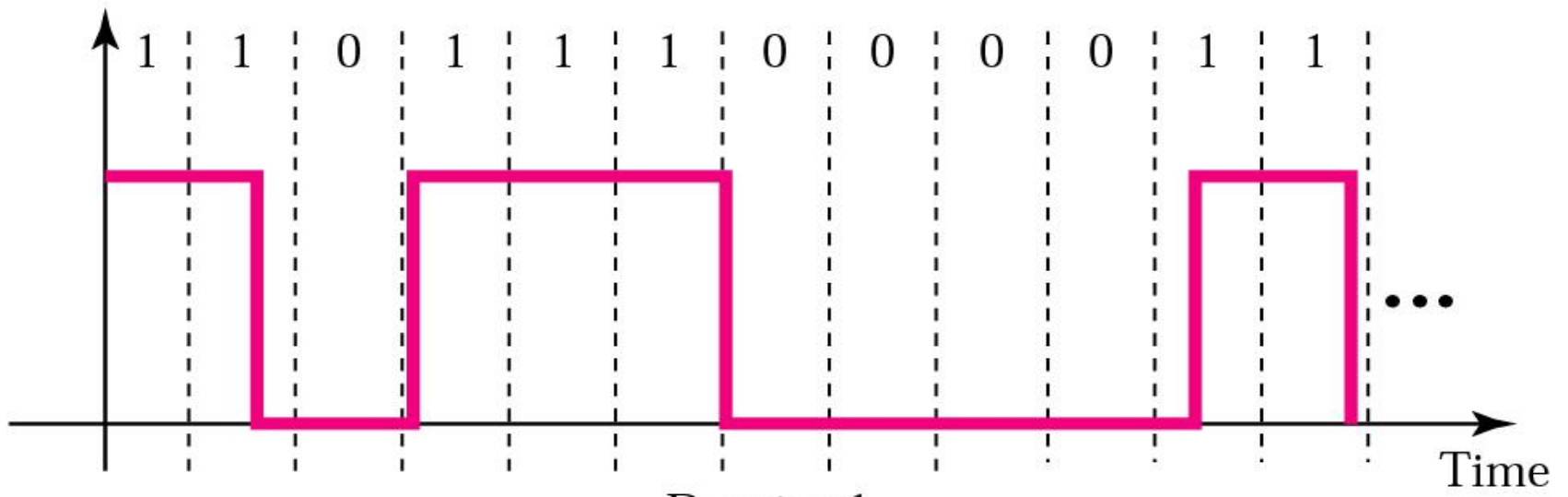
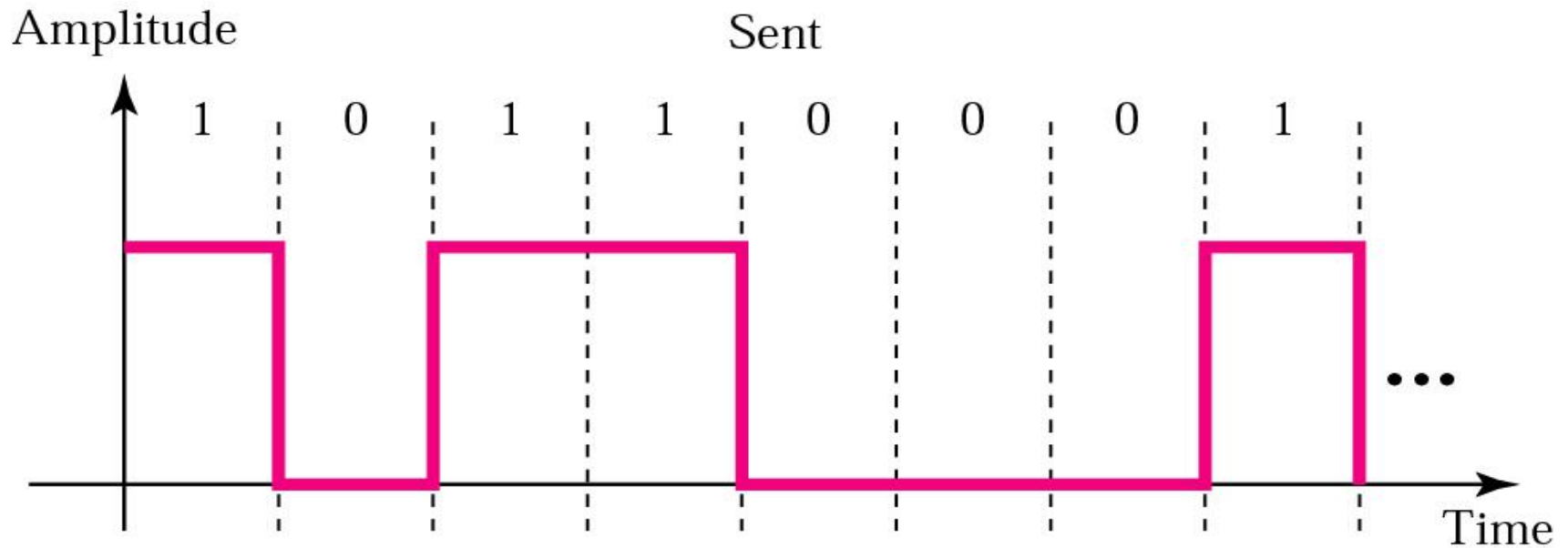
■ Manchester

- ◆ Transition in **middle** of each bit period
- ◆ Transition serves as **clock** and **data**
- ◆ Low to high represents 1, High to low represents 0, used by IEEE 802.3 LAN(Classic Ethernet)



Back to ←
Bit rate & Baud rate

Loss of Synchronization



Received
Computer Networks, Physical Layer

Baseband transmission

■ Clock Recovery

- ◆ the receiver must know **when one symbol ends and the next symbol begins** to correctly decode the bits
- ◆ Strategies:
 - **send a separate clock** signal to the receiver
 - **mix the clock signal with the data signal**

■ Bandwidth Efficiency

- ◆ **Bandwidth is a limited resource**, even for wired channels
- ◆ One strategy for using limited bandwidth more efficiently is to use **more than two signaling levels**.

■ Balanced Signals

- ◆ Signals that **have as much positive voltage as negative voltage** even over short periods of time
- ◆ Have **no DC** electrical component
 - Strongly attenuate
 - **Capacitive coupling**(电容耦合) passes only the AC portion of a signal.

Baseband transmission

Clock Recovery

Strategies:

- send a separate clock signal to the receiver;
- mix the clock signal with the data signal

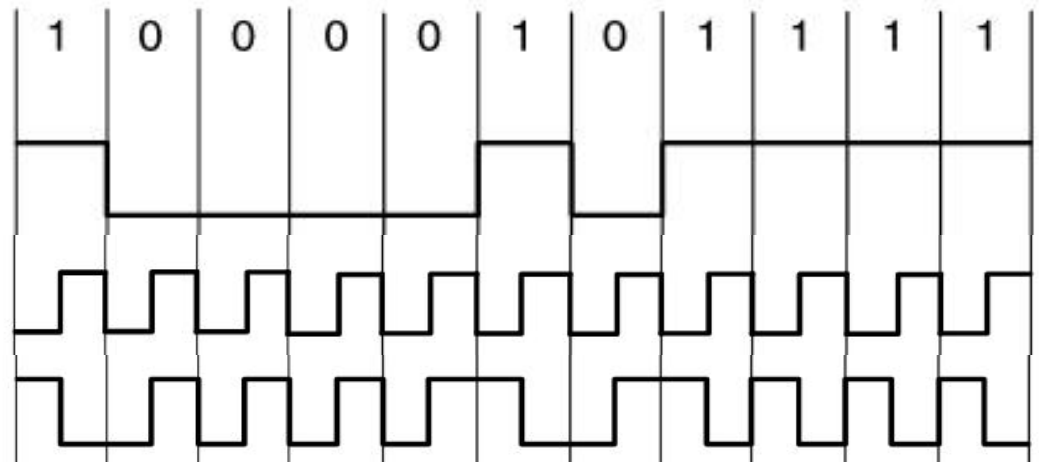
Manchester

(a) Bit stream

(b) Non-Return to Zero (NRZ)

(Clock that is XORed with bits)

(d) Manchester



$$X \oplus 0 = X, \quad X \oplus 1 = \bar{X}$$

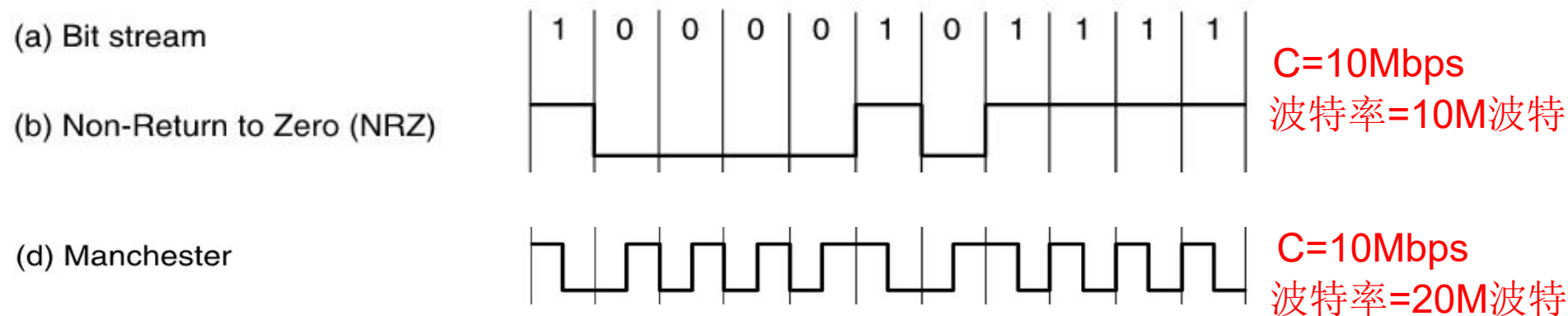
$$\text{Clock} \oplus \text{NRZ} = \text{Manchester}$$

- ◆ 4B/5B never be a run of more than three consecutive 0s

Baseband transmission

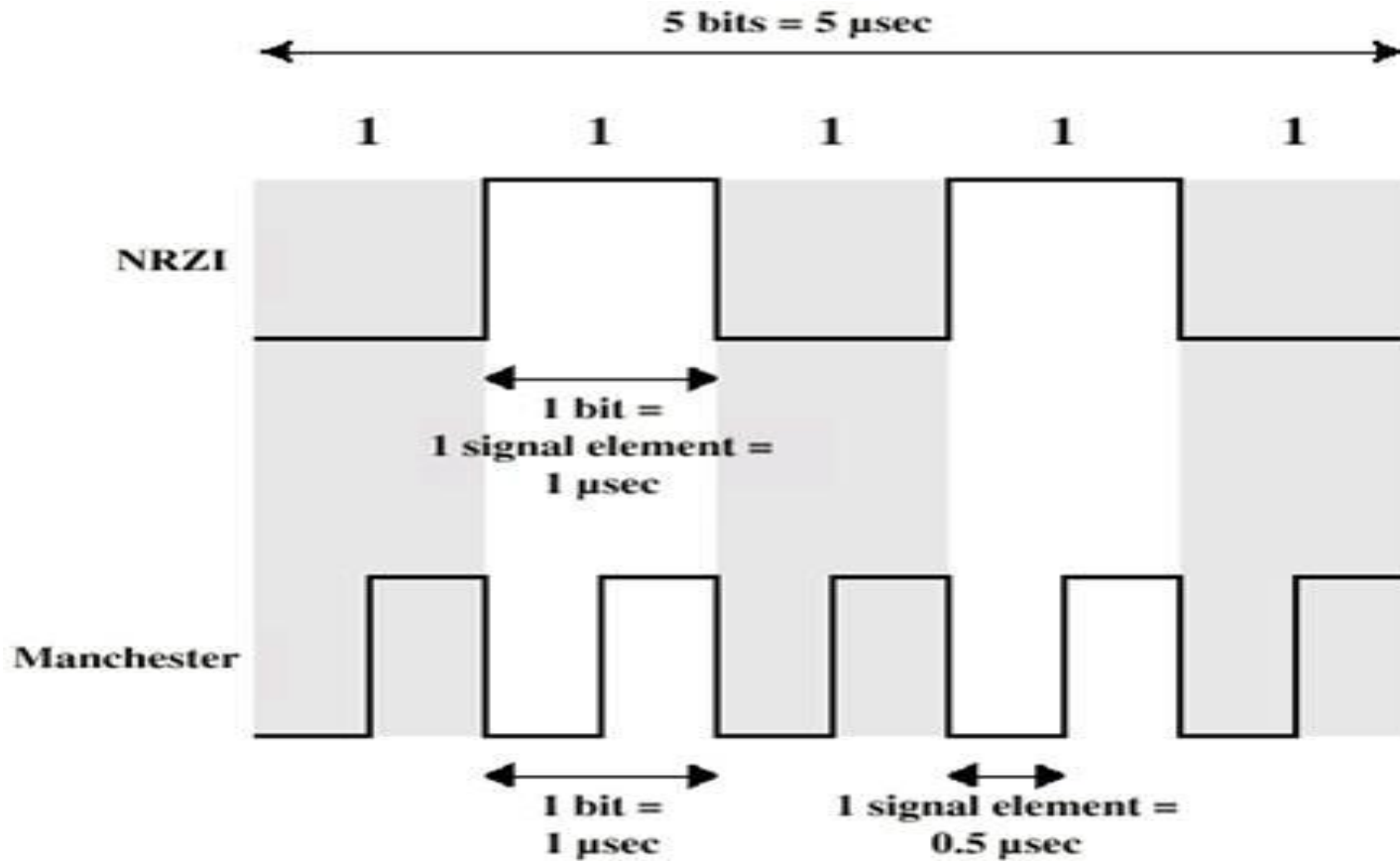
Bandwidth Efficiency

- ◆ NRZ need a bandwidth of at least $B/2$ Hz when the bit rate is B bits/sec (Nyquist rate: $C=2H\log_2 V$) (2bit/Hz)



- ◆ The bandwidth efficiency of Manchester is 50%
- ◆ the number of signal levels does not need to be a power of two

Manchester



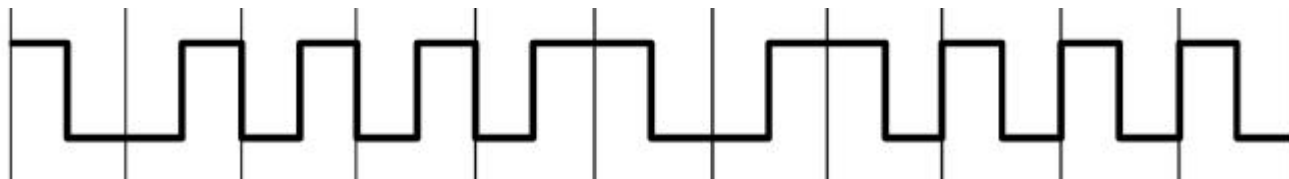
Baseband transmission

■ Balanced Signals

- ◆ **Bipolar encoding**: use two voltage levels to represent a logical 1.(AMI)



- ◆ **Manchester**



- ◆ **NRZ**
- ◆ **Using mapping to achieve balance. 8B/10B**

2.5.2 Passband Transmission--Modulation

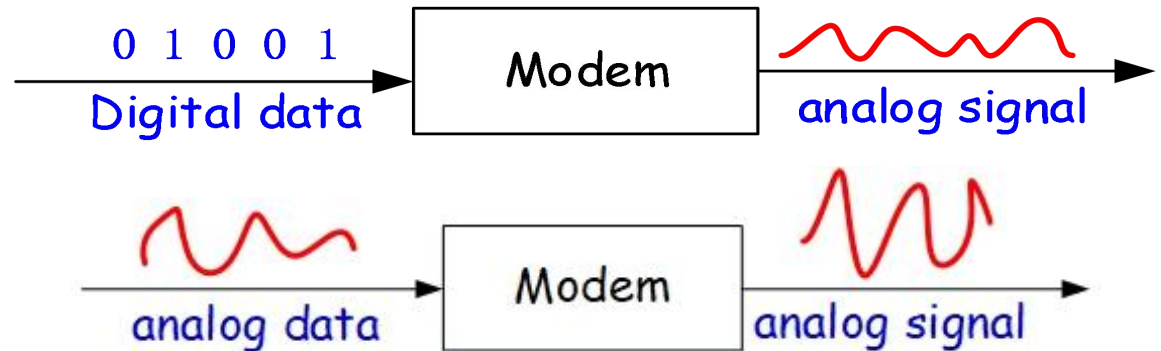
■ Placing a signal in a given frequency band is useful

- ◆ the frequency of transmission media does not start at zero
- ◆ for wireless channel: sending very low frequency signals is not practical
- ◆ for wires: different kinds of signals can coexist on one channel

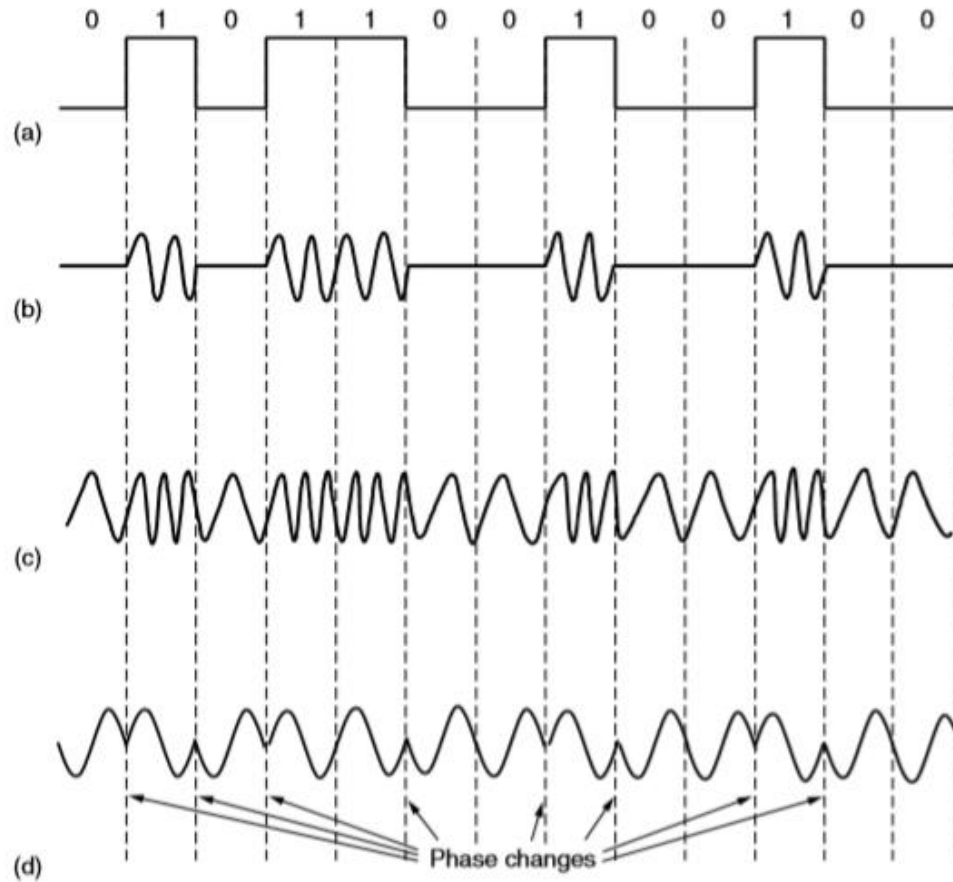
■ Digital modulation is accomplished by modulating a carrier signal sits in the passband.

■ modulate the amplitude, frequency, or phase of the carrier signal.

■ device: Modem



Modulations(调制)



(a) A binary signal

(b) Amplitude shift keying(ASK)

(c) Frequency shift keying(FSK)

(d) Phase shift keying(PSK)

Combination of modulation techniques

■ Modems use a combination of modulation techniques to transmit multiple bits per baud

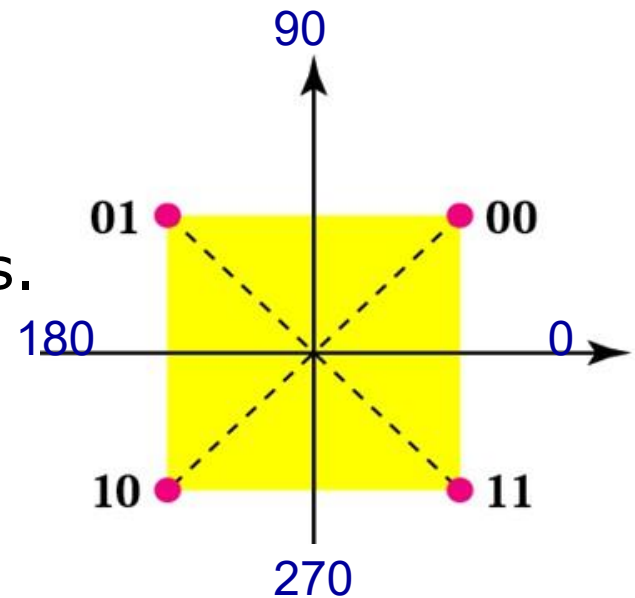
◆ **QPSK**: **Q**uadrature **P**hase Shift Keying-正交相移键控

◆ **QAM**: **Q**uadrature **A**mplitude Modulation-正交幅度调制

■ Constellation Diagram

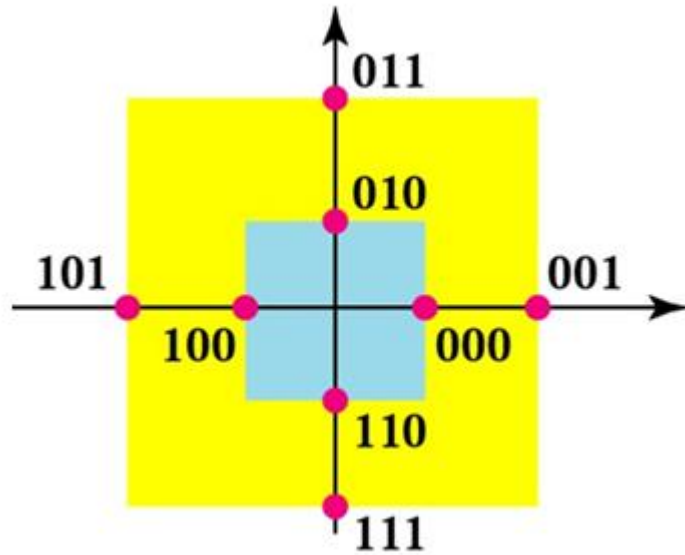
◆ The **phase** of dot is indicated by the angle of a line from it to the origin makes with the positive X-axis.

◆ The **amplitude** of dot is the distance from the origin.

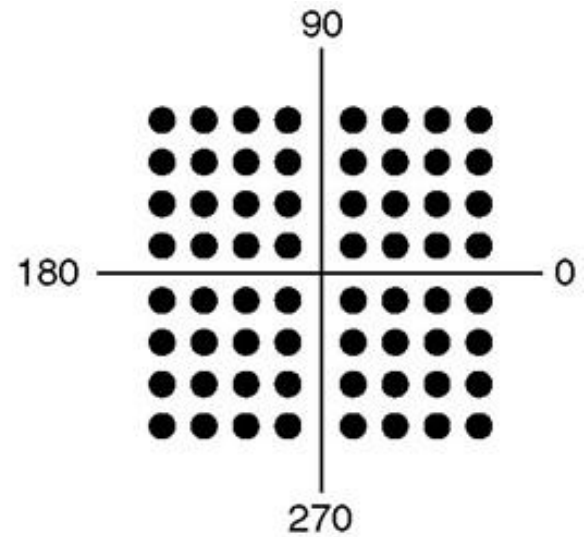


QPSK(4-PSK)
1 amplitude
4 phases

QAM



8QAM



64QAM

(补充)Digitized Analog Signals--Source Coding

■ Procedures

◆ Sampling (采样)

- According to **Nyquist theorem**, the sampling rate must be at least 2 times the highest frequency(Low-pass signal).

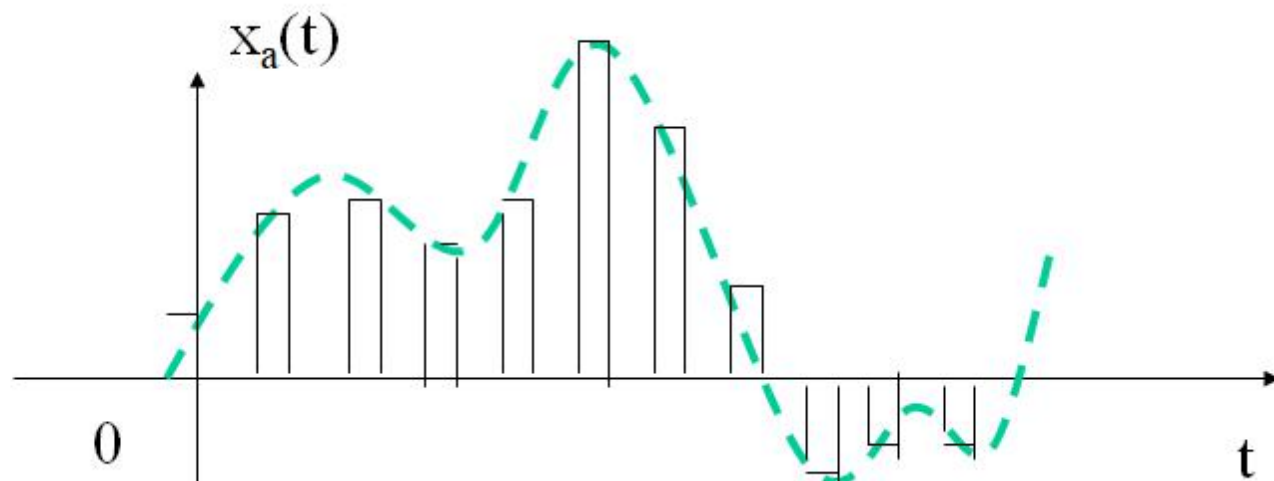
◆ Quantizing (量化)

- Dividing signal strength into levels linearly or logarithmically
- **A Law** algorithm: Used in European and China
- **Mu Law(μ -Law)** algorithm:Used in North America and Japan

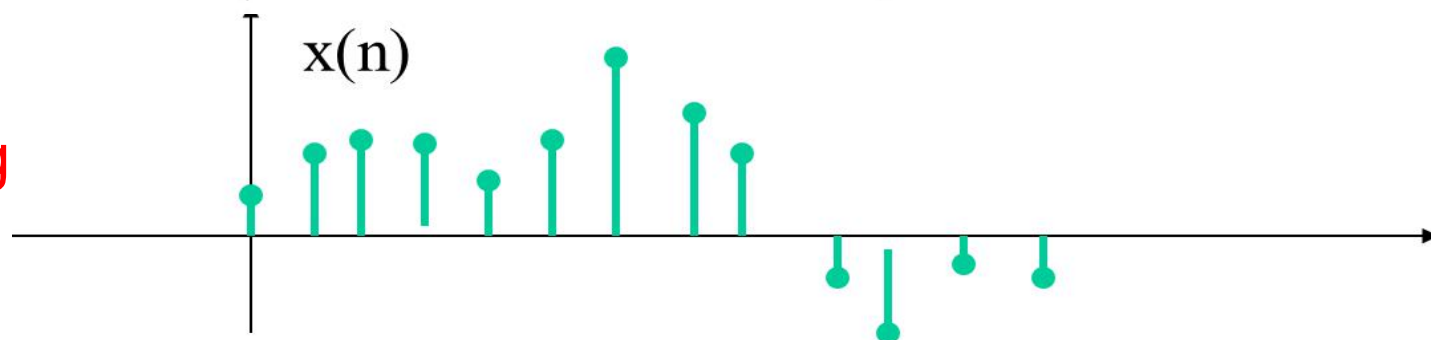
◆ Encoding (编码)

■ Device: Codec编码解码器

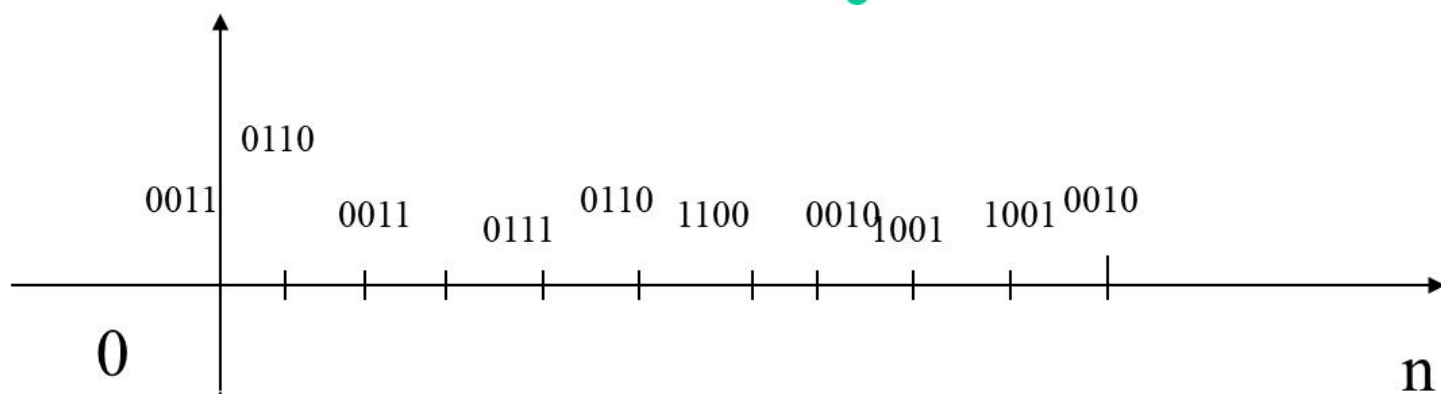
Sampling



Quantizing



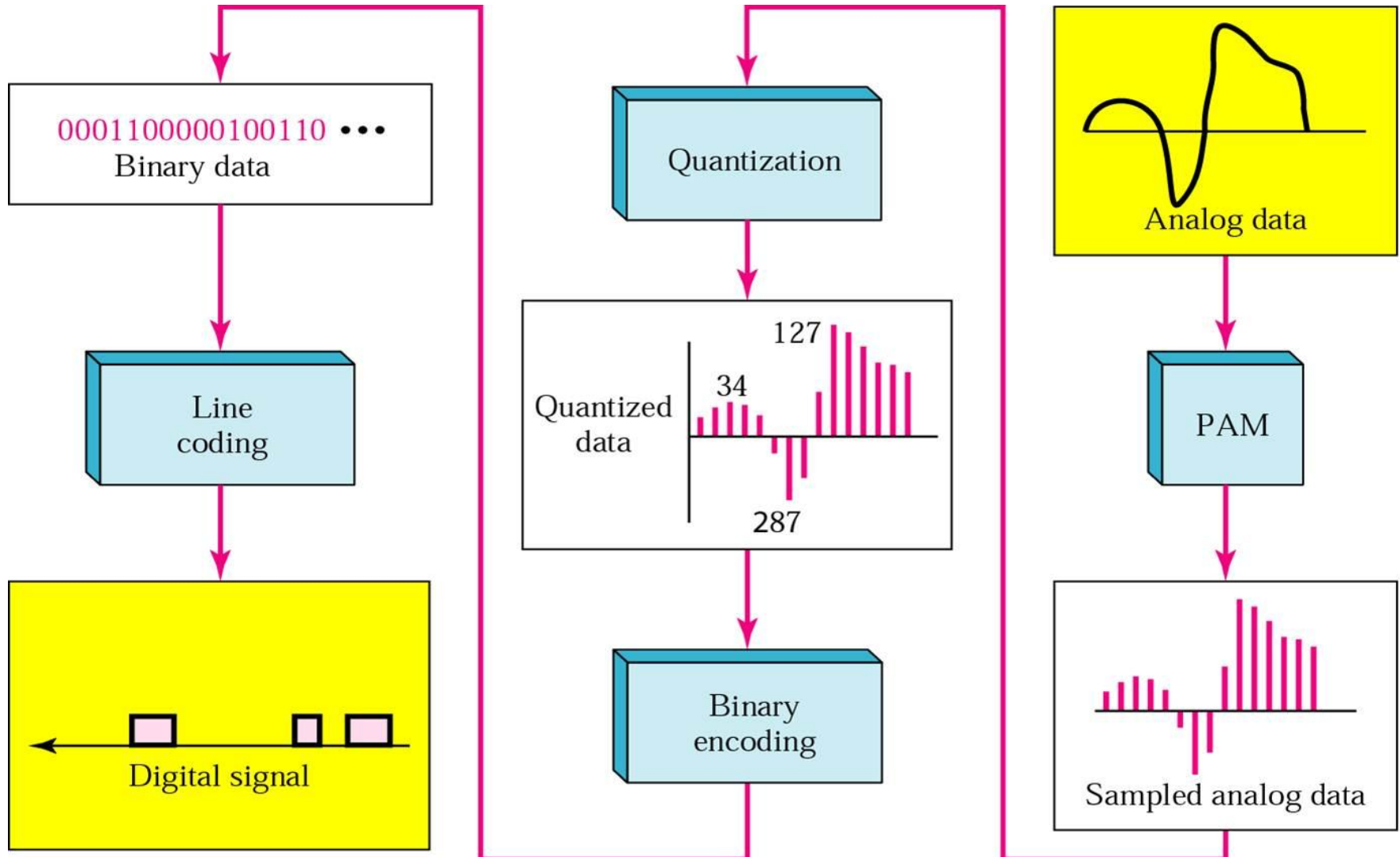
Encoding



$x(n)$ 的二进制数

From analog signal to PCM digital code

■ 数字化语音信号 **PCM**: Pulse Code Modulation (脉冲编码调制)



PCM

- 已知：语音信号频率范围约4KHz,转化为数字信号，每个采样值用8bit表示，不考虑压缩，请问对应的数字信号的带宽是多少？

- Sampling 采样频率：8KHz, 即调制速率是8Kbaud

- Quantizing & Encoding

$$C = \text{调制速率} * 8 = 8K * 8 = 64Kb/s$$

小结

■ 传输介质

◆ 传输介质特性：带宽、时延、最大传输距离、抗干扰性

...

◆ 有线传输介质：原理、结构、分类、特性、应用

◆ 无线传输介质：特性、分类、应用

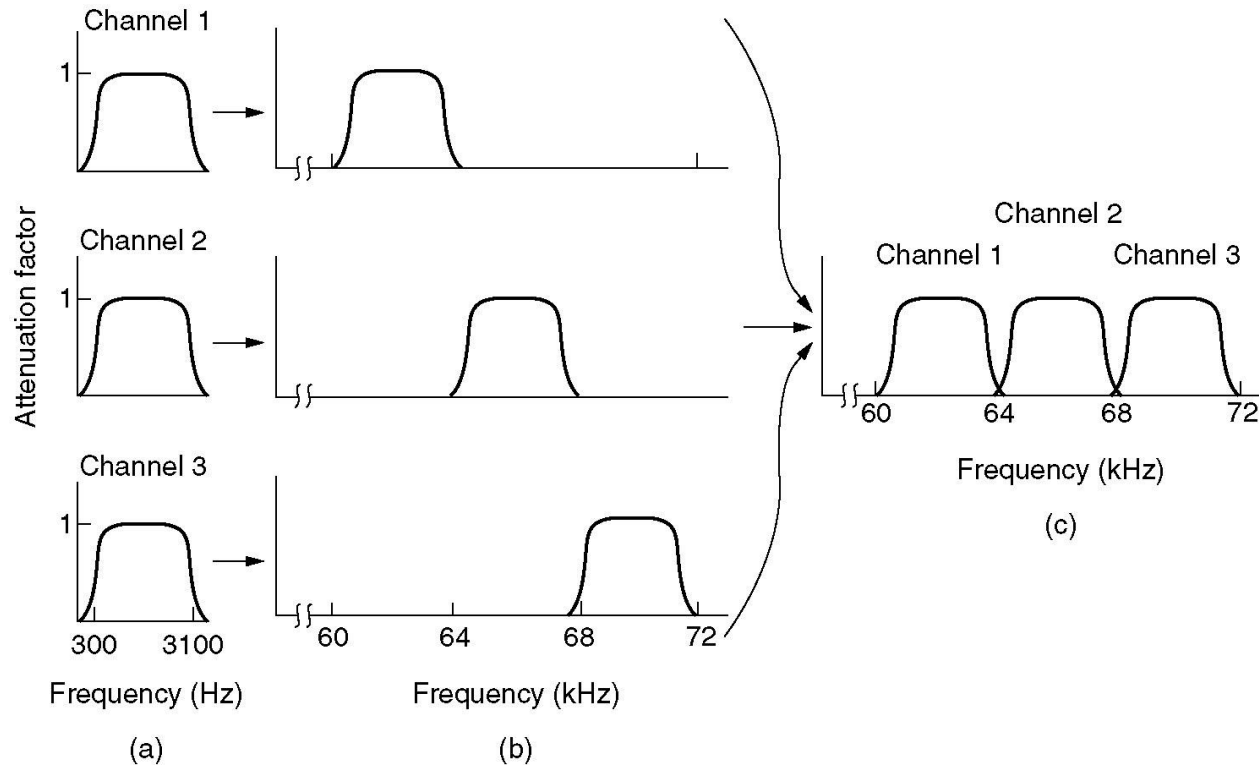
◆ 通信卫星

2.5.3 Multiplexing (复用)

- “refer to a process where multiple analog message signals or digital data streams are combined into one signal” (*Wikipedia*)
 - ◆ TDM (时分复用)
 - ◆ FDM (频分复用)
 - ◆ WDM (波分复用)
 - ◆ CDM (码分复用)



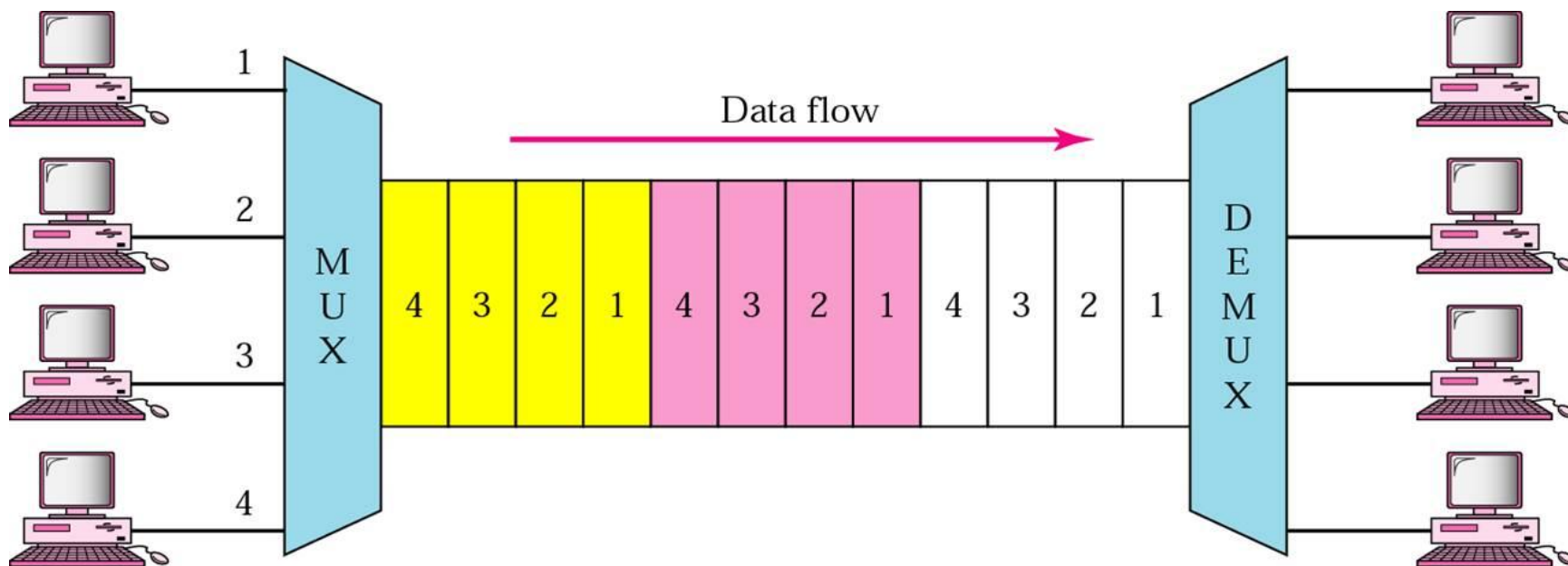
Frequency Division Multiplexing



- (a) The original bandwidths.
- (b) The bandwidths raised in frequency.
- (b) The multiplexed channel.

2.5.4 Time Division Multiplexing

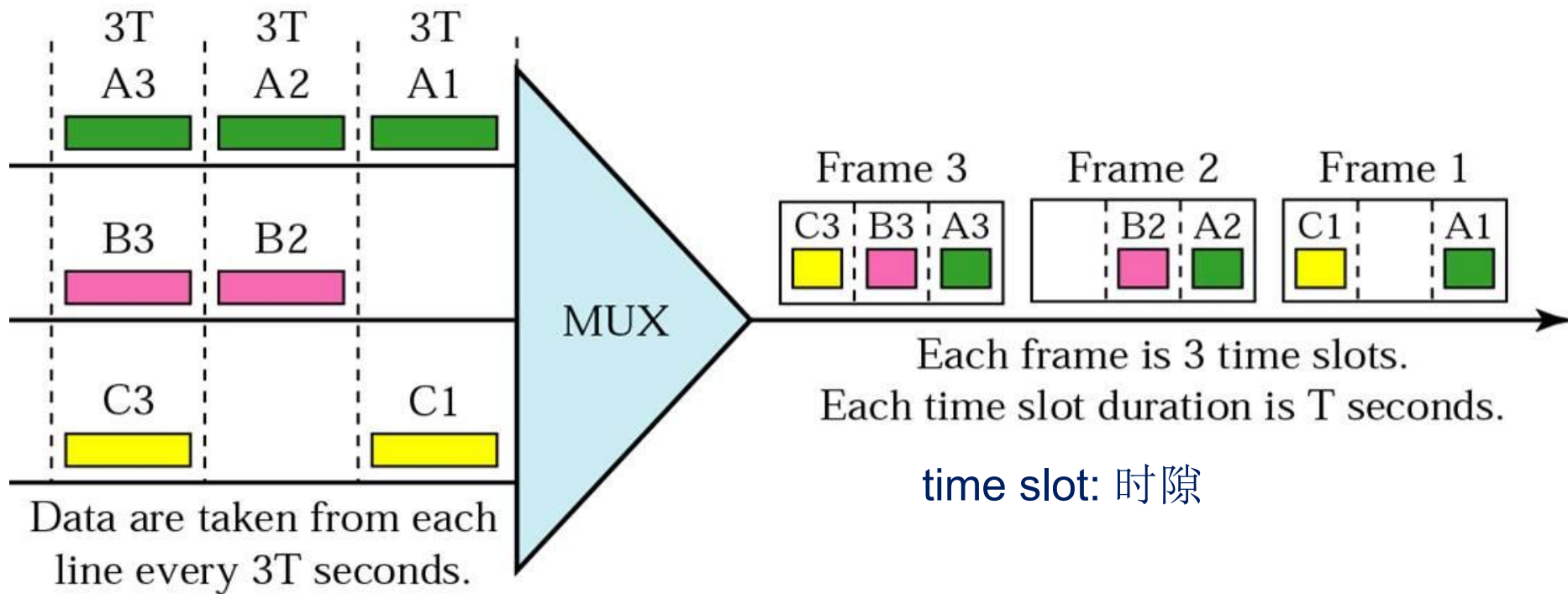
- Signals are transferred apparently simultaneously as sub-channels in one communication channel, but are **physically taking turns** on the channel.
- The time domain is divided into several recurrent **timeslots** of fixed length, one for each sub-channel.



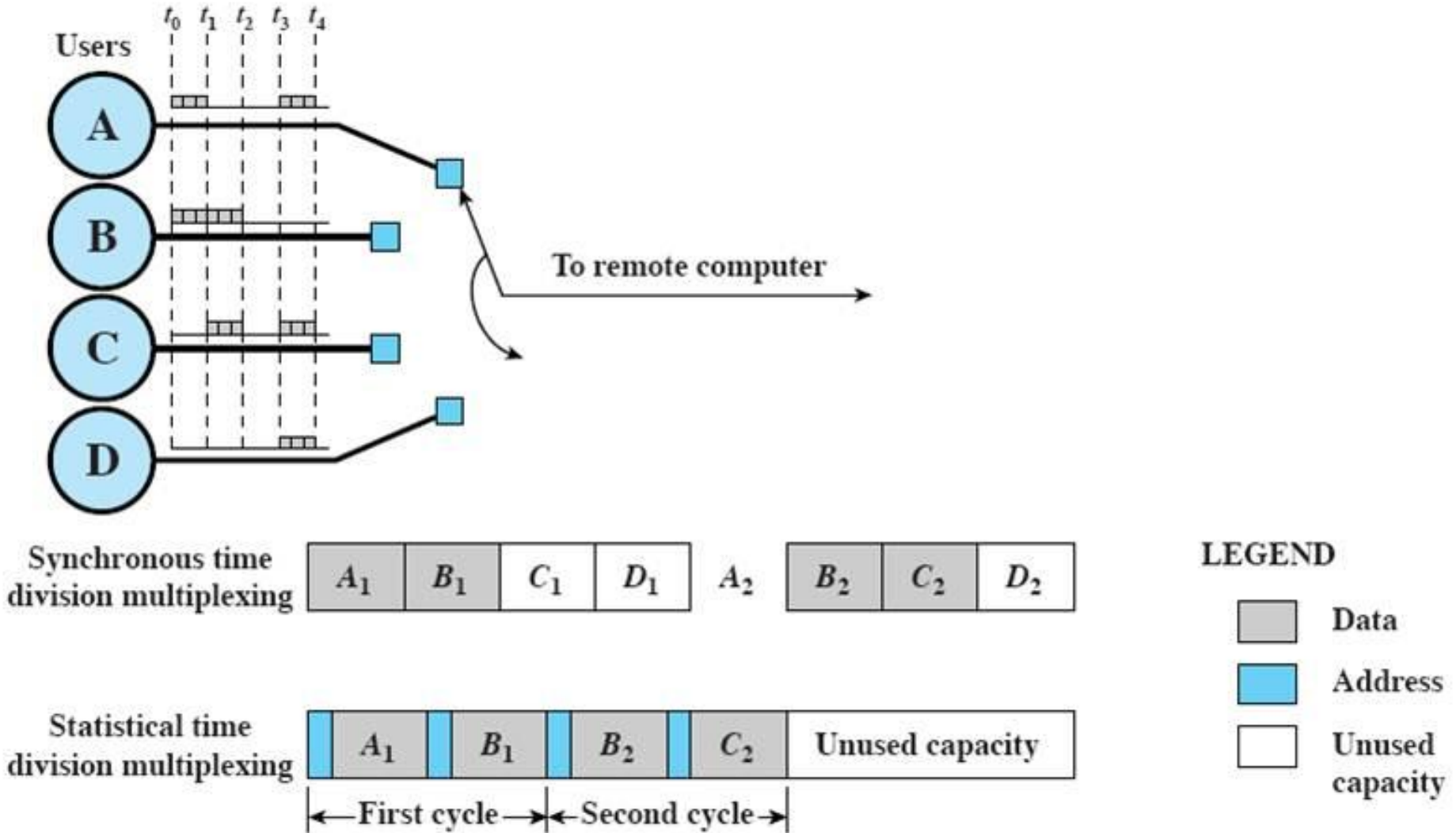
[Forouzan]

TDM Frames (复用帧)

- May cause waste of resources

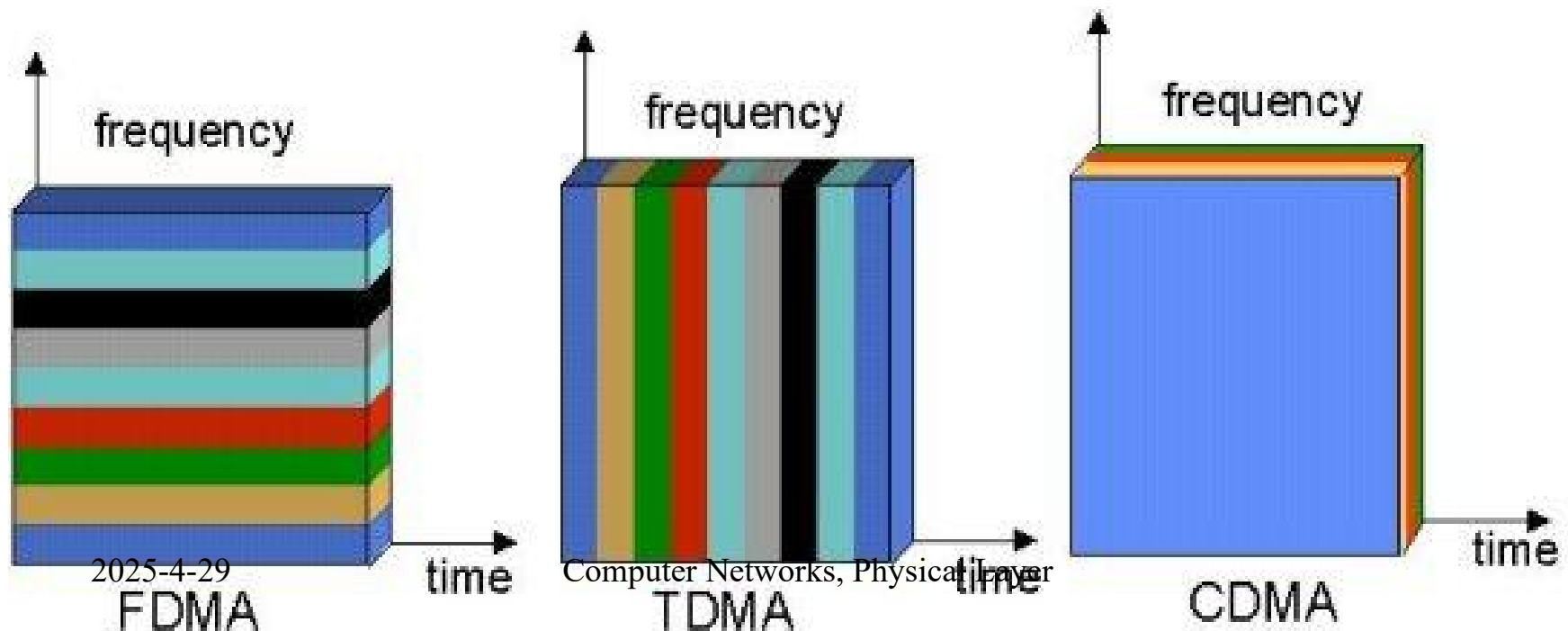


Synchronous TDM vs. Statistical TDM



2.5.5 Code Division Multiplexing

- A form of spread spectrum communication
- Code Division **Multiple Access**: Each station can transmit over the **entire frequency spectrum** all the time.
- Multiple simultaneous transmissions are separated using coding theory.



Summary (1)

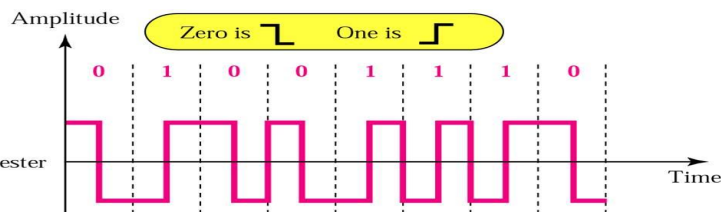
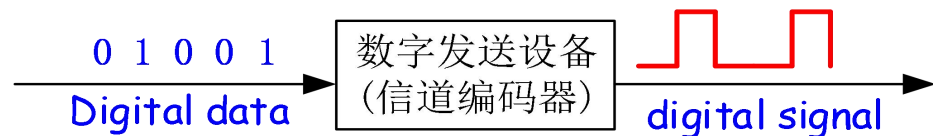
Line code--Baseband transmission

- 目标：便于接收方正确接收道载有数据的信号，提高纠错能力和抗干扰能力。

- 评价指标

- Clock Recovery(便于时钟同步)
- Bandwidth Efficiency(节省带宽资源)
- Balanced Signals(消除直流分量)

- 编码技术：NRZ, Manchester, AMI



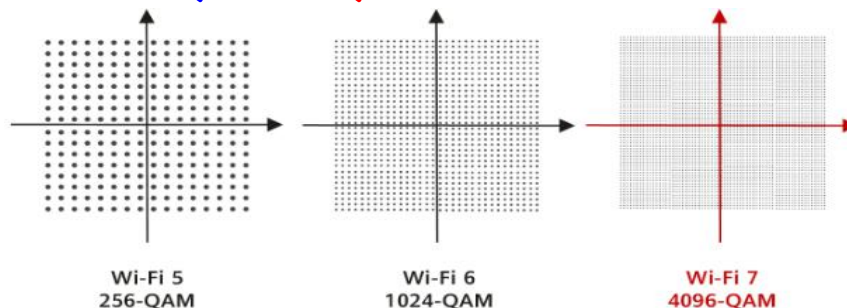
Modulation--Paseband transmission

- 目标：改变基带信号的频谱，以适应信道特性



- 技术：ASK, FSK, PSK, QPSK, QAM

- 设备：Modem



Q: QAM 的阶数越高越好吗?

Summary(2)

Digitized Analog Signals--source coding

◆ 将信源的模拟信号进行数字化

◆ 技术

➤ Sampling(Nyquist theorem)

➤ Quantizing(A-law, μ -law)

➤ Encoding

◆ 设备: CodeC

◆ PCM(64kbps)

Multiplexing technology

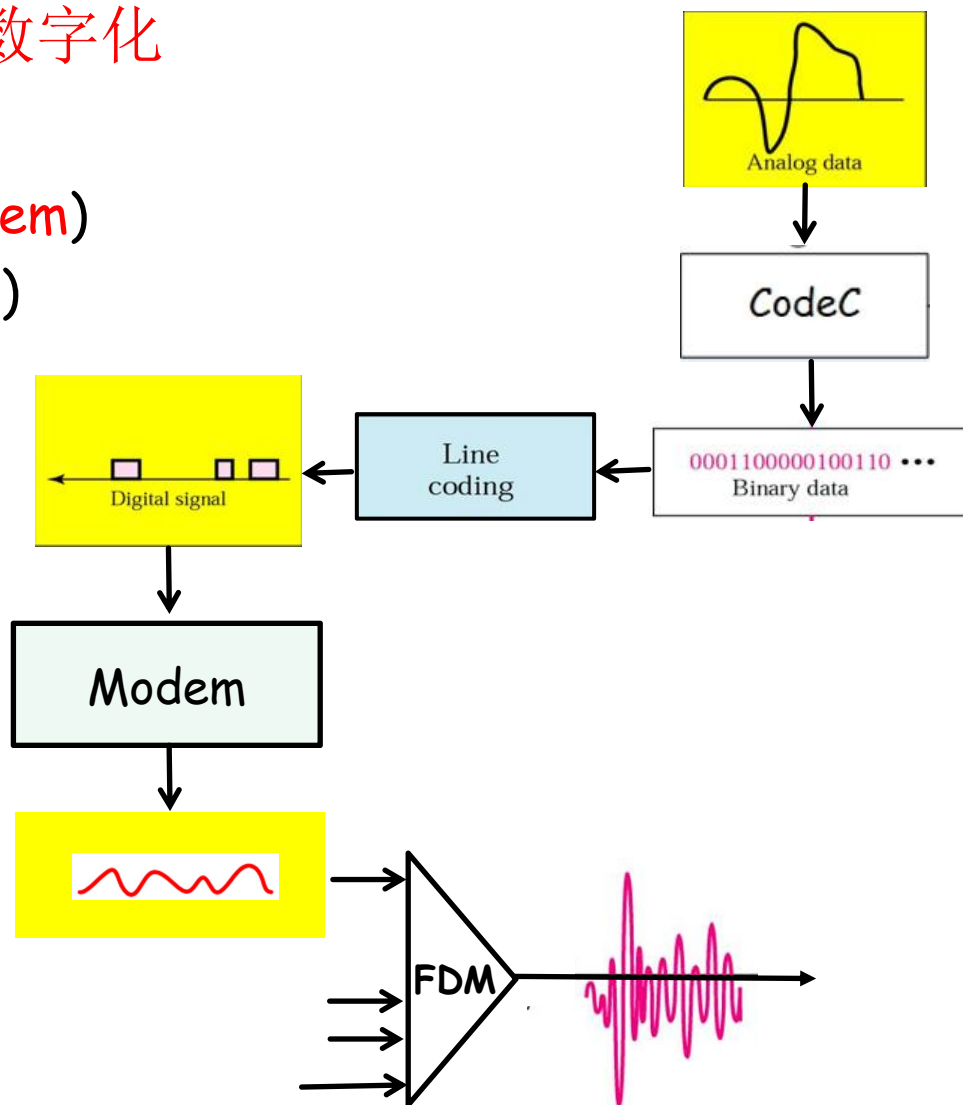
◆ 提高信道利用率

◆ 技术

➤ FDM,TDM

➤ CDM,WDM

◆ 设备: Multiplexer

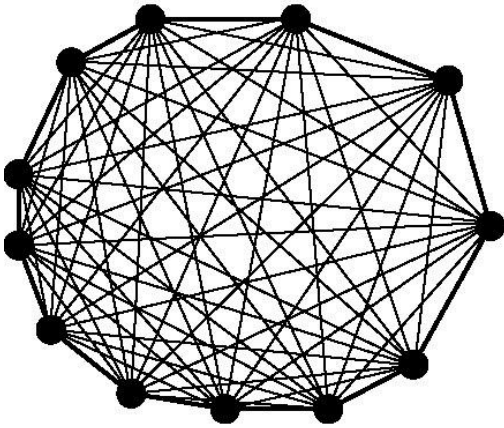


Outline

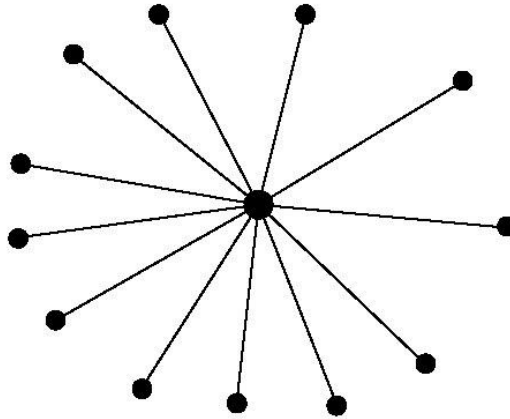
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- (补充) Protocol Example: RS-232

2.6.1 Structure of the Telephone System

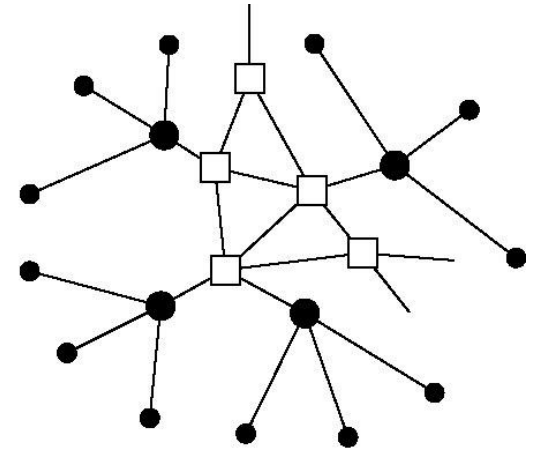
Structure of the Telephone System



(a)



(b)



(c)

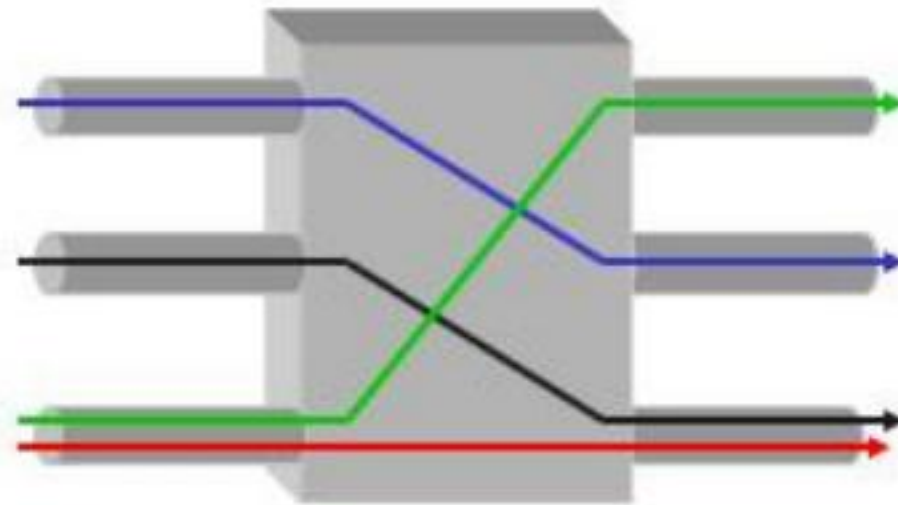
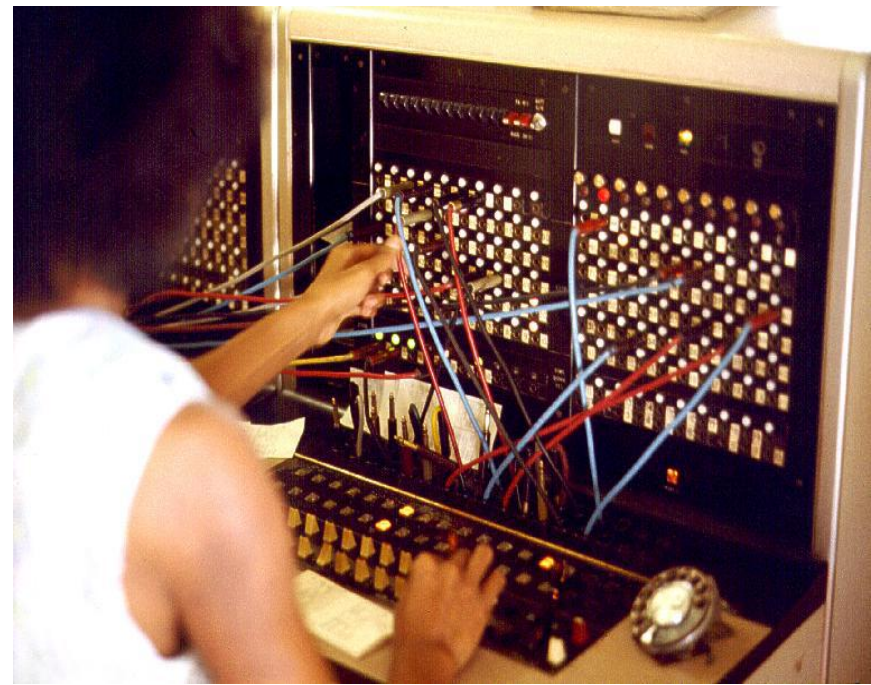
(a) Fully-interconnected network.

(b) Centralized switch.

(c) Two-level hierarchy.

What is Switching?(交换)

- In telephone system, switches **interconnect** **telephone subscriber lines** or virtual circuits of digital systems to establish telephone calls between subscribers.

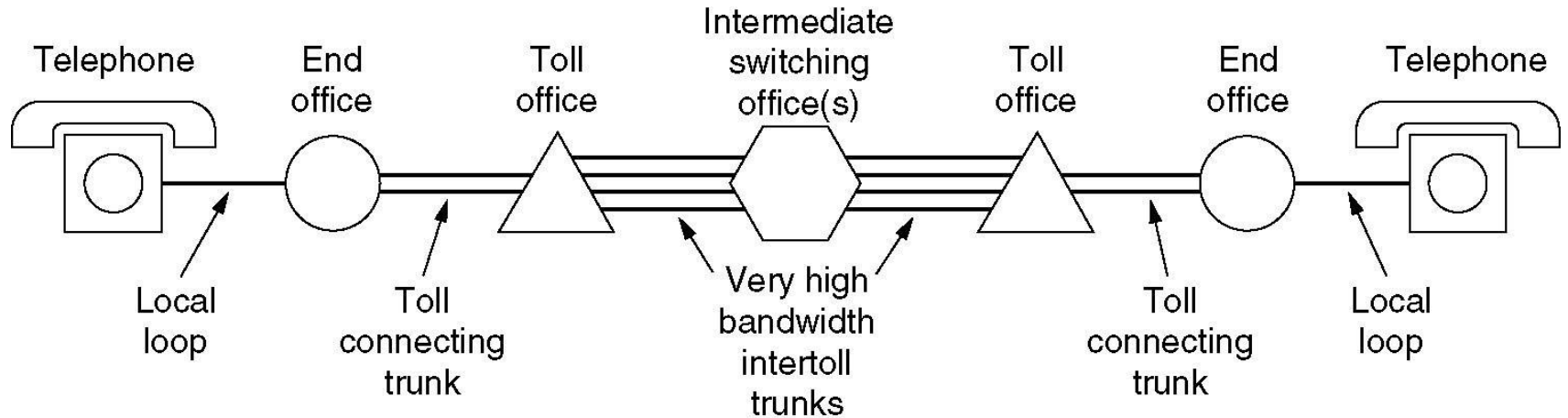


Incoming links **Node (Switch)** Outgoing links

Computer Networks, Physical Layer

Structure of Telephone System

■ A typical circuit route for a long-distance call

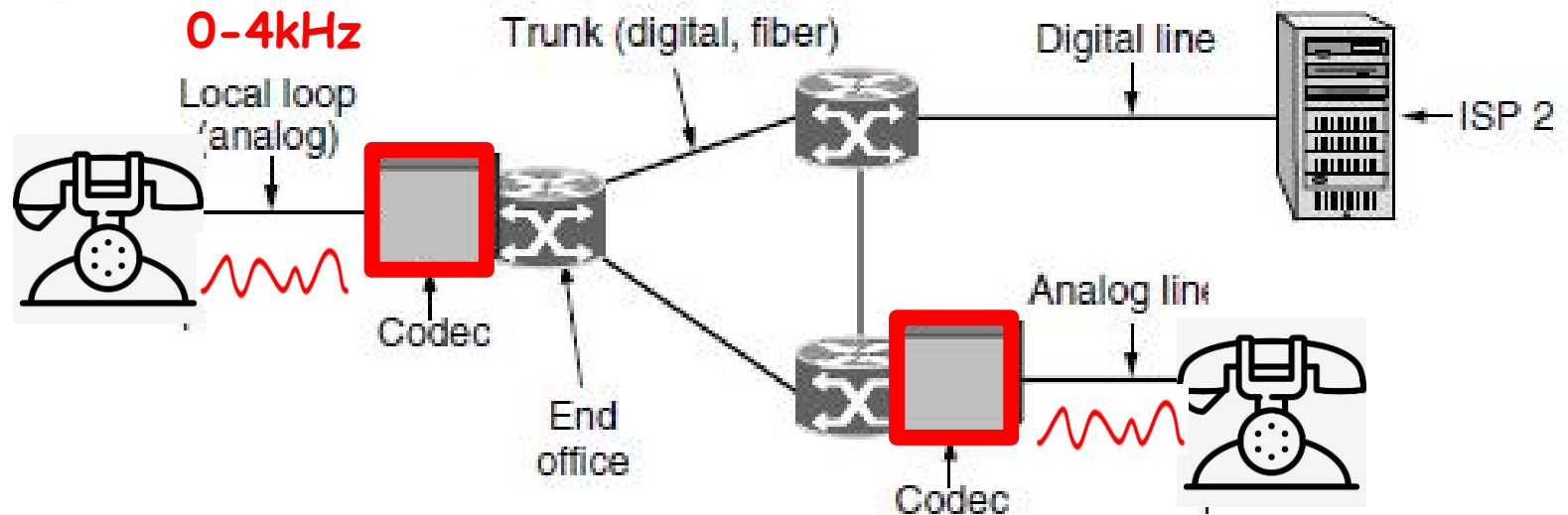


- Local loops 本地环路
 - Analog twisted pairs going to houses and businesses
- Trunks 中继 (intertoll trunk, toll connecting trunk)
 - Digital fiber optics connecting the switching offices
- Switching offices 交换局
 - End office端局、Tandem office汇接局、Toll office长途局

2.6.3 The Local Loop: Modems, ADSL and Fiber

Digital and Analog Transmission in PSTN

The use of both analog and digital transmissions for a telephone to telephone call. Conversion is done by the **Codecs**.



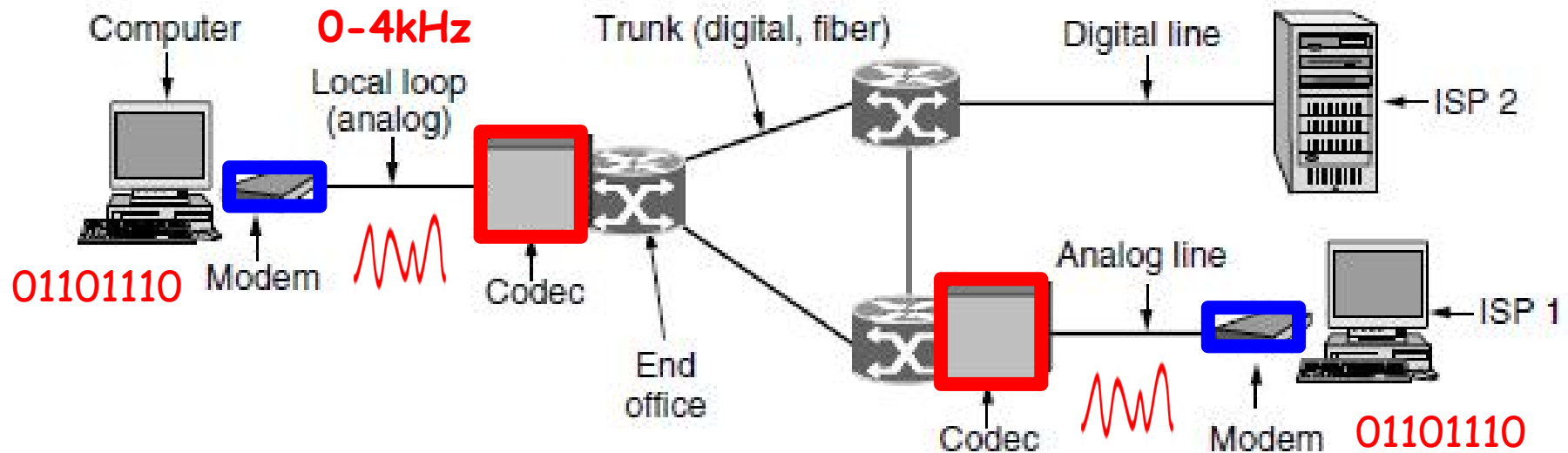
Voice signal: 300Hz-3400Hz

Codec(编码解码器):

- 8000 samples per second, 1 sample = 8 bits (非线性编码)

Digital and Analog Transmission in PSTN

The use of both analog and digital transmissions for a computer to computer call. Conversion is done by the **Modems and Codecs**.



Codec(编码解码器):

- 8000 samples per second, 1 sample = 8 bits (非线性编码)
- PCM $8000 * 8 = 64000$ bits per second

Modems

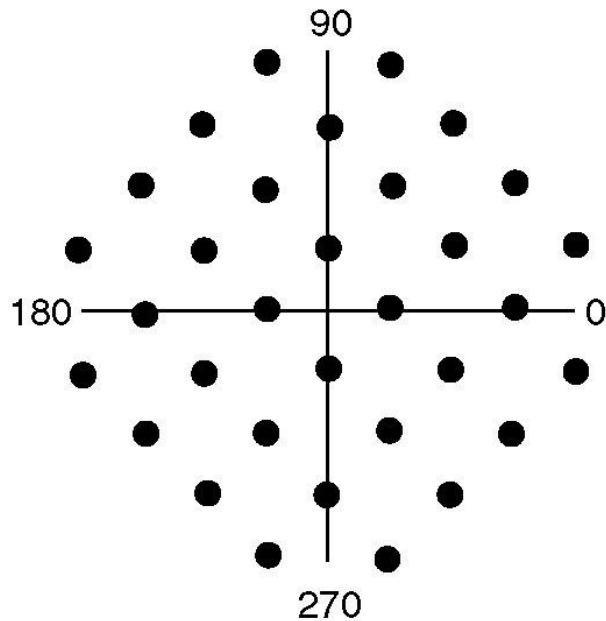
■ Modems use a combination of modulation techniques to transmit multiple bits per baud (multiple amplitudes and multiple phase shifts)

◆ **QAM** : Quadrature Amplitude Modulation-正交幅度调制

■ **Modems sample 2400 times/sec** and focus on getting more bits per sample

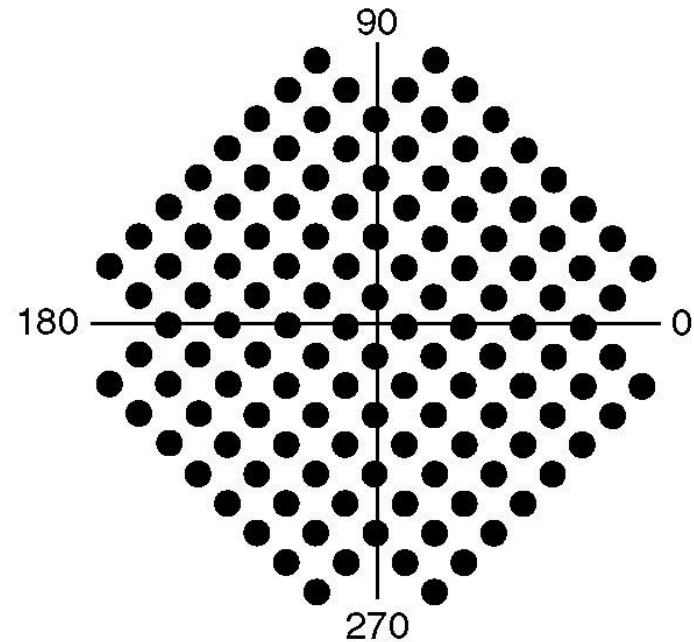
- ◆ V.32: $2400 \text{ baud} * (5-1) \text{ bits} = 9.6\text{Kbps}$
- ◆ V.32bis: $2400 \text{ baud} * (7-1) \text{ bits} = 14.4\text{Kbps}$
- ◆ V.34: $2400 \text{ baud} * 12 \text{ bits} = 28.8\text{Kbps}$
- ◆ V.34bis: $2400 \text{ baud} * 14 \text{ bits} = 33.6\text{Kbps}$

V.32 & V.32bis



(a)

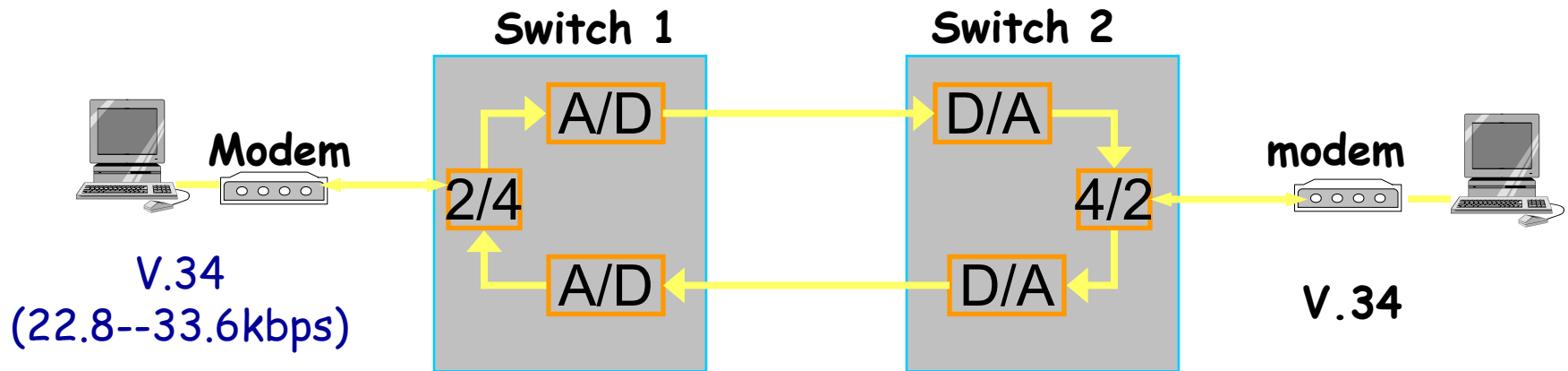
(a) V.32 for 9600 bps.



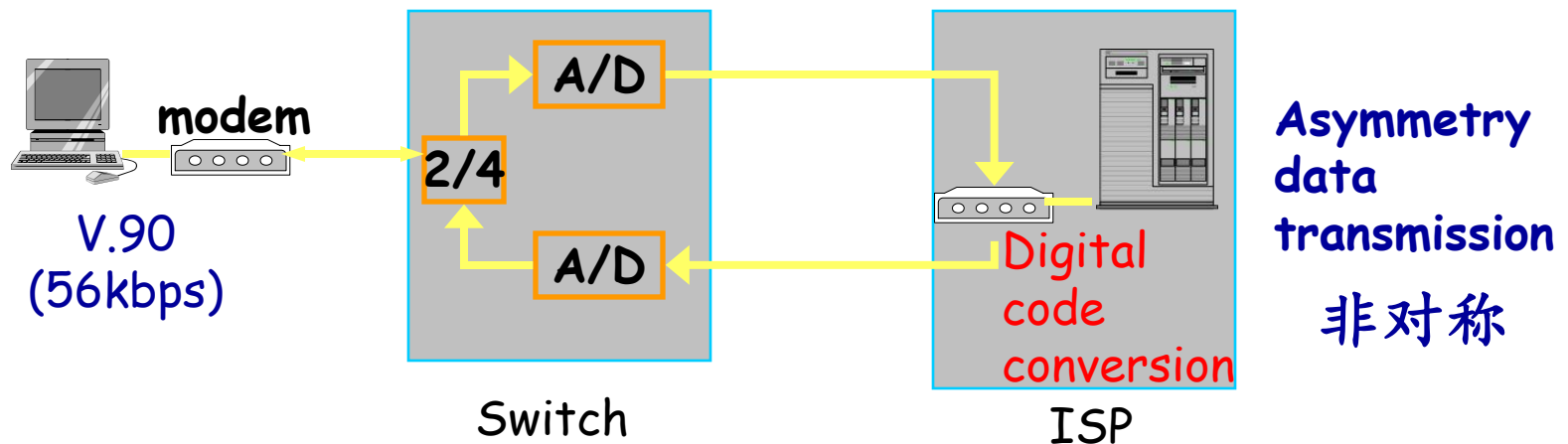
(b)

(b) V32 bis for 14,400 bps.

Maximum Data Rate of Dial-up

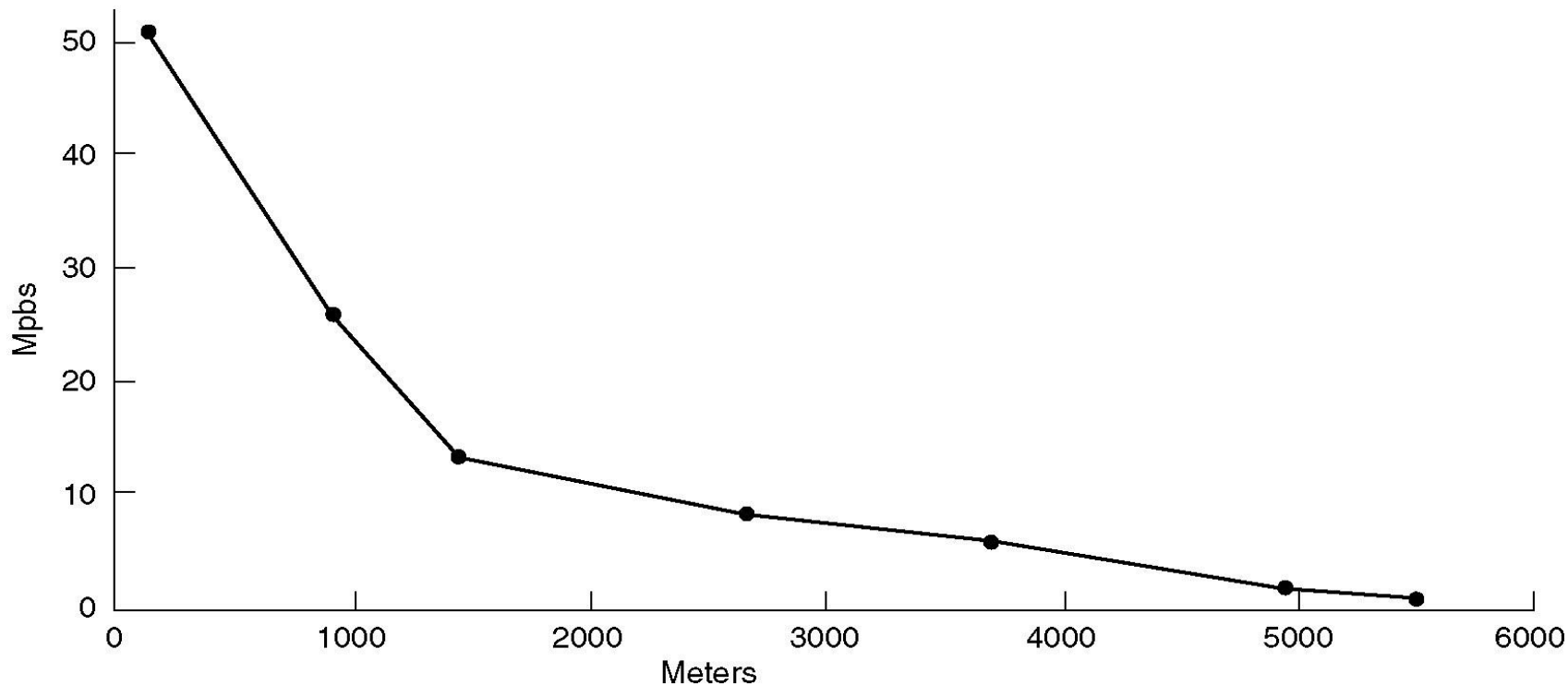


Noise comes from local loops. **Shannon** limit of telephone system is about 35kbps



DSL: Digital Subscriber Lines (数字用户线)

Bandwidth versus distances over UTP3 for DSL

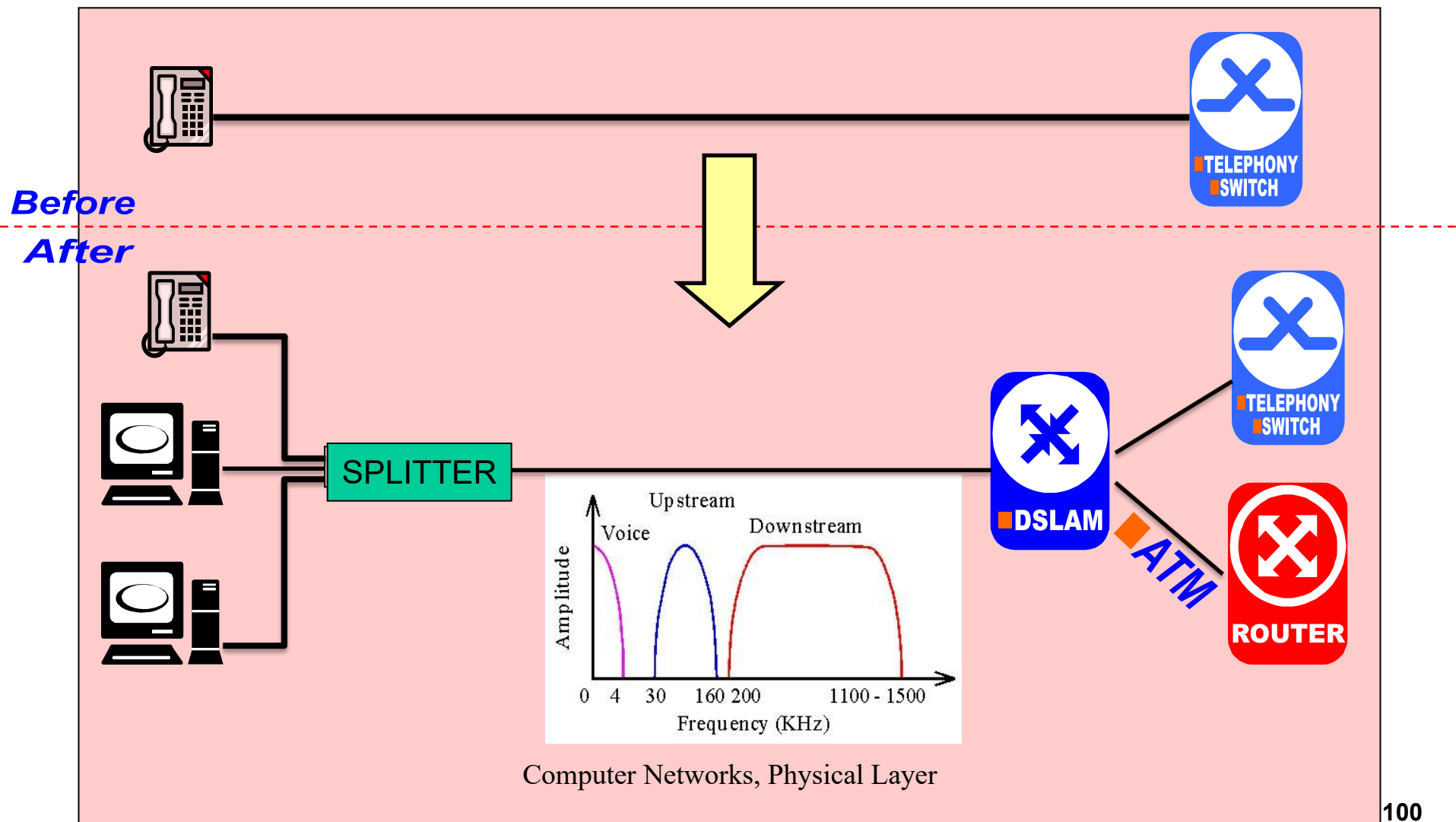


- DSL在较短距离内可提供较高带宽
- 随着距离增加，带宽急剧下降

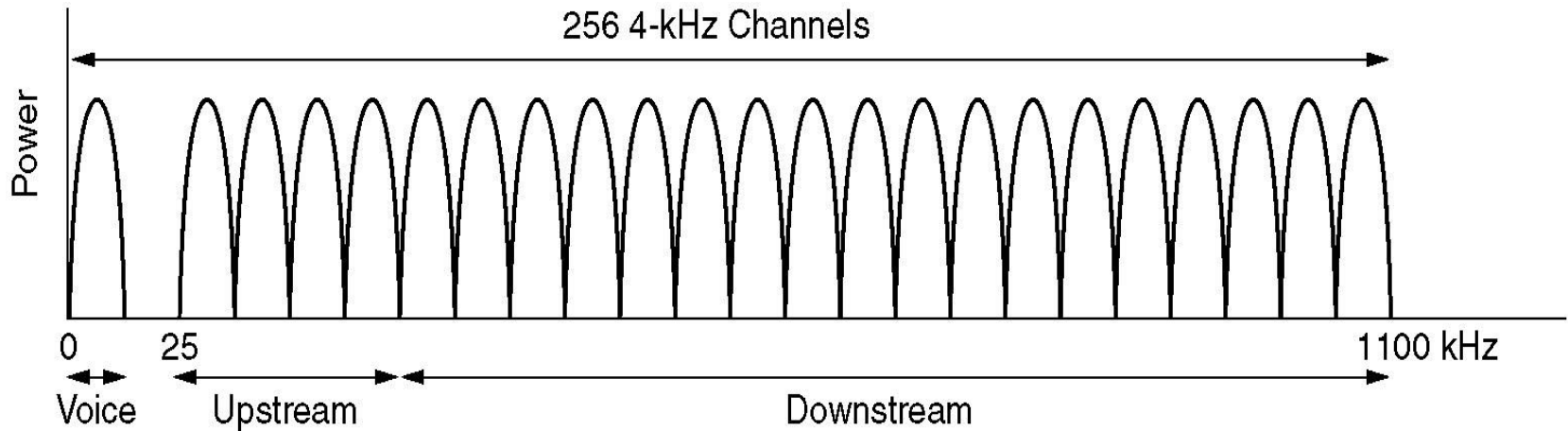
xDSL Services

- Goal of xDSL services
 - ◆ The services must work over the existing **UTP3** local loops
 - ◆ They must **not affect** customers' **existing telephones** and fax machine
 - ◆ They must be much **faster than 56kbps**
- **ADSL: Asymmetric DSL**(非对称数字用户线)

ADSL接入



ADSL: Discrete Multitone Modulation



- Divide 1.1 MHz into 256 channels of 4312.5 Hz
- Channel 0 for POTS(电话)
- 1-5 not used, to keep voice signal and data signals from interfering with each other
- 250 channels
 - ◆ Upstream control (1)
 - ◆ Downstream control (1)
 - ◆ User data (248)

FDM(频分复用)

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ADSL Modulation Scheme

- Within each channel, a modulation scheme similar to V.34 is used
 - ◆ Sampling rate is 4000 baud
 - ◆ The line quality in each channel is monitored and data rate adjusted continuously
 - ◆ The actual data are sent with QAM(正交幅度调制), with up to 15 bits/baud
- Typical data rate
 - ◆ Standard service: 1M bps(downstream), 256 kbps(upstream)
 - ◆ Improved service: 4Mbps, 1Mbps
 - ◆ Premium service: 8 Mbps, 2M kbps

Review

■ PSTN



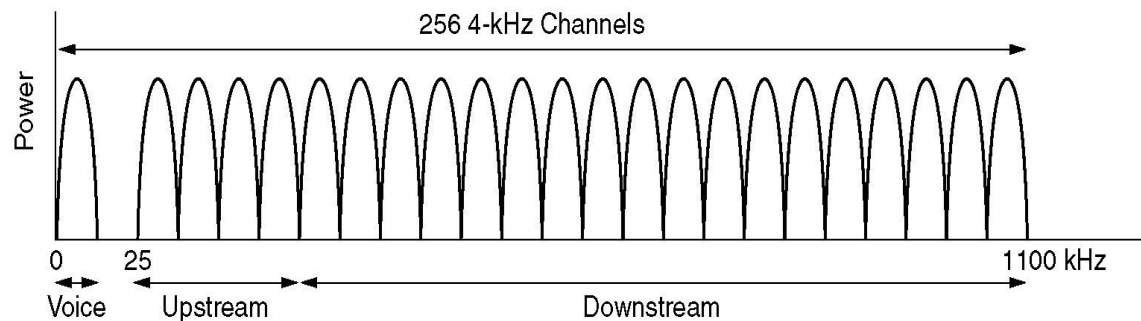
- ◆ Local loops: Analog twisted pairs (UTP3)
- ◆ Trunks: Digital fiber optics
- ◆ Devices: Modem, CodeC

■ xDSL

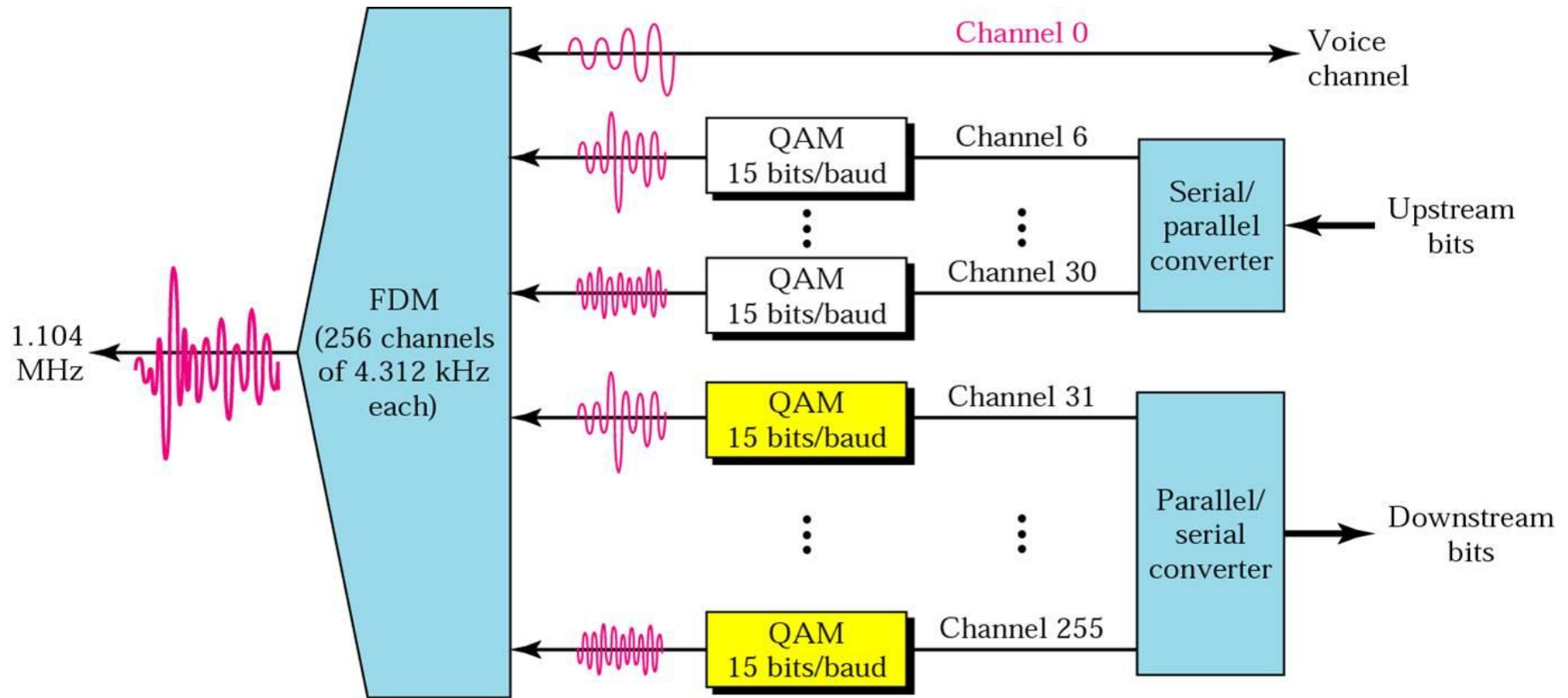
- ◆ UTP3 local loops, not affect telephones, faster than 56kbps

■ ADSL

- ◆ FDM

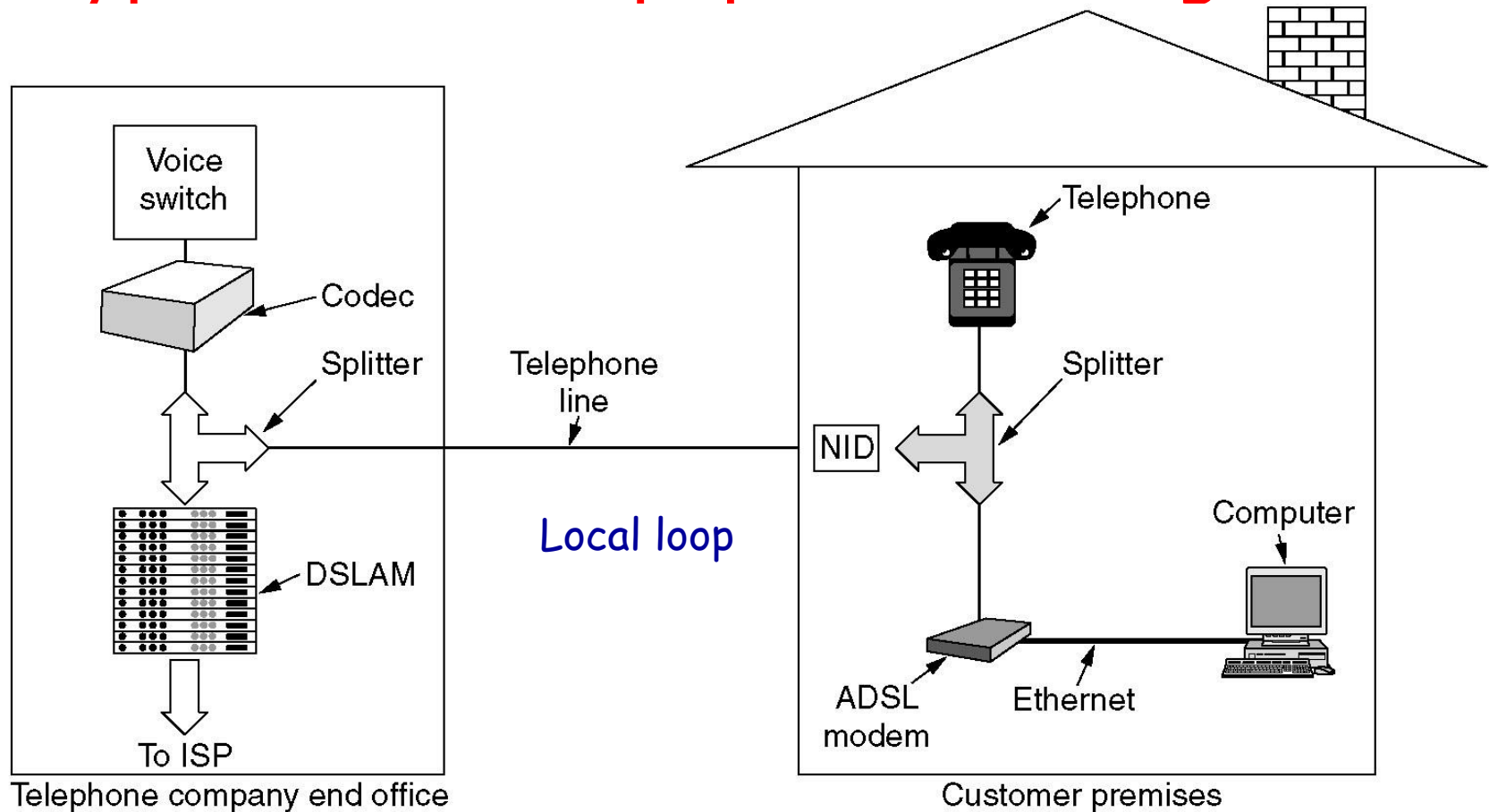


DMT: Discrete Multitone Modulation (离散多音调制)



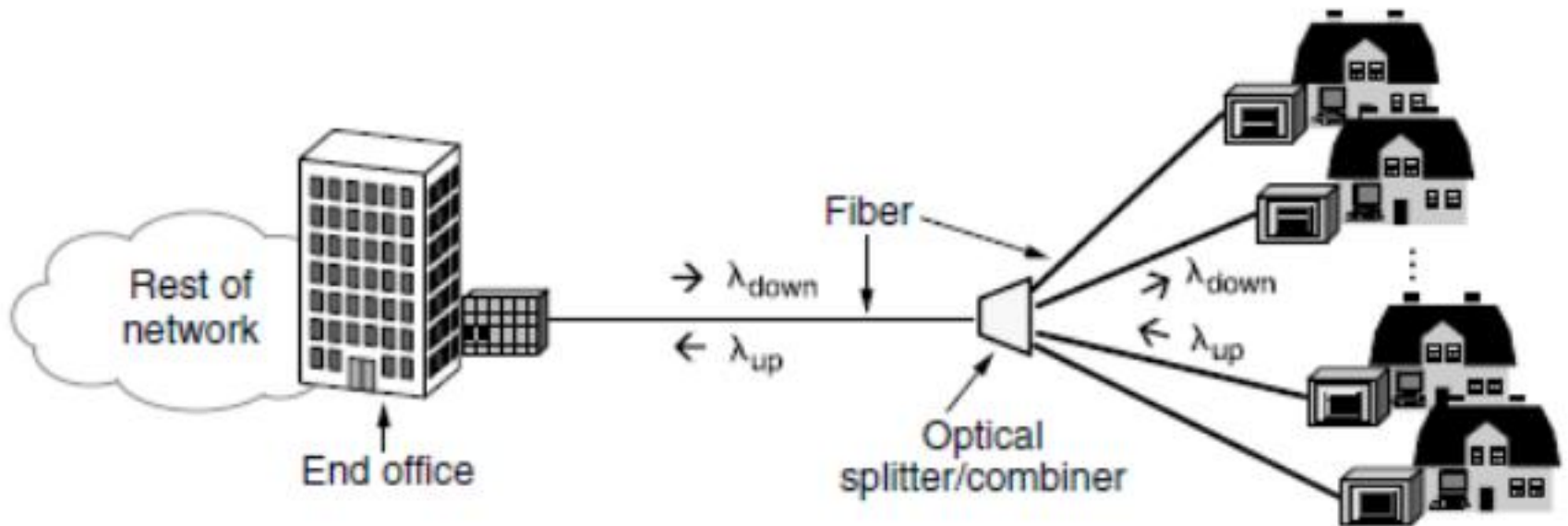
QAM : Quadrature Amplitude Modulation-正交幅度调制

A typical ADSL equipment configuration



- Splitter
- DSLAM: Digital Subscriber Line Access **Multiplexer**
- NID: Network Interface Device

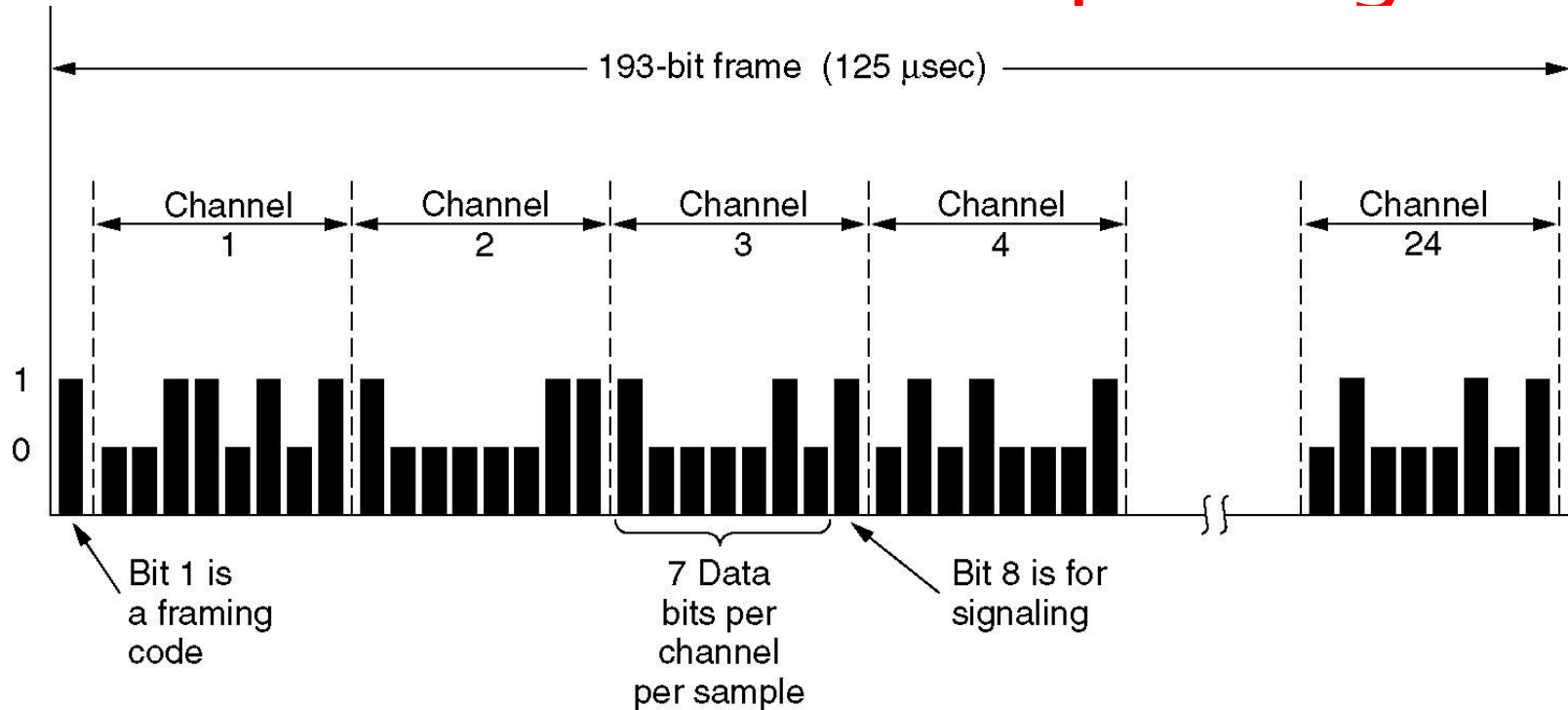
FTTH (光纤入户)



PON(Passive Optical Network) 无源光网络

2.6.4 Trunks and Multiplexing

Time Division Multiplexing



■ **Codec:** digitizing voice signals

◆ **PCM:** 8000/s, $T=1/8000=125\mu\text{sec}$, $C=64\text{kbps}$

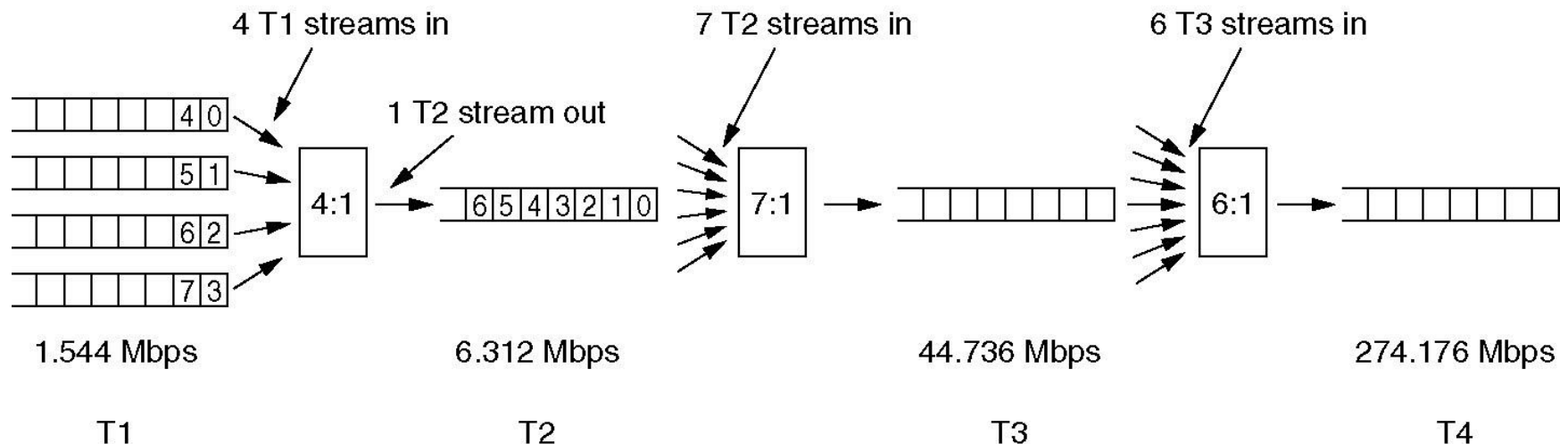
■ **Multiplexing**

◆ T1 --- 1.544 Mbps 24路PCM信号, 8kbps for signaling

◆ E1 --- 2.048Mbps 30/32路PCM信号

Time Division Multiplexing (2)

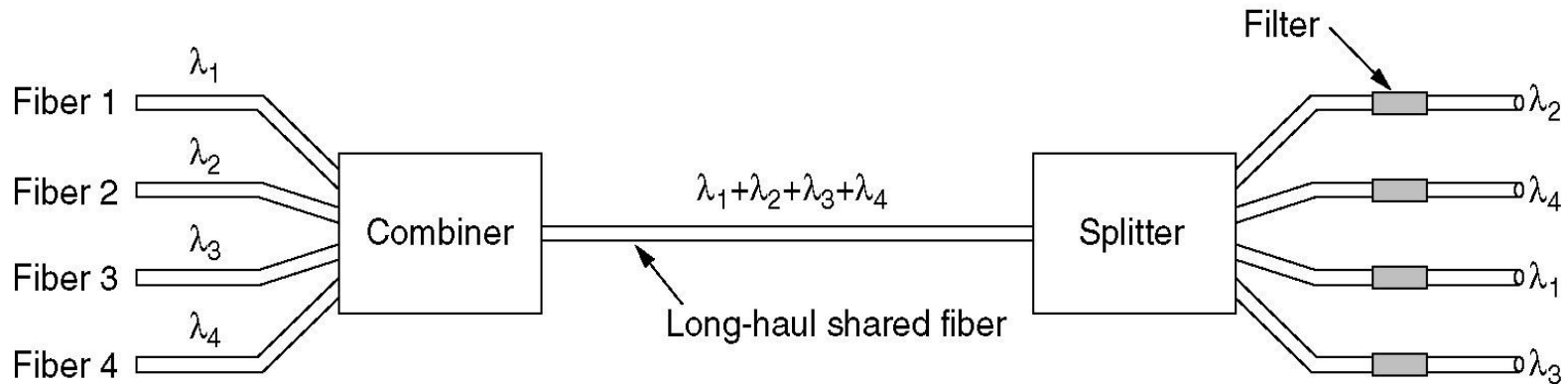
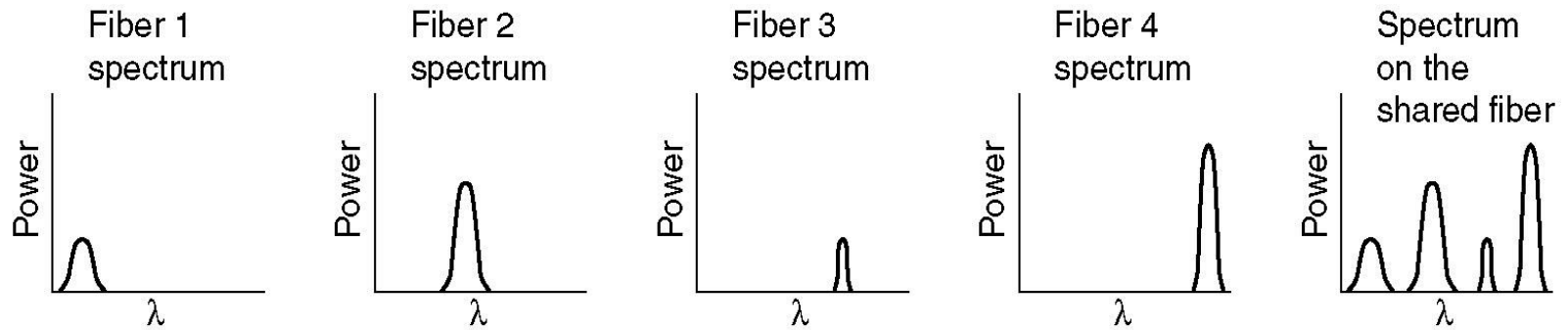
Multiplexing T1 streams into higher carriers.



Optical TDM system

- ◆ SONET Synchronous Optical NETWORK, 1985
- ◆ SDH Synchronous Digital Hierarchy, ITU recommendations, 1989

Wavelength Division Multiplexing



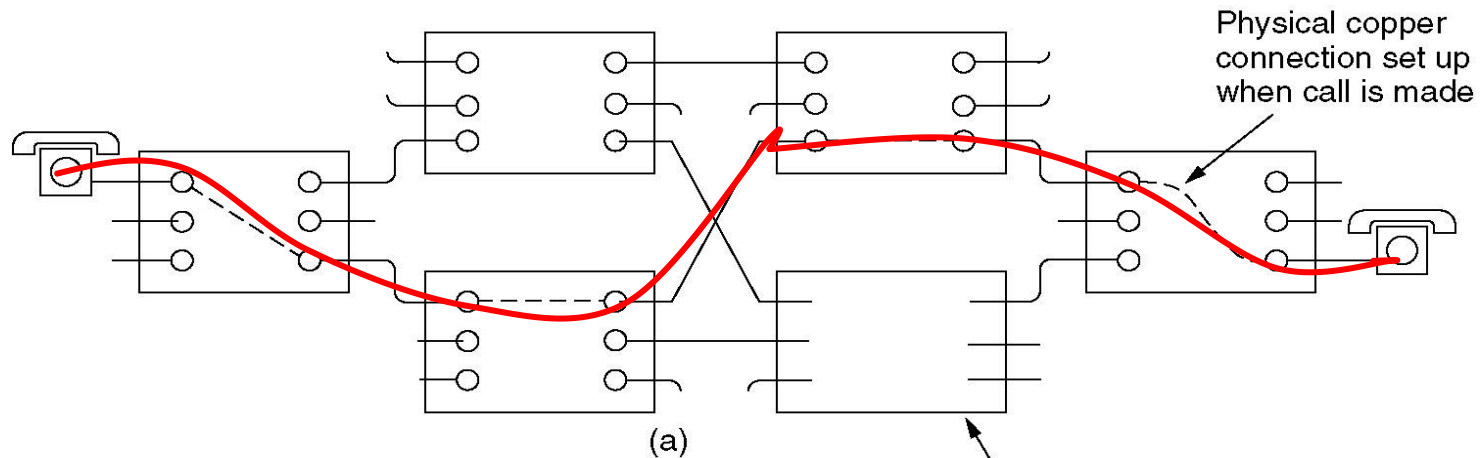
WDM (Wavelength Division Multiplexing).

DWDM (Dense WDM).

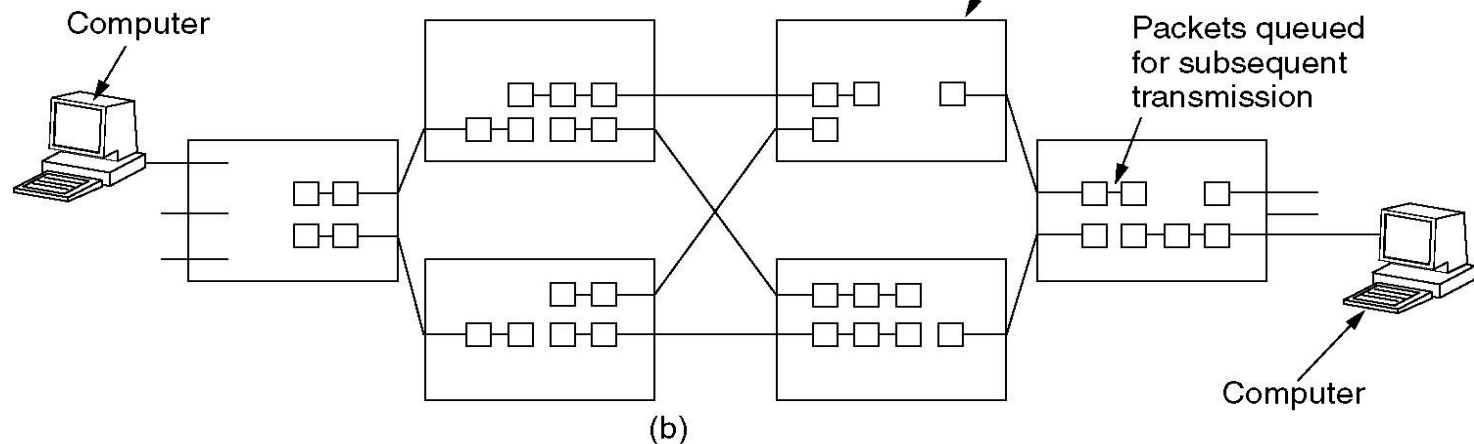
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2.6.5 Switching: Circuit Switching vs. Packet Switching

Circuit Switching vs. Packet Switching



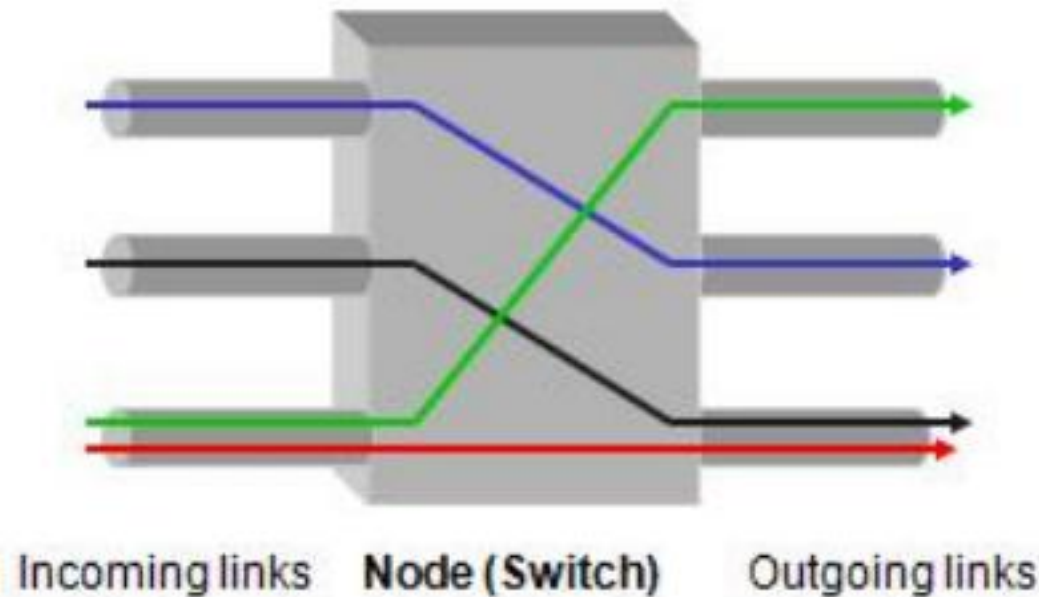
Circuit switching



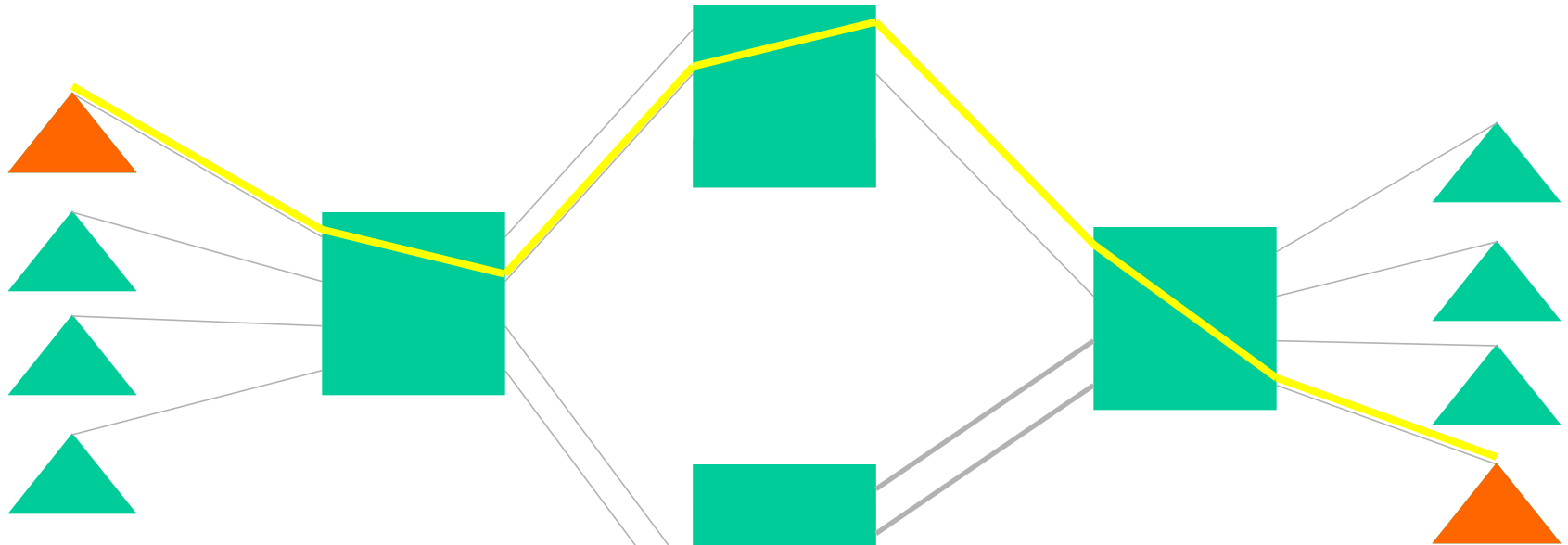
Packet switching

Circuit switching (电路交换)

- Example: Telephone network
- Dedicated communication path between 2 stations
- Resources allocated in advance
- 3 phases: circuit establishment, data transfer and circuit termination



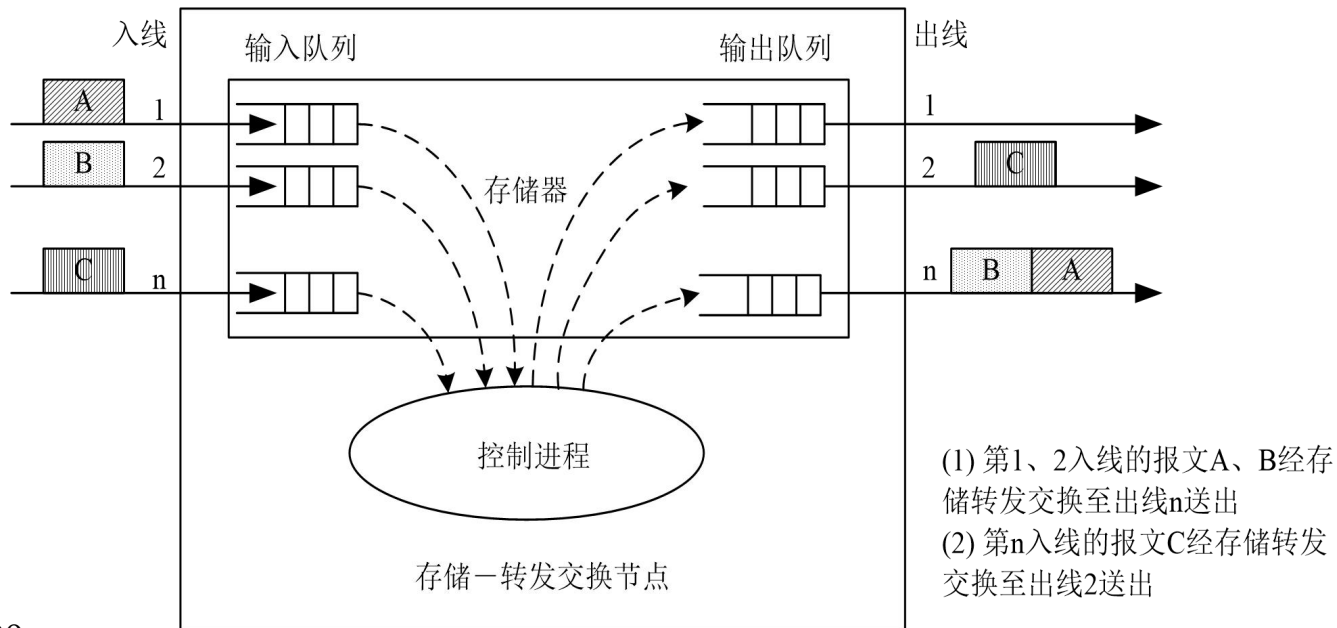
Demo of Circuit Switching



- Strength
 - Small delay, good quality
 - Simple control in switches
- Weakness
 - Time consumed in call set up
 - Fixed bandwidth, not suitable for various rate services
 - Channel waste when both parties keep silent

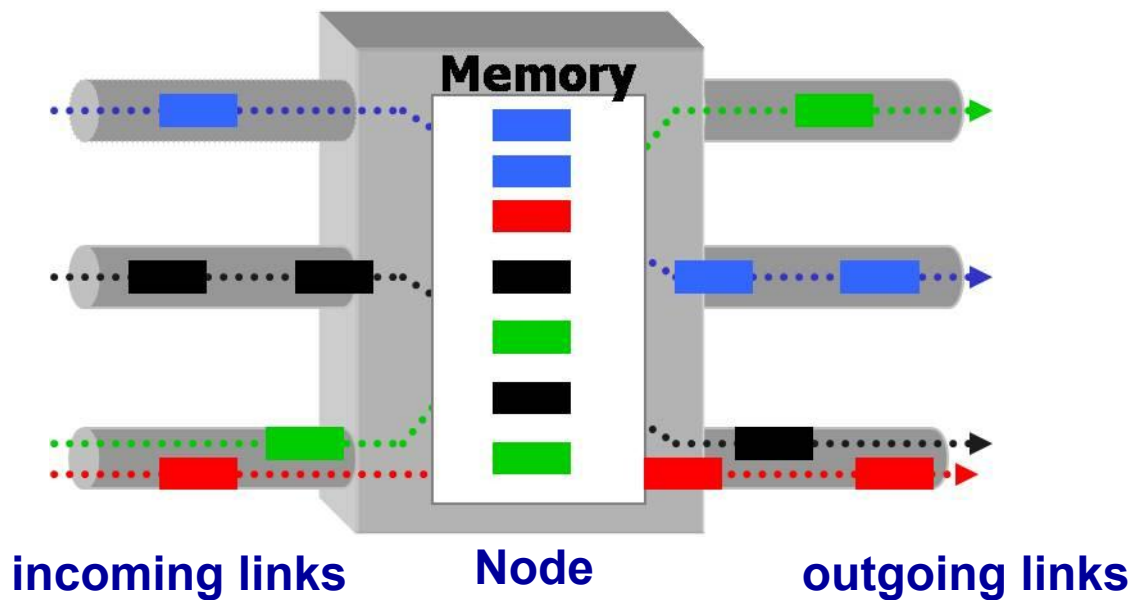
Message Switching (报文交换)

- Example: Post office
- Message transmission and switching
- At each node the **entire message** is received, stored, and then forwarded to the next node (Store-and-Forward, 存储-转发)

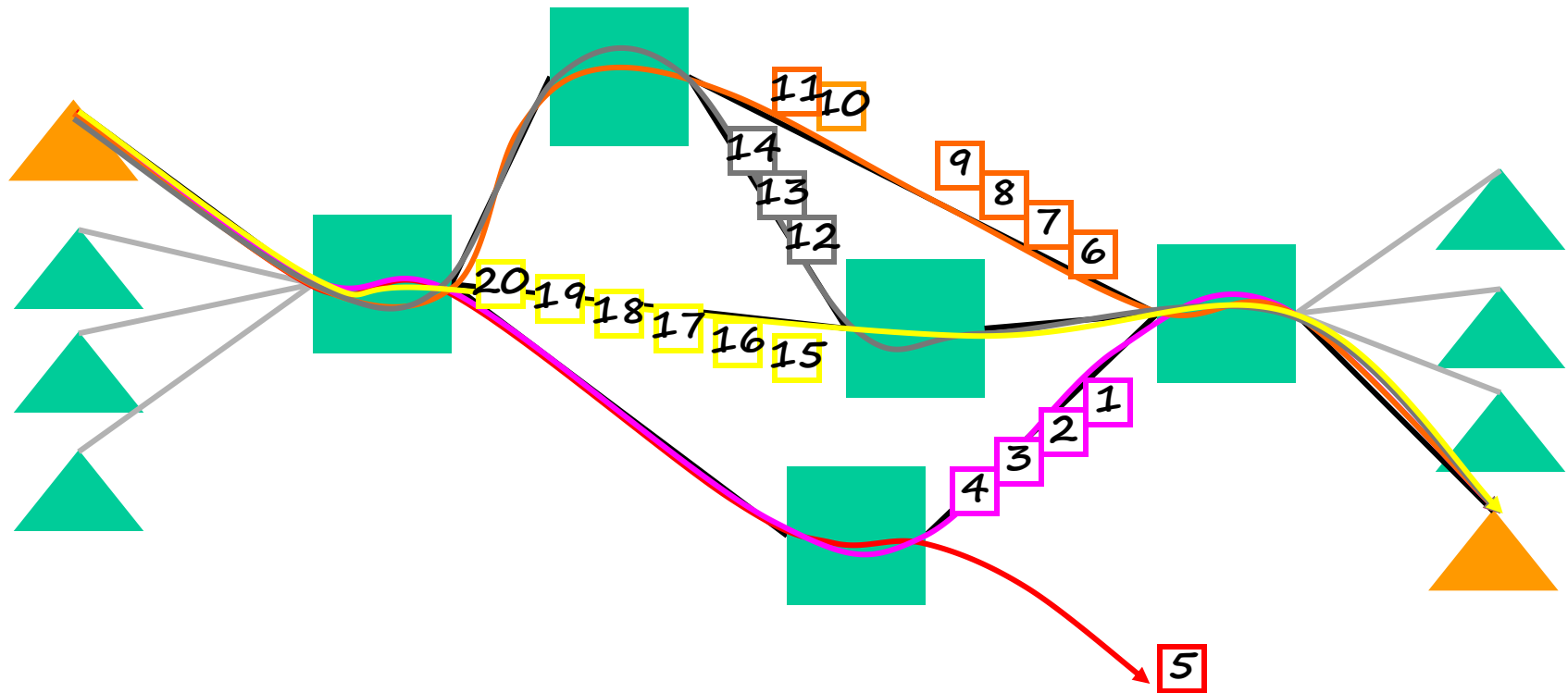


Packet Switching (分组交换)

- Data sent as formatted bit-sequences (**Packets**)
- At each node the entire packet is received, stored briefly, and then forwarded to the next node (**Store-and-Forward**, 存储-转发)
- **Dynamic** resource allocation
- Each **packet** is **independently** routed



Demo of Packet Switching



Basic Operation of Packet Switching

- Data transmitted in small packets
 - ◆ Longer messages split into series of packets(reduce delay)
 - ◆ Each packet contains a portion of user data plus some control information
- Control information
 - ◆ Routing (addressing) information
- Packets are received, stored briefly (buffered) and past on to the next node
 - ◆ Store and forward

Packet Switching: Good & Bad

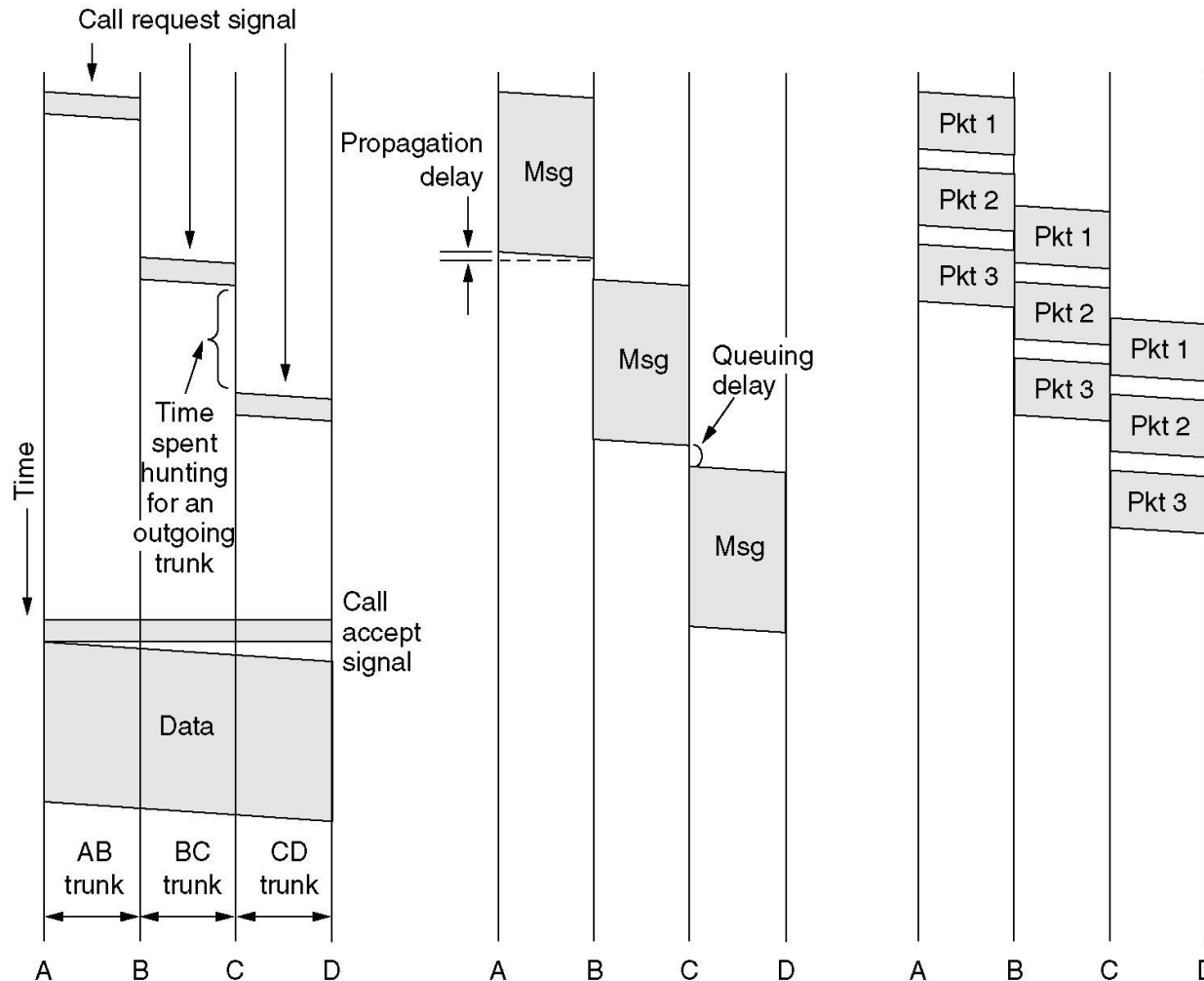
■ Strength

- ◆ Line efficiency
 - Single node to node link can be **shared** by many packets over time
 - Packets queued and transmitted as fast as possible
- ◆ Data rate conversion(速率适配)
 - Each station connects to the local node at its own speed
 - Nodes **buffer data** if required to equalize rates
- ◆ **Packets are accepted even when network is busy**(排队制)
 - Delivery may slow down
 - **Priorities** can be used

■ Weakness

- ◆ **Complex** network
- ◆ **Larger delay**
- ◆ **No QoS guarantee**

Timing of CS, MS and PS



(a) Circuit switching (b) Message switching (c) Packet switching

报文交换

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Circuit switched and Packet-switched: Comparison

Item	Circuit-switched	Packet-switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
When can congestion occur	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes

Charging	Per minute	Per packet
----------	------------	------------

Outline

- 2.0 Basic Concepts on Data Communications
- 2.1 Theoretical Basis for Data Communication
- 2.2 Guided Transmission Media(Self-study)
- 2.3 Wireless Transmission(Self-study)
- 2.4 Communication Satellites(Self-study)
- 2.5 Digital Modulation and Multiplexing
- 2.6 The Public Switched Telephone Network
- (补充) Protocol Example: RS-232

Physical Interface Features

- **Mechanical features**(机械特性):指明接口所用接线器的形状和尺寸、引线数目和排列、固定和锁定装置等
- **Electrical features**(电气特性):指明在接口电缆的各条线上出现的电压的范围
 - Encoding, Data rate, cable length
- **Functional features**(功能特性):指明某条线上出现的某一电平的电压表示何种意义
- **Procedure features**(过程特性): 指明对于不同功能的各种可能事件的出现顺序

Physical Interface: RS-232



- EIA RS-232 (ITU-T V.24)
 - ◆ First issued by EIA in 1969
 - ◆ EIA-232-C EIA-232-D EIA-232-E
- Interface standard between DTE and DCE
 - ◆ DTE: Data Terminal Equipment, eg. computers
 - ◆ DCE: Data Communications Equipment, eg. modems
- **Serial Transmission** (串行传输)
- **Asynchronous Transmission** (异步传输)

RS-232 Specifications

- Mechanical features

- 25-pin

- Electrical features

- NRZL encoding

- $< -3V$ = binary 1, $> 3V$ = binary 0

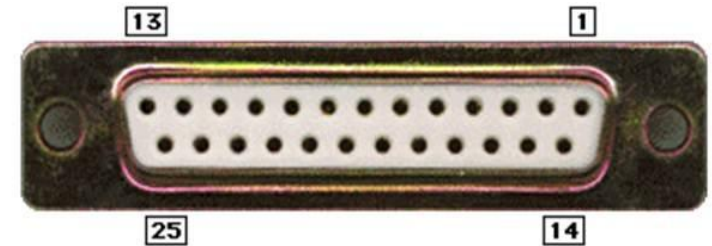
- Data rate: 20Kbps

- cable length < 15 m

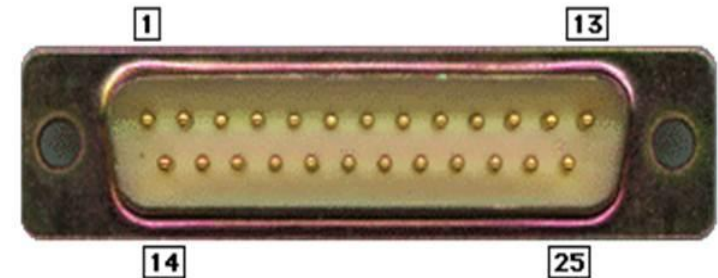
- Functional features: pin definition

- Procedure features: protocol,

eg full-duplex



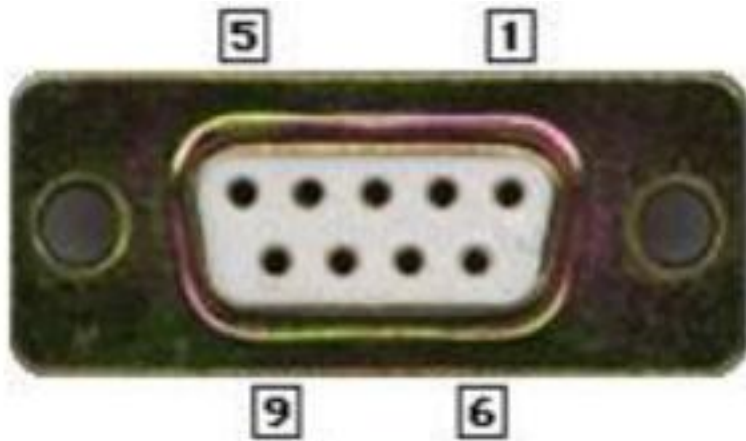
DB-25 Female (DTE side)



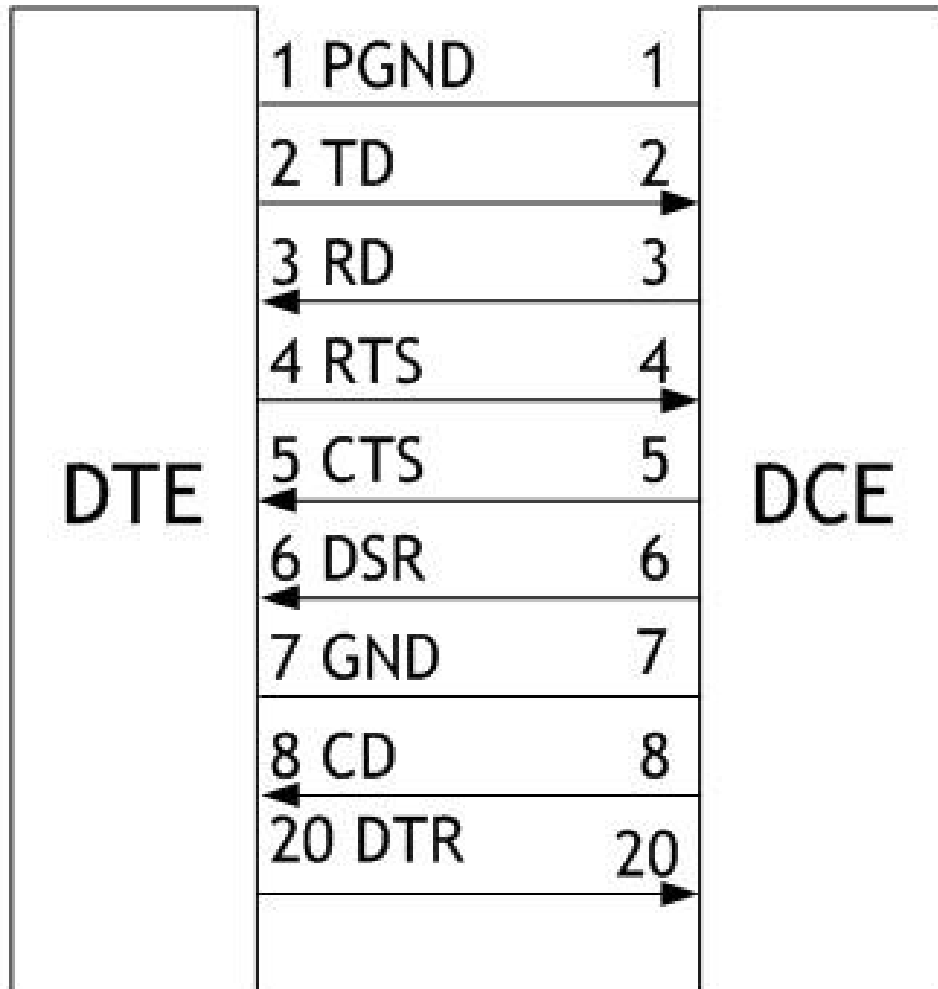
DB-25 Male (DCE side)

RS-232 DB-9 Connectors

- Limited RS 232
- Used in practice

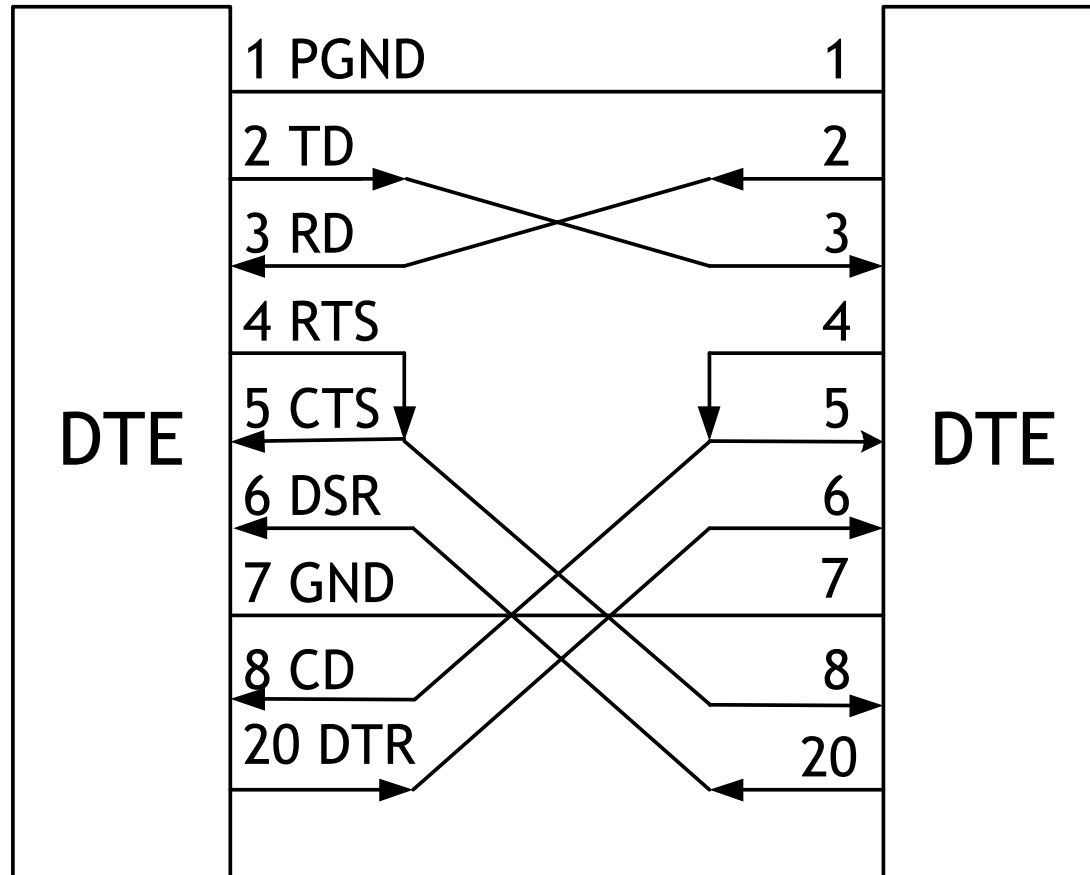


RS-232 : Functional Specifications



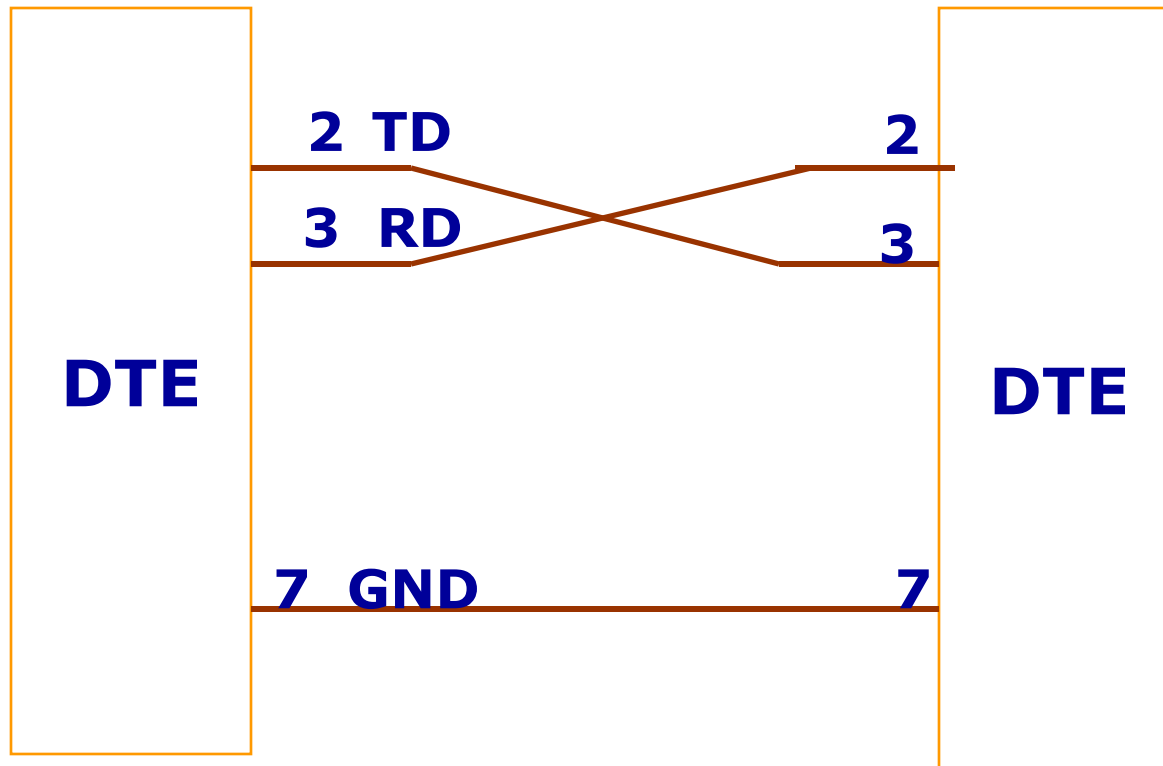
- DTE: data terminal equipment
- DCE: data communications equipment
- TD: Transmitted data
- RD: Received data
- RTS: Request to send
- CTS: Clear to send
- DSR: data set ready
- CD: Data Carrier detection
- DTR: Data Terminal Ready

Virtual MODEM



Allows DTE to DTE direct communication

Virtual MOEM(2)



Physical Interface: Ethernet

IEEE802.3, 10BaseT

- ◆ Data rate: 10Mbps, Twist pair as transmission media

- ◆ Mechanical features

 - RJ-45 Interface

- ◆ Electrical features

 - Manchester Encoding

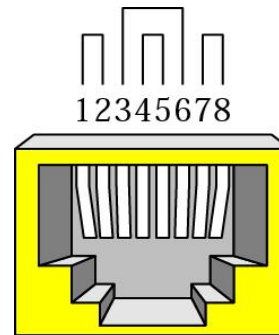
 - 2.5v, -2.5v

- ◆ Functional features

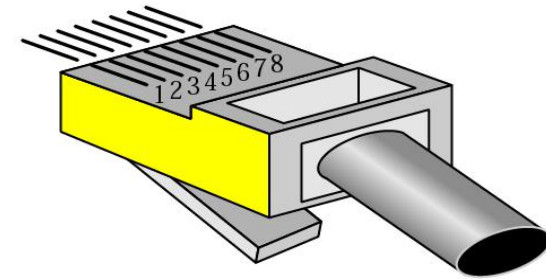
 - 1 pair for sending(pin 1&2), 1 pair for receiving(pin 3&6)

- ◆ Procedure features

 - Full duplex communication



RJ-45 Female



RJ-45 Male

Key Points of Physical Layer

- Concepts
 - ◆ Data vs. Signal, Analog vs. Digital
 - ◆ Bandwidth, Channel capacity
 - ◆ Baud rate vs. Bit rate
 - ◆ Transmission delay vs. Propagation delay
- Nyquist theorem and Shannon theorem
- Transmission media
- Baseband transmission vs Passband transmission
- Modulation: ASK, PSK, FSK, QPSK & QAM
 - ◆ ADSL Modem
- Line coding: NRZ, Manchester
 - ◆ CodeC: A/D and D/A
- Digitized Analog Signals: Sampling, Quantizing, Encoding
 - ◆ eg. PCM
- Multiplexing: FDM, WDM, TDM and CDMA
- Circuit Switching vs. Packet Switching

Terms

缩写	英文全称	中文
	analog signal	模拟信号
	Asynchronous Transmission	异步传输
ADSL	Asymmetric Digital Subscriber Line	非对称数字用户线
	bandwidth	带宽
	base station	基站
	baud	波特
BER	Bit Error Rate	比特误码率
	bit rate	数据率(比特率)
	bluetooth	蓝牙
	broadband	宽带
	cable headend	线缆头端
CATV	Community Antenna Television	社区天线电视(有线电视)
	cellular networks	蜂窝网络
	channel	信道
2025-4-29	channel capacity	信道容量

缩写	英文全称	中文
	circuit	电路
CDMA	Code Division Multiple Access	码分多址
CODEC	Coder-Decoder	编码解码器
	chip sequence	码片序列
CS	Circuit Switching	电路交换
	Coaxial Cable	同轴电缆
	cut-through switching	直通交换
	data link layer	数据链路层
	dial-up	拨号
	digital signal	数字信号
DCE	Data Communications Equipment	数据通信设备
	Differential Manchester	差分曼彻斯特
DMT	Discrete MultiTone	离散多音(调制)
DSL	Digital Subscriber Line	数字用户环线
DSLAM	Digital Subscriber Line Access Multiplexer	数字用户环线接入复用器

缩写	英文全称	中文
DTE	Data Terminal Equipment	数据终端设备
	error detection	差错检测
	Ethernet	以太网
FDM	Frequency Division Multiplexing	频分复用
	fiber	光纤
	frame	帧
FTTH	Fiber to the Home	光纤到户
	full duplex	全双工
GEO	Geosynchronous orbit	地球同步轨道
GPS	Global Positioning System	全球定位系统
GSM	Global System for Mobile communications	全球移动通信系统
	half duplex	半双工
	harmonics	谐波
	HUB	集线器

缩写	英文全称	中文
ISM	Industrial, Scientific, Medical	工业、科学、医药（频段）
ISP	Internet Service Provider	互联网服务提供商
LAN	Local Area Network	局域网
	Line coding	线路编码
	Magnetic Media	磁介质
MAN	Metropolitan Area Network	城域网
	Manchester Encoding	曼彻斯特编码
	Message Switching	报文交换
MODEM	Modulator-Demodulator	调制解调器
	microwave	微波
NFC	Near Field Communication	近场通信
NRZ	Nonreturn-to-Zero	不归零
	Nyquist theorem	奈奎斯特定理

缩写	英文全称	中文
	Packet Switching	分组交换
PAD	Personal Digital Assistant	个人数字助手(平板电脑)
PAN	Personal Area Network	个域网
PCM	Pulse Code Modulation	脉冲编码调制
	periodic signal	周期性信号
	physical layer	物理层
	physical medium	物理媒体/介质
PLC	Power Line Communication	电力线通信
	Port	端口
	parallel transmission	并行传输
PSK	Phase Shift Keying	相移键控
	power-line networks	电力线网络
	propagation delay	传播时延
PSTN	Public Switched Telephone Network	公众交换电话网络

缩写	英文全称	中文
	Radio	无线电
QAM	Quadrature Amplitude Modulation	正交幅度调制
QPSK	Quadrature Phase Shift Keying	正交相移键控
RFID	Radio Frequency Identification	射频识别
	satellite	卫星
	serial transmission	串行传输
	Shannon theorem	香农定理
SMS	Short Message Service	短消息业务
	sensor networks	传感网
	signal	信号
	simplex	单工
	statistical multiplexing	统计复用
STP	Shielded Twisted Pair	屏蔽双绞线
	store-and-forward switching	存储转发交换
	subnet	子网

缩写	英文全称	中文
	switch	交换机
	switched Ethernet	交换式以太网
	Synchronous Transmission	同步传输
TDM	Time Division Multiplexing	时分复用
	Transmission Media	传输媒体(介质)
	transmission delay	发送时延
	transport layer	传输层
UTP	Unshielded Twisted Pair	非屏蔽双绞线
VoIP	Voice over IP	IP话音
VPN	Virtual Private Networks	虚拟专用网
WAN	Wide Area Network	广域网
WCDMA	Wideband Code Division Multiple Access	宽带码分多址接入
WDM	Wave Division Multiplexing	波分复用
WLAN	Wireless Local Area Network	无线局域网

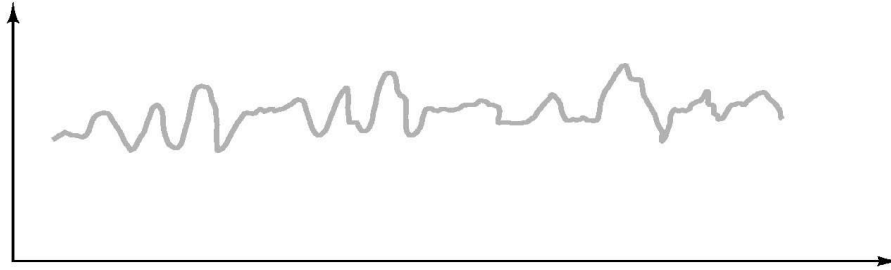
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 - ◆ “Data and Computer Communications”, William Stallings, which are marked with *[Stallings]*
 - ◆ “计算机网络”, 谢希仁, 第五版, 电子工业出版社, 2008年, which are marked with *[谢]*

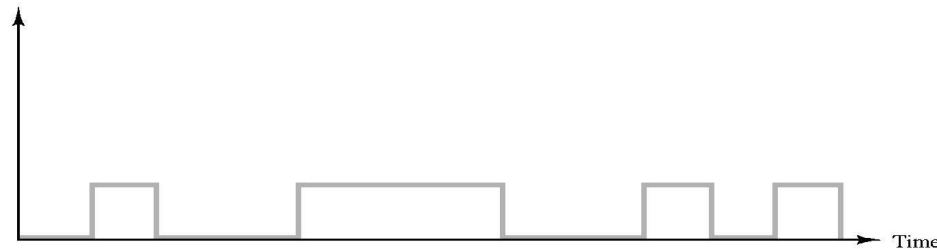
Self-study

Signals: Analog vs. Digital

- **Signal(信号):** 数据的电编码或电磁编码，数据以信号的形式传播。
 - ◆ a function $s(t)$ that varies with time (t stands for time)
- **Analog: varies continuously** (模拟信号)
 - ◆ Example: voltage representing audio (analog phone call)

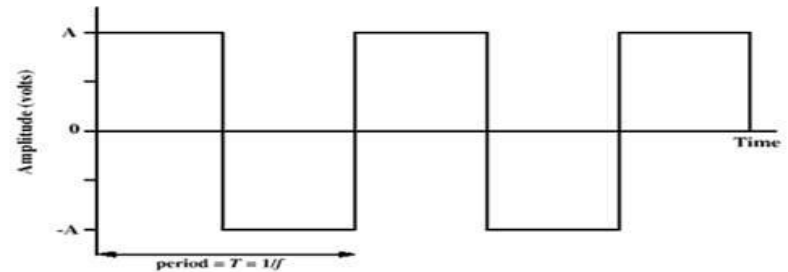
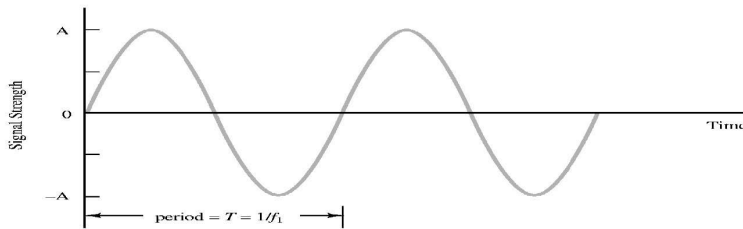


- **Digital: discrete values; varies abruptly** (数字信号)
 - ◆ Example: voltage representing 0s and 1s

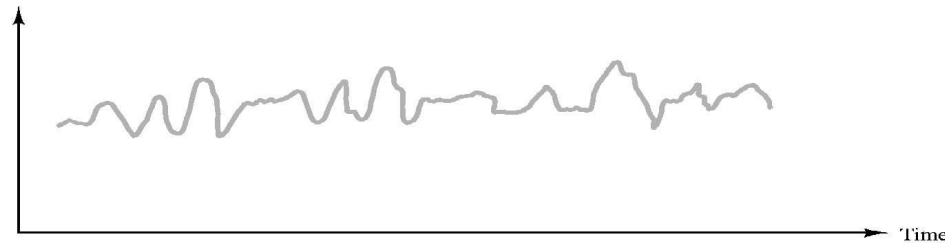


Signals: Periodic vs. Aperiodic

- Periodic (周期性) : repeat over and over again, once per period
 - ◆ 周期 Period (T) is the time it takes to make one complete cycle
 - ◆ 频率 Frequency (f) is the inverse of period, $f = 1/T$; measured in hertz(hz)



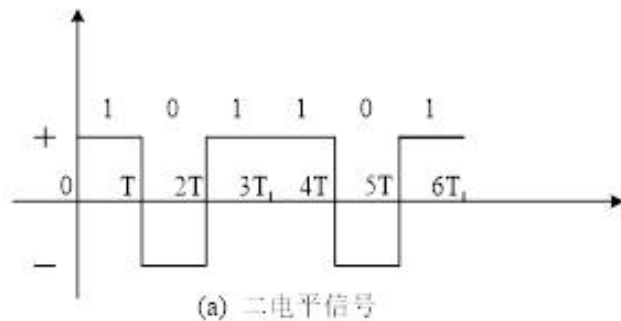
- Aperiodic (非周期性) : don't repeat according to any particular pattern



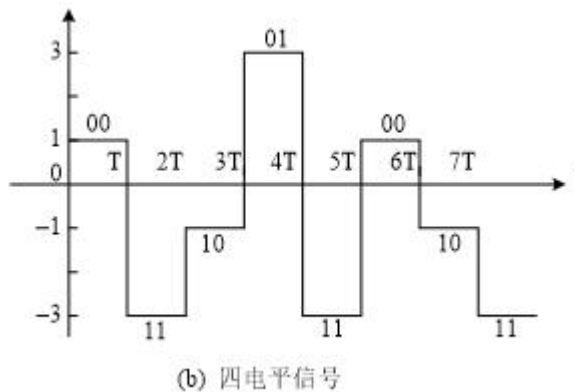
Bits vs. Signal units

- **Signal units(码元)**: 承载信息的基本信号单元, 一个码元能承载的信息量 D 多少, 是由脉冲信号所能表示的有效状态个数 V 决定的。
- **Bits(比特)**: 二进制表示的信息量的基本单位
- 一码元携带的信息量的计算公式:

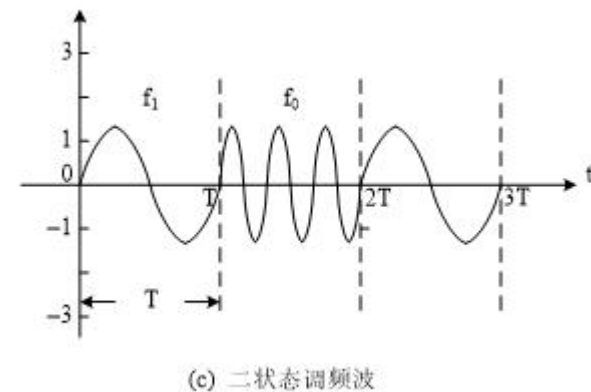
$D = \log_2 V$, V 表示一个脉冲信号所能表示的有效状态数



1码元携带1比特信息



1码元携带2比特信息



1码元携带1比特信息

推导带宽和最大数据率的关系： Fourier Transform（傅里叶变换）

- Any **periodic signal** $g(t)$ with period $T (=1/f)$ can be constructed by summing a (possibly infinite) number of sines and cosines

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- To construct signal $g(t)$ we need to compute the values 振幅 $a_0, a_1, \dots, b_0, b_1, \dots$, and 常数 c
- Energy: proportional to root-mean-square amplitude $\sqrt{a_n^2 + b_n^2}$
- Often the magnitude of the a_n 's and b_n 's get smaller as the frequency $(2\pi n f)$ gets higher (频率越高，能量越小)

Bandwidth-Limited Signals

- All transmission facilities diminish different Fourier components by different amounts
 - ◆ Physical media attenuate (reduce) different harmonics at different amounts
 - ◆ Any transmission system has a limited band of frequencies
 - The width of the frequency range transmitted without being strongly attenuated is called the bandwidth
 - The bandwidth is a physical property of the transmission medium.

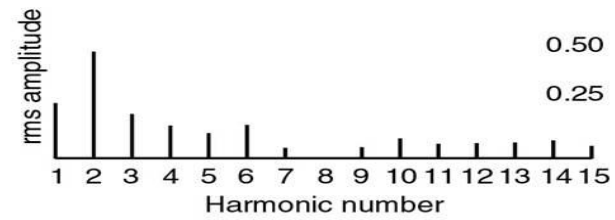
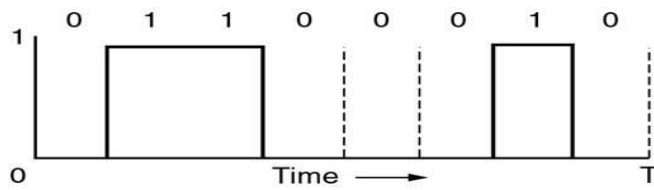
Bandwidth-Limited Signals

- **Key point:** a “reasonable reconstruction” can be often be made from just the first few terms (harmonics谐波) (用几个低次谐波就可以重构信号),
- Tough the more harmonics the better the reconstruction

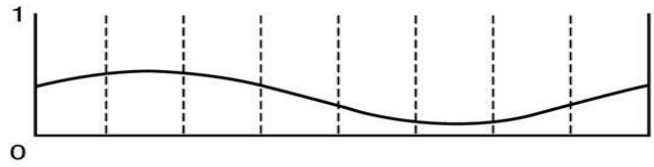
- Example: Transmission of ASCII character “b” (0110 0010)

- Fourier analysis:

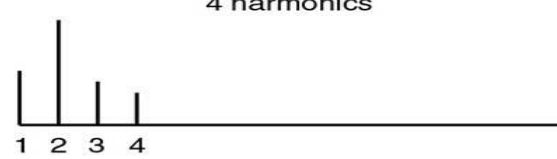
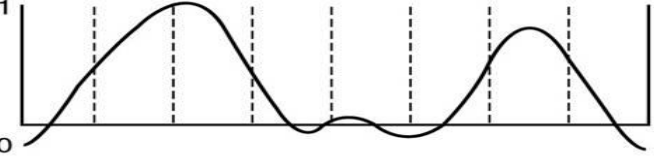
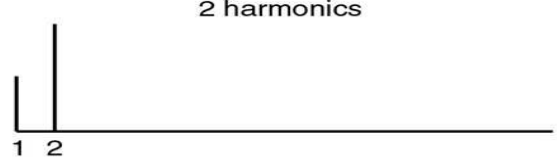
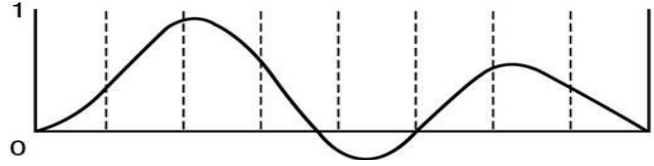
$$a_n = \frac{1}{\pi n} \left[\cos\left(\frac{\pi n}{4}\right) - \cos\left(\frac{3\pi n}{4}\right) + \cos\left(\frac{6\pi n}{4}\right) - \cos\left(\frac{7\pi n}{4}\right) \right]$$
$$b_n = \frac{1}{\pi n} \left[\sin\left(\frac{3\pi n}{4}\right) - \sin\left(\frac{\pi n}{4}\right) + \sin\left(\frac{7\pi n}{4}\right) - \sin\left(\frac{6\pi n}{4}\right) \right]$$



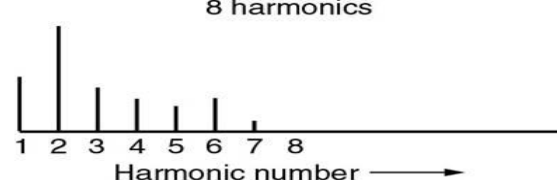
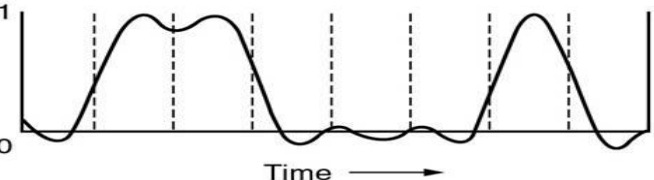
(a)



(b)



(d)



(e)

Relation between data rate and harmonics

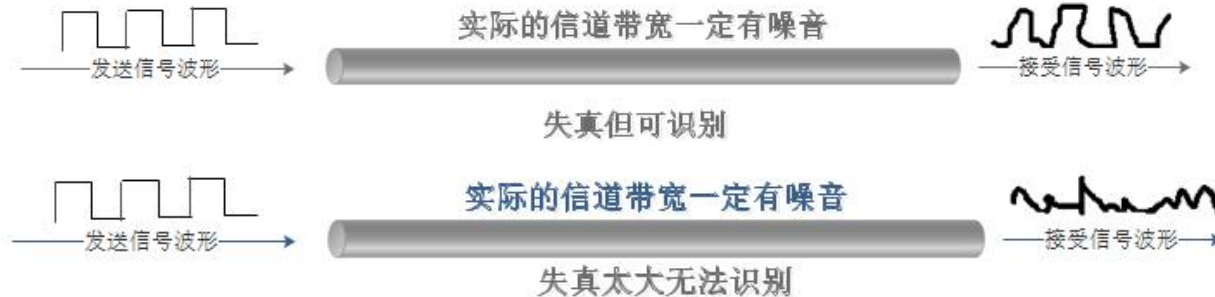
b	$T=8/b$	$f=1/T=b/8$	$3000/f$
Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

- Voice grade channel: 3000Hz
- Conclusion: Higher data rates require more bandwidth. Limiting bandwidth limits data rate.

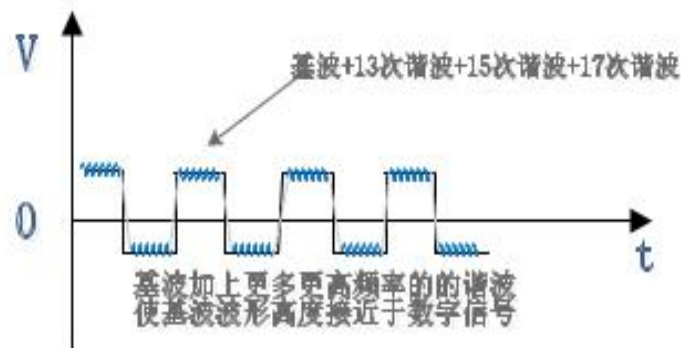
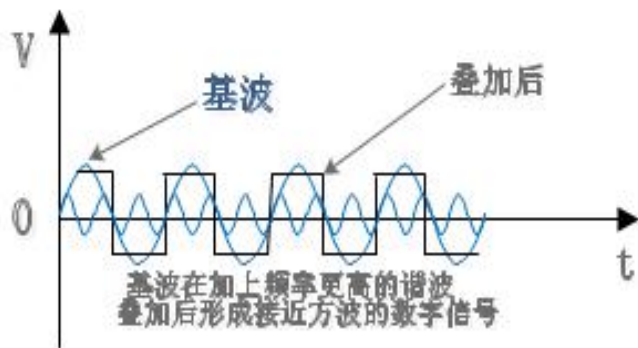
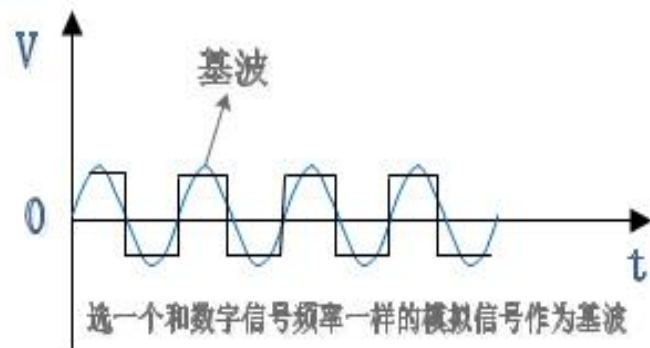
2025-4-29

补: 数字信道极限容量

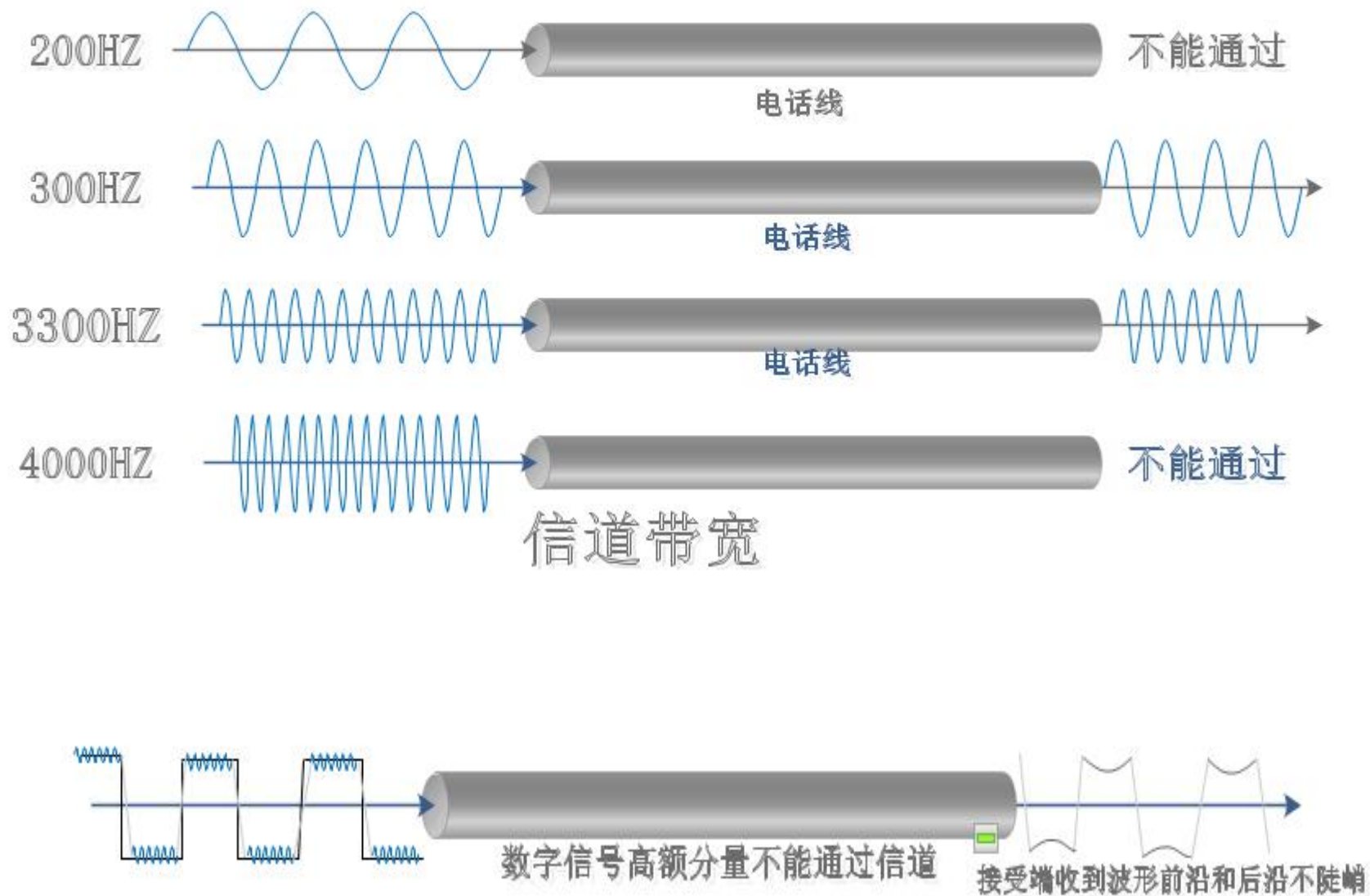
- **Digital communication advantages:** 接收端只要能够从失真的波形中识别出原来的信号，则该失真对通信质量就没有影响。



- **码元传输速度越高、信号传输的距离越远、噪声干扰越大、传输媒体质量越差，在信道的接收端，波形的失真就越严重。**



数字信号由模拟信号谐波而成



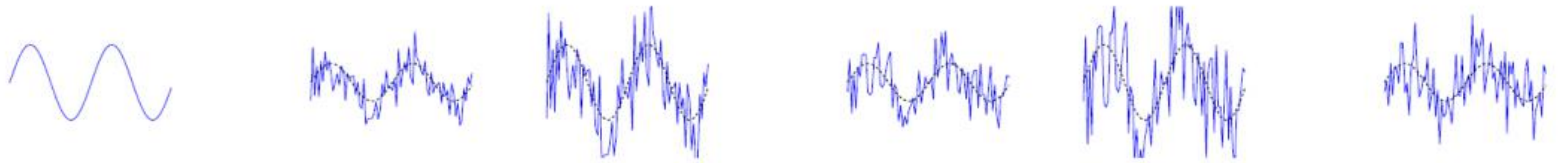
Amplification vs Regeneration

When a signal travels through a channel, it suffers **channel impairment**. Since their negative effects increase with the distance, special equipment called **repeaters** are inserted along the way.

- In analog systems continuously-varying waveforms are transmitted. In order to preserve the transmitted waveforms, repeaters essentially **filter**, **equalise** and **amplify** the signal.
- In digital systems sequences predefined waveforms (symbols) are transmitted. Repeaters **regenerate** such waveforms.

物理层网络互联设备--repeater

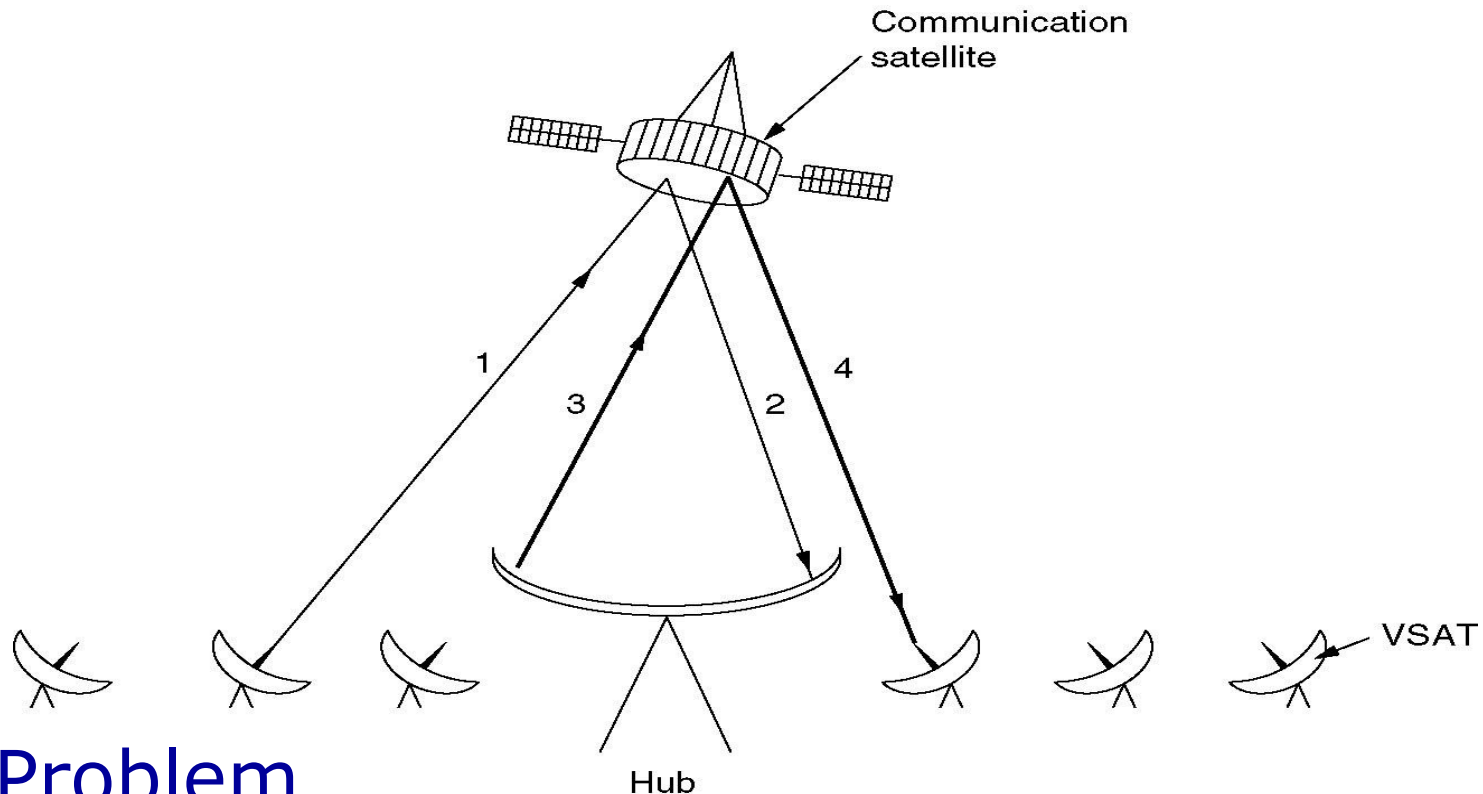
Analog World:
Amplification



Digital World:
Regeneration



GEO: VSATs Using a Hub



■ Problem

- ◆ Micro-station **do not have enough power** to communicate directly with one another via the satellite

LEO: Iridium

■ LEO

- ◆ The ground stations do not need much power
- ◆ Round-trip delay is only a few ms

■ Iridium

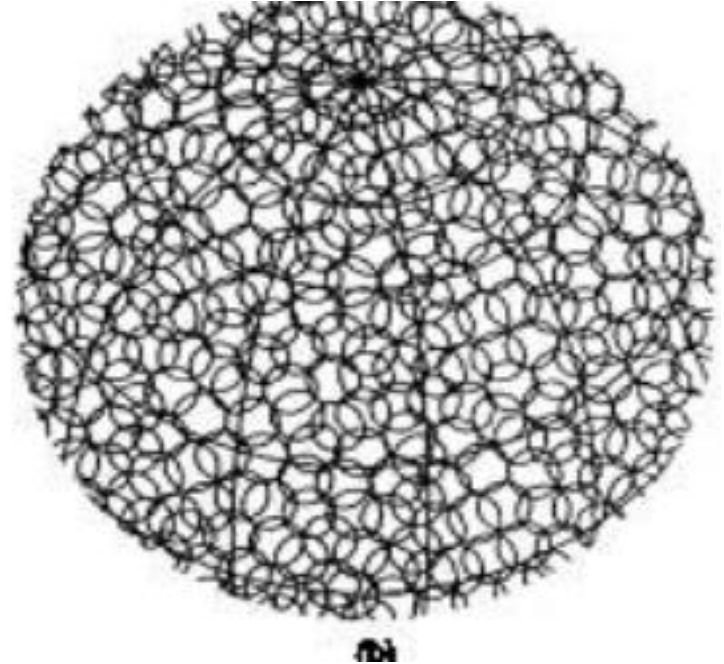
- ◆ Altitude: 750km
- ◆ 6 satellite necklaces, $11 \times 6 = 66$ satellites
- ◆ Each satellite has a **max** of 48 cells (spot beams)
- ◆ **Relaying in space** : Communication between distant customers takes place **in space**.

■ 实施情况

- ◆ 90年策划，97年发射，98年运营，99年破产，2001年重新开始提供服务

2025-4-29

Low-Earth Orbit Satellites: Iridium (铱星)



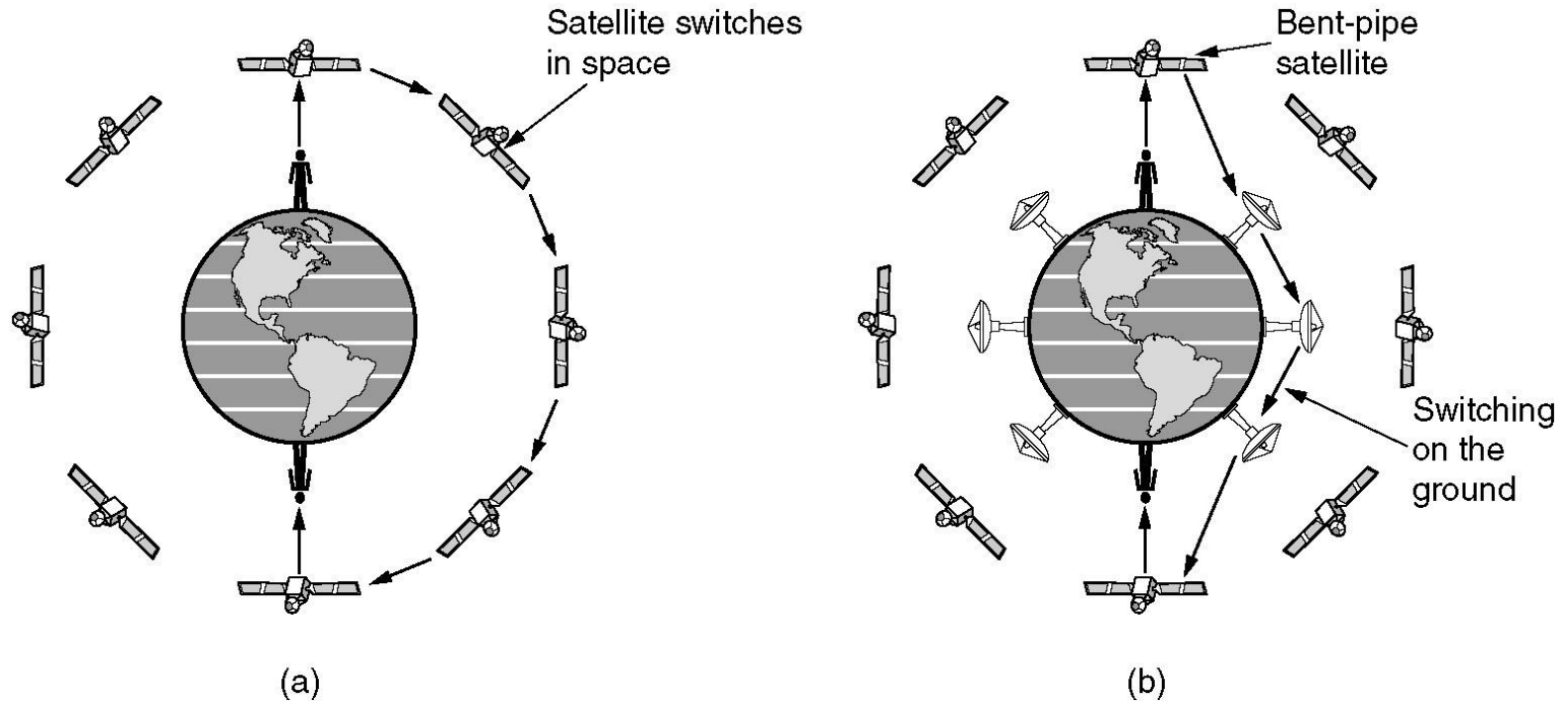
(a) The Iridium satellites form six necklaces around the earth. $6 \times 11 = 66$ 颗卫星

(b) 3840 channels

LEO: Globalstar

- 48 LEOs satellites
- Relaying on the ground
 - ◆ puts much of the complexity on the ground, ease to manage
 - ◆ lower-powered telephones can be used (a few milliwatts)

Low-Earth Orbit Satellites: Globalstar (全球星)



(a) Relaying in space.

(b) Relaying on the ground.

4B/5B

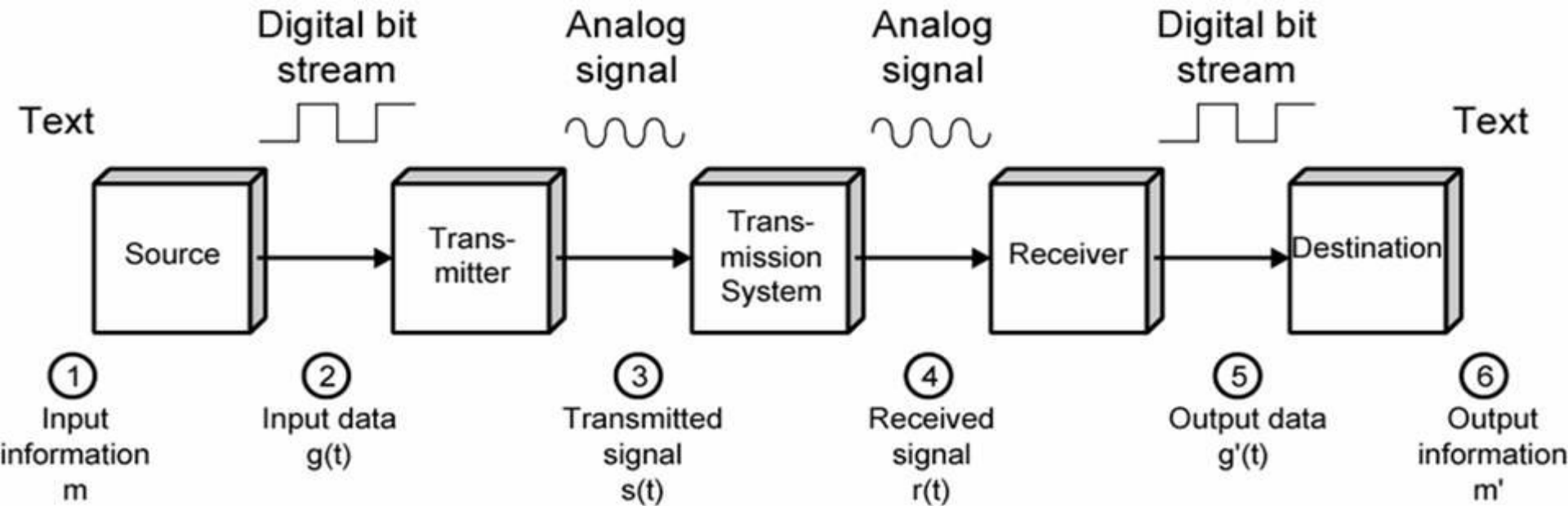
- Every 4bits is mapped into a 5bits pattern with a fixed translation table
- The five bit pattern are chosen so that there will never be a run of more than three consecutive 0s

Data(4B)	Codeword(5B)	Data(4B)	Codeword(5B)
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

8B/10B

- The 8 bits are split into a group of 5 bits, which is mapped to 6 bits, and a group of 3 bits, which is mapped to 4 bits.
- The 6-bit and 4-bit symbols are concatenated as a group.

Simplified Data Communications Model



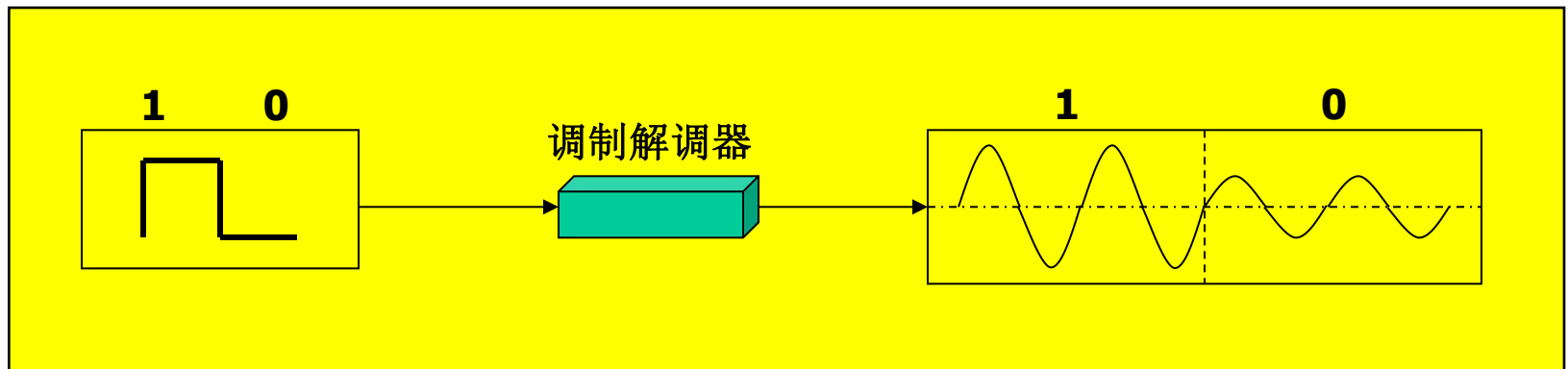
(b) Example

2ASK

■ 2ASK基本原理

- ◆ 正弦载波的幅度随二进制基带信号的变化而变化的数字调制，即源信号为“1”时，发送载波，源信号为“0”时，发送0电平

■ 2ASK信号的时域波形

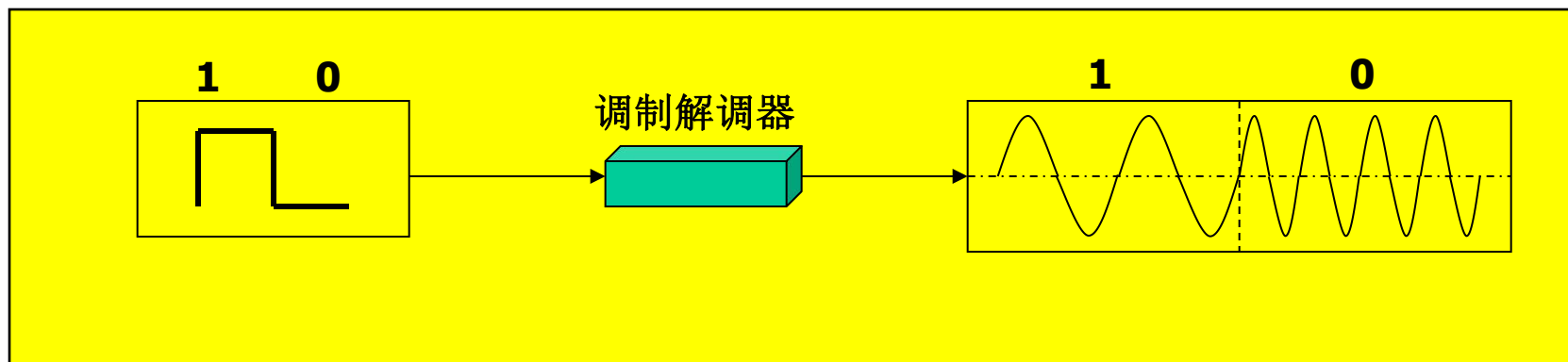


2FSK

■ 2FSK基本原理

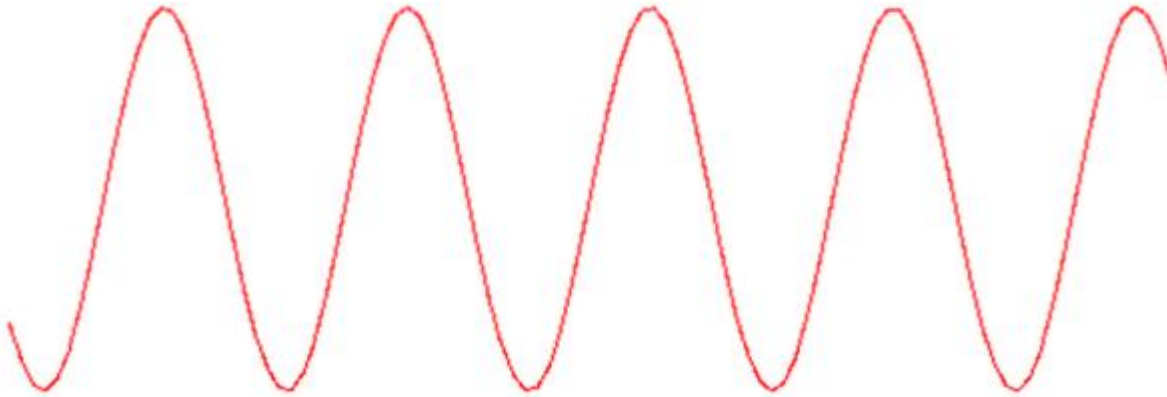
- ◆ 正弦载波的频率随二进制基带信号在**f1**和**f2**两个频点间变化，产生**2FSK**信号

■ 2FSK信号的时域波形

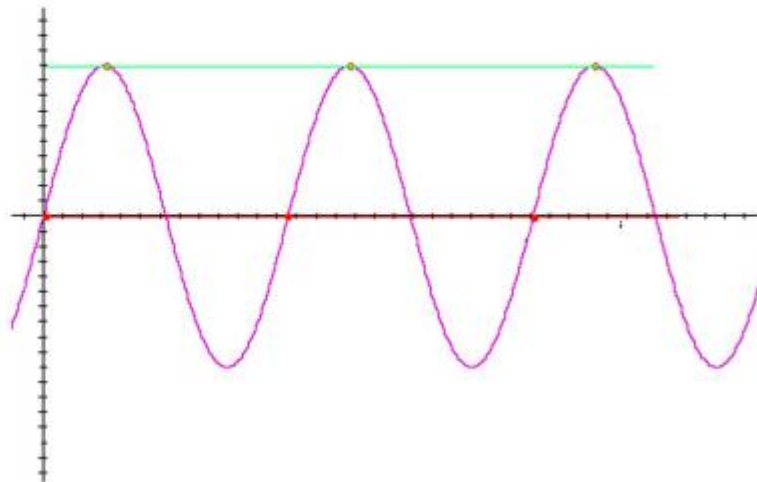


Nyquist's Sampling Theorem

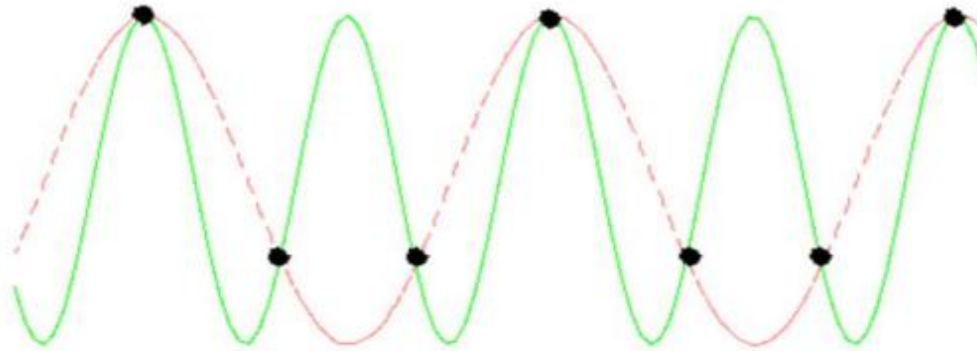
- ◆ A sine wave:



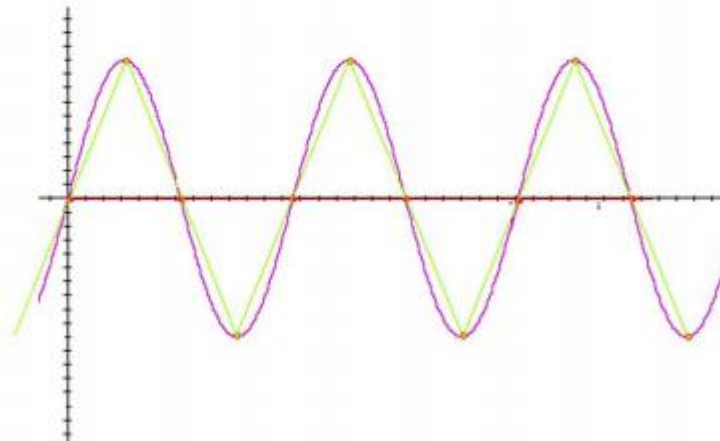
If we sample at 1 time per cycle, we can think the reconstructed signal as a constant:



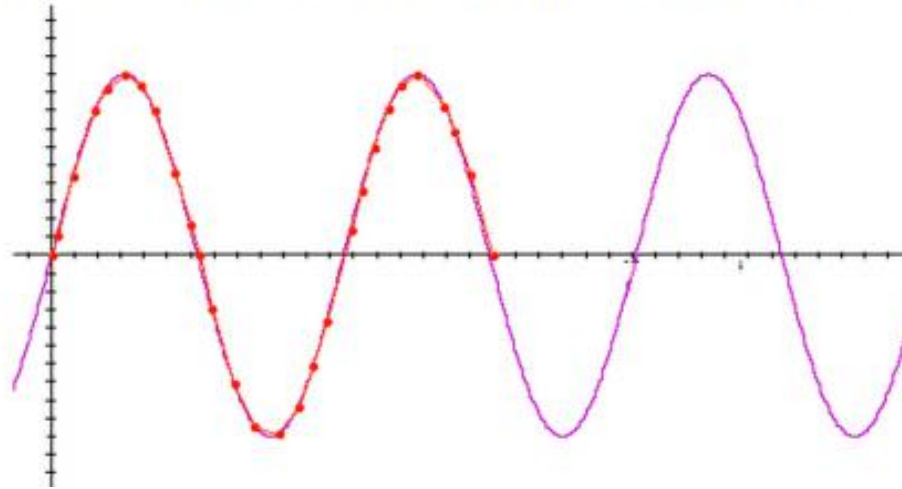
If we sample at 1.5 times per cycle, we can think it's a lower frequency sine wave:



If we sample at twice the frequency, the **Nyquist Rate**, we start to make some progress in terms of being able to replicate the signal:



For lossless digitisation, the sampling rate should be at least twice the maximum frequency responses. Indeed, many times more the better:

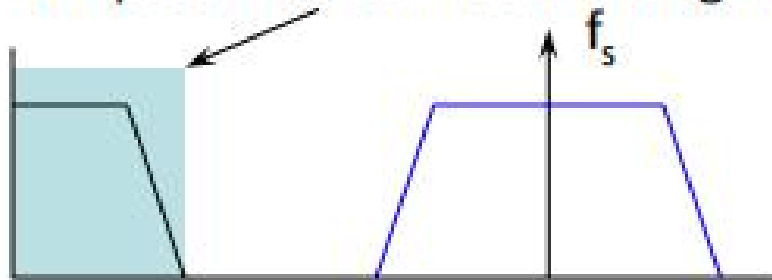


Nyquist's Sampling Theorem

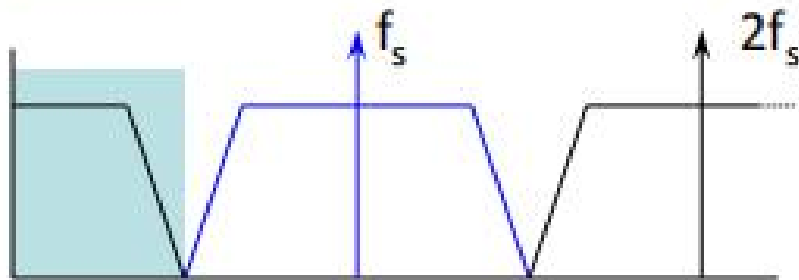


original signal

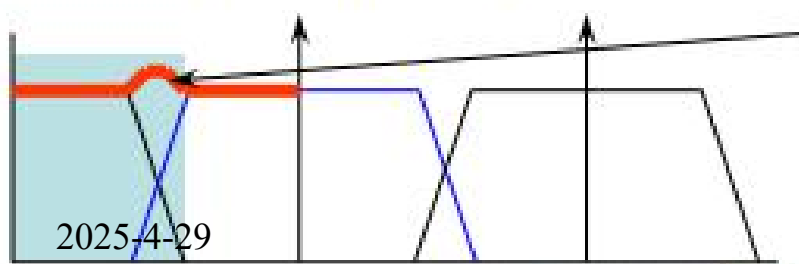
low-pass filter can recover original signal



signal sampled with $f_s > 2 f_{\max}$



signal sampled with $f_s = 2 f_{\max}$



recovery not possible - spectra interfere

signal sampled with $f_s < 2 f_{\max}$
aliasing distortion occurs